Biochemistry



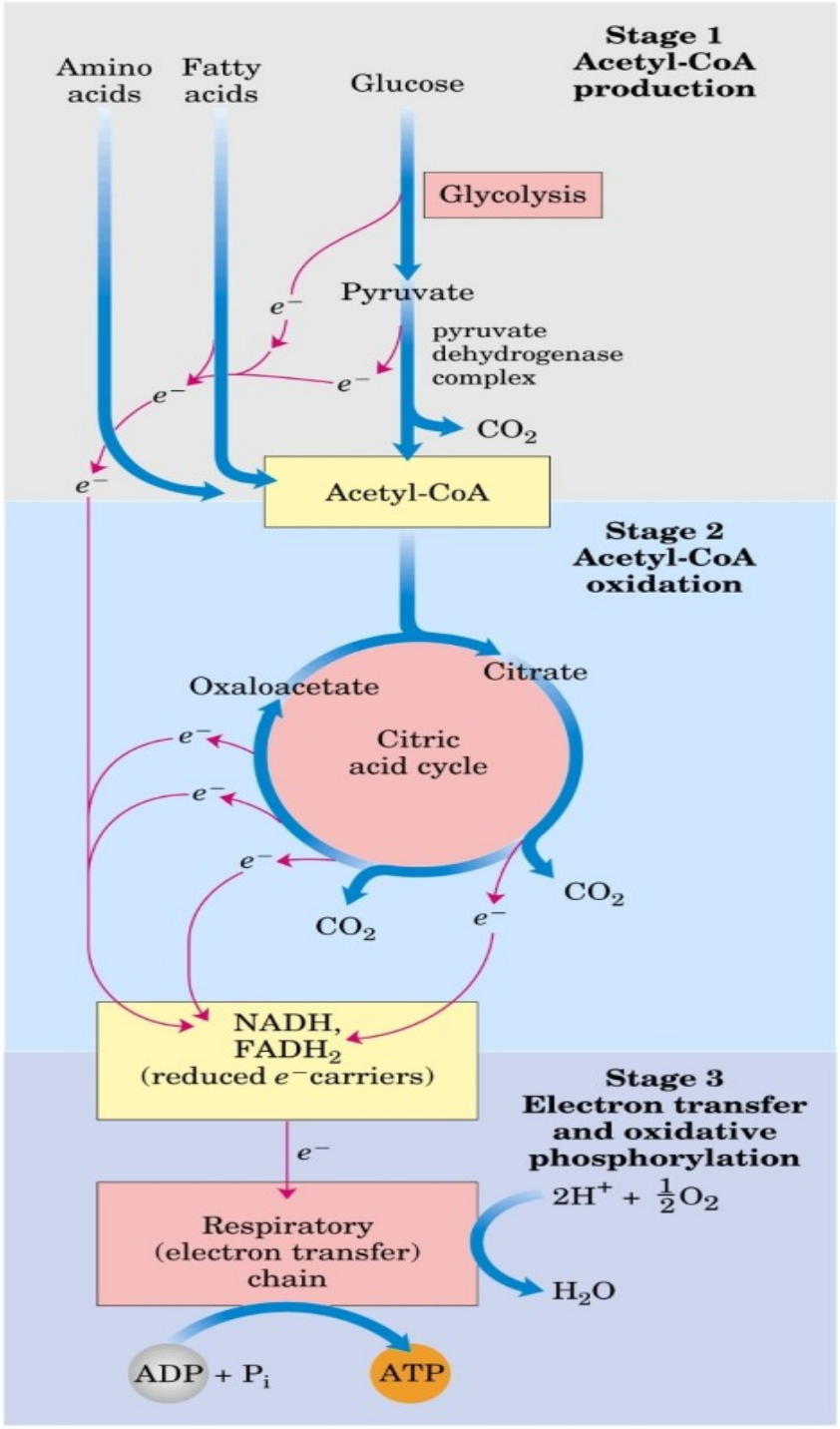
Section L
Respiration and energy

Guodong Yang

Biochemistry and Molecular Biology, College of Life Sciences

gdyang@sdau.edu.cn

13854808762


Stage 1
Acetyl-CoA
production
Amino
acids
Fatty
acids
Glucose
Glycolysis
Pyruvate
pyruvate
dehydrogenase
complex
e^-
e^-
e^-
e^-
CO_2
Acetyl-CoA
Stage 2
Acetyl-CoA
oxidation
Citrate
Oxaloacetate
Citric
acid cycle
e^-
e^-
e^-
e^-
CO_2
CO_2
NADH,
$FADH_2$
(reduced e^- carriers)
Stage 3
Electron transfer
and oxidative
phosphorylation
e^-
$2H^+ + \frac{1}{2}O_2$
Respiratory
(electron transfer)
chain
H_2O
ADP + P_i
ATP

Biochemist

1. Cells need energy to do all their work

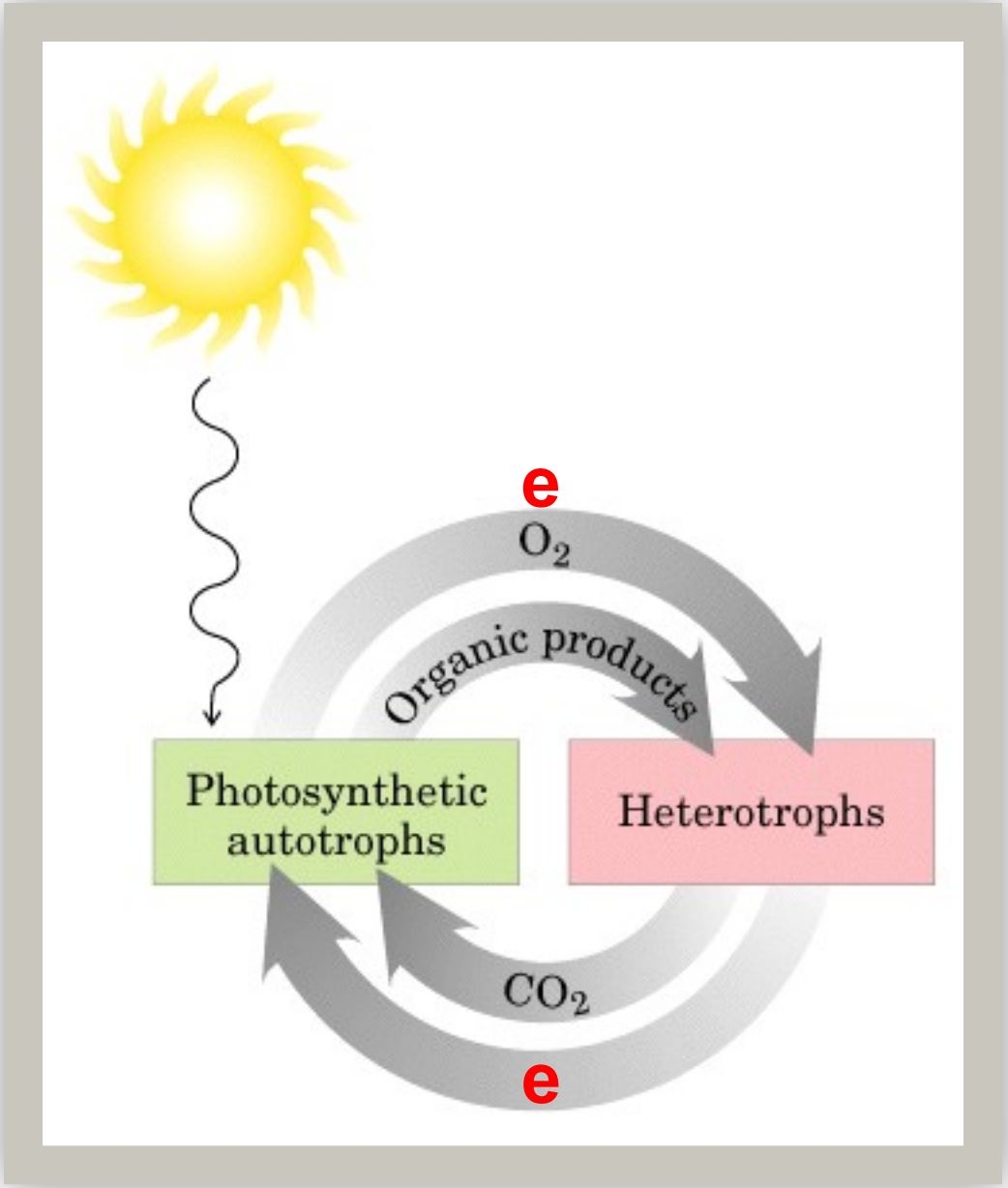
- To generate and maintain its highly ordered structure (biosynthesis of macromolecules).
- To generate all kinds of movement.
- To generate concentration and electrical gradients across cell membranes.
- To maintain a body temperature.
- To generate light in some animals.

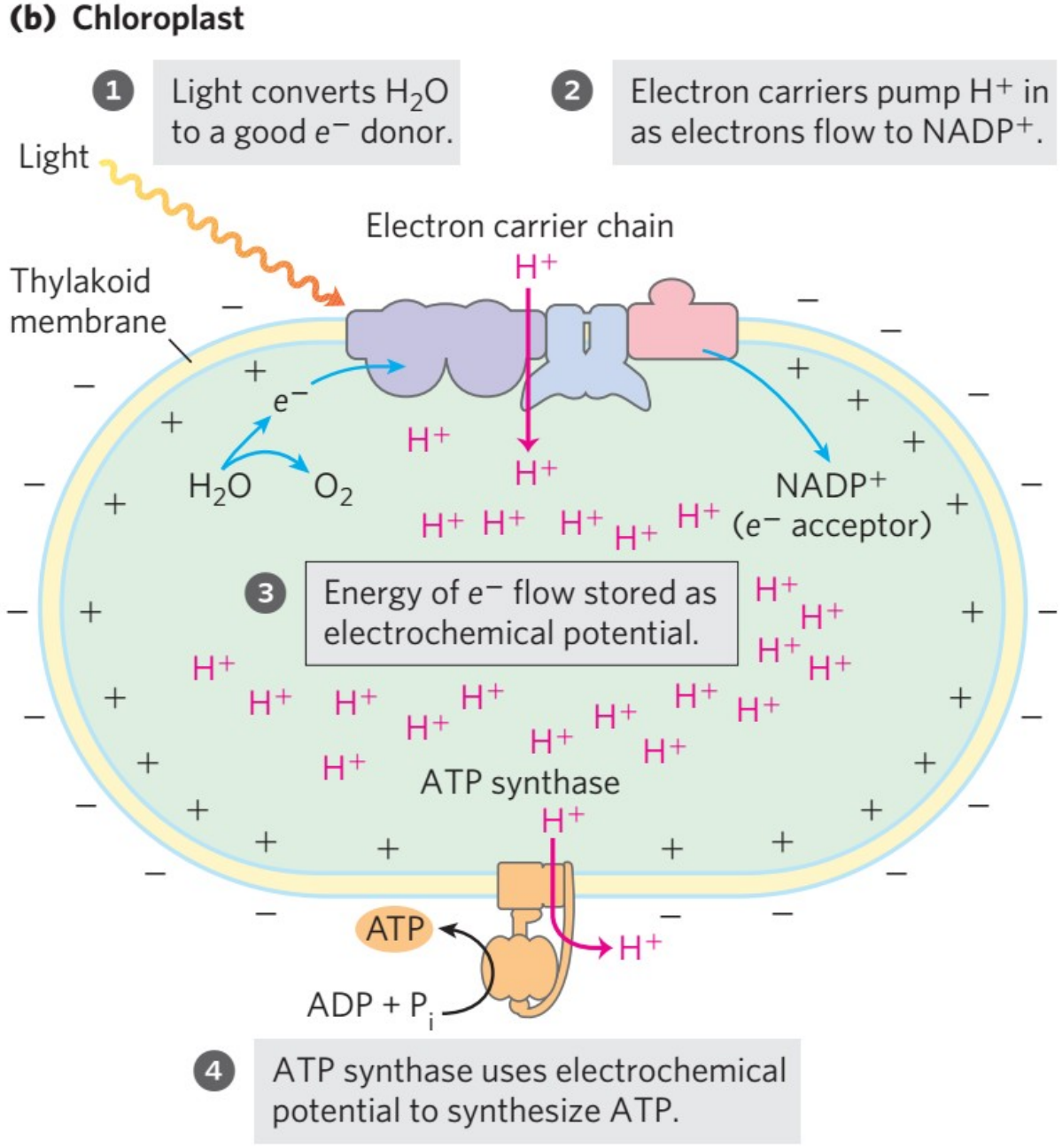


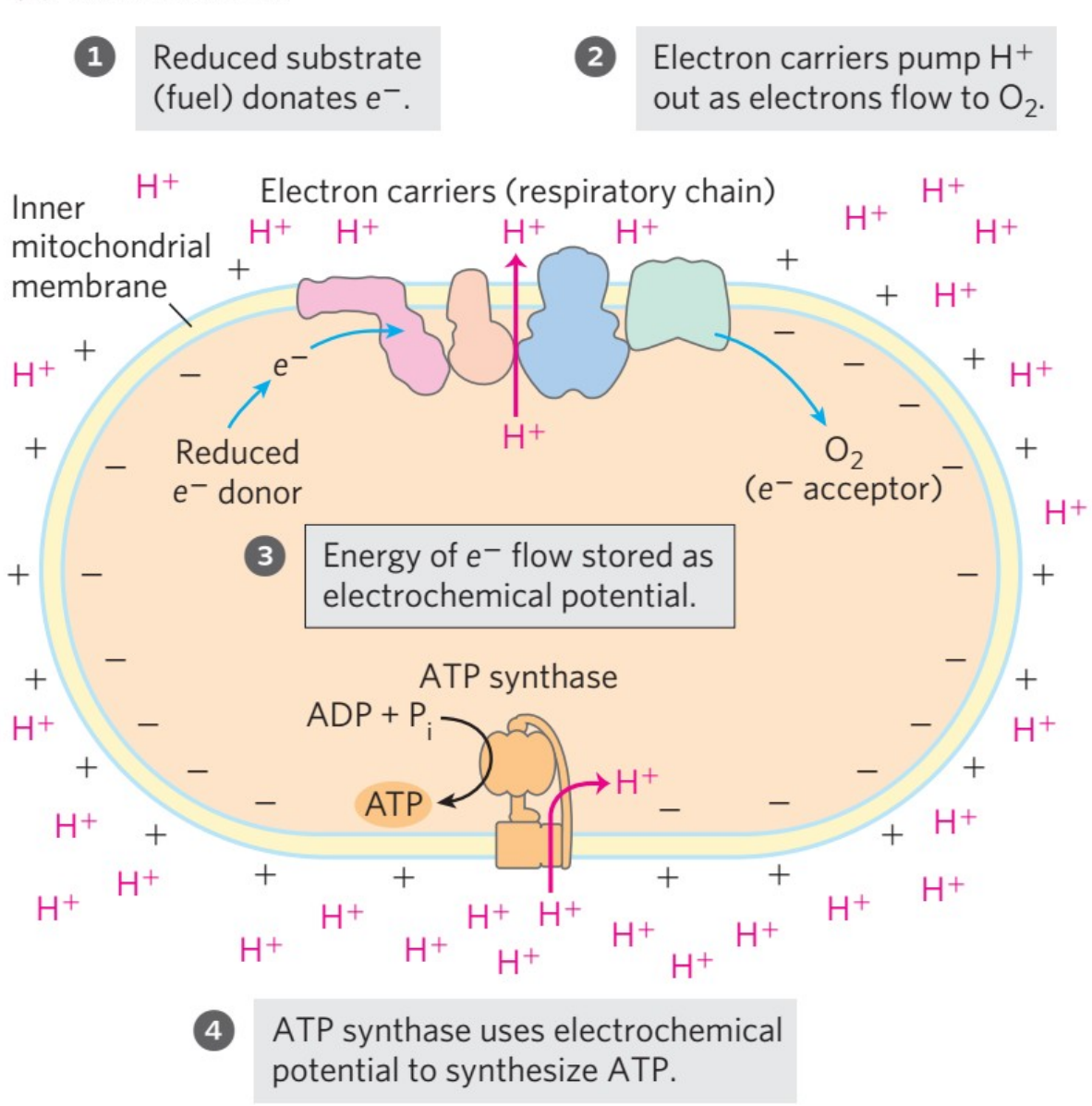


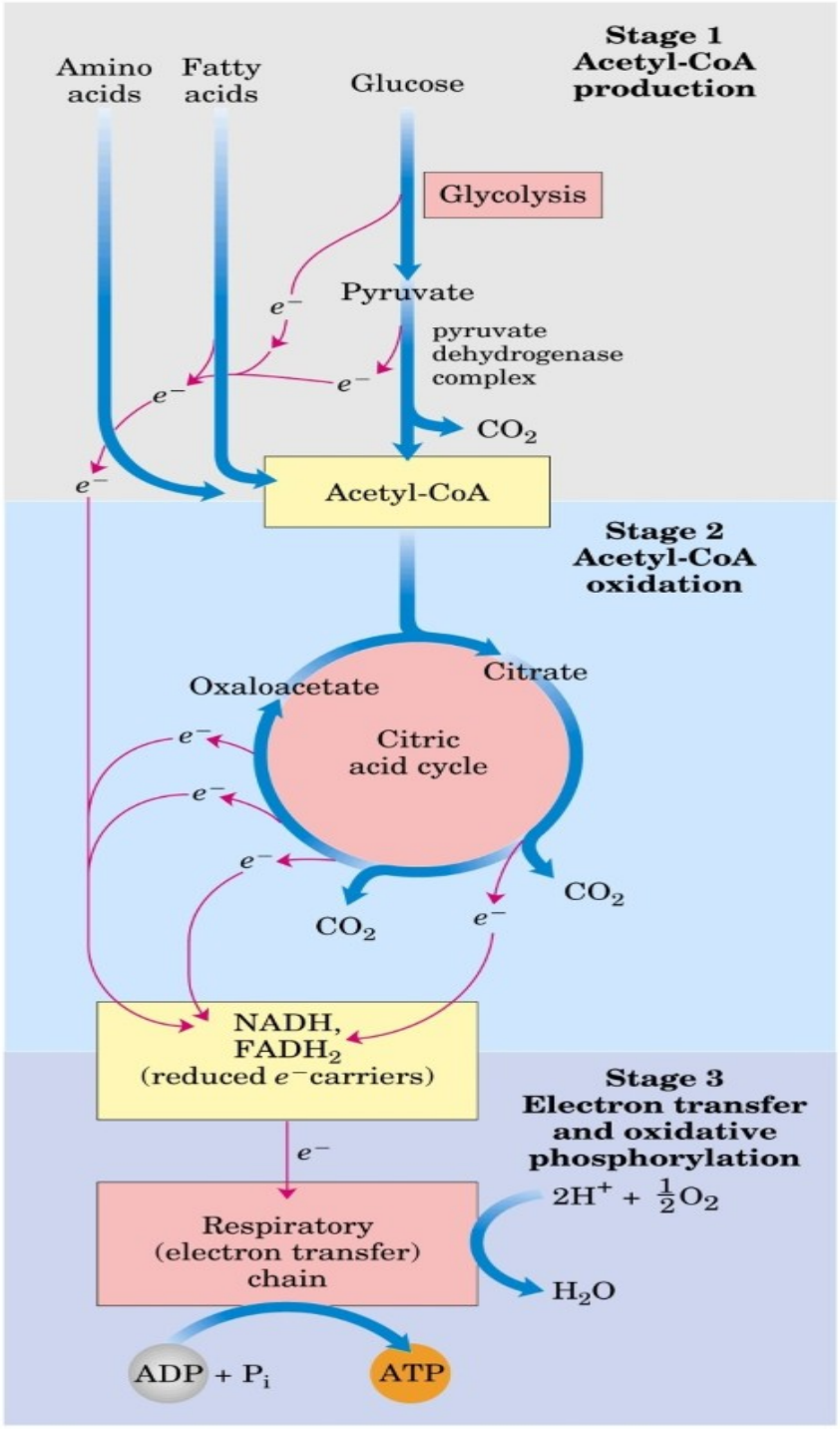
Firefly

Biochemistry


e
O_2
Organic products
Photosynthetic autotrophs
Heterotrophs
CO_2
e

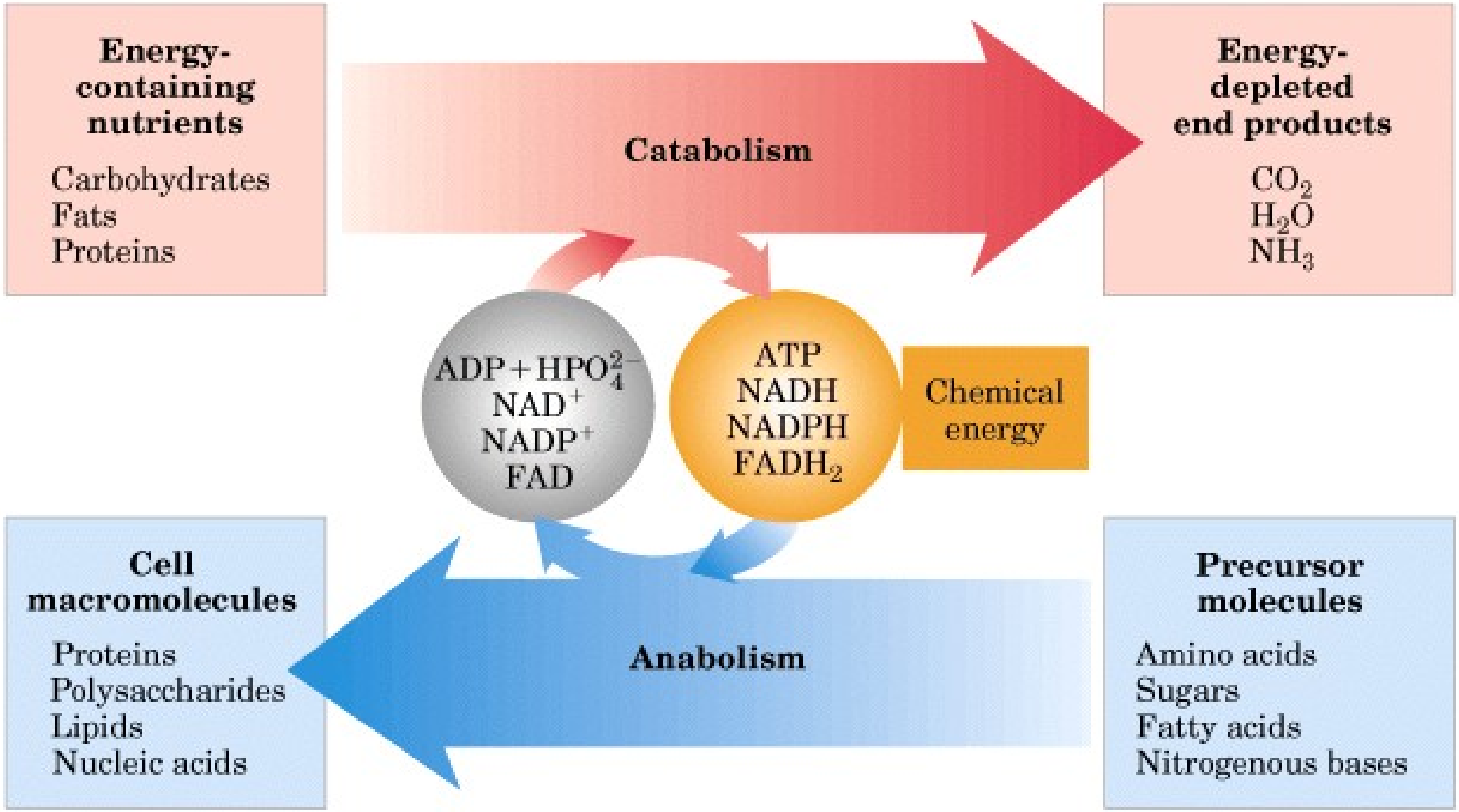

(b) Chloroplast
1 Light converts H_2O to a good e^- donor.
2 Electron carriers pump H^+ in as electrons flow to $NADP^+$.
Light
Electron carrier chain
Thylakoid membrane
e^-
H_2O
O_2
$NADP^+$ (e^- acceptor)
3 Energy of e^- flow stored as electrochemical potential.
ATP synthase
ATP
ADP + P_i
4 ATP synthase uses electrochemical potential to synthesize ATP.

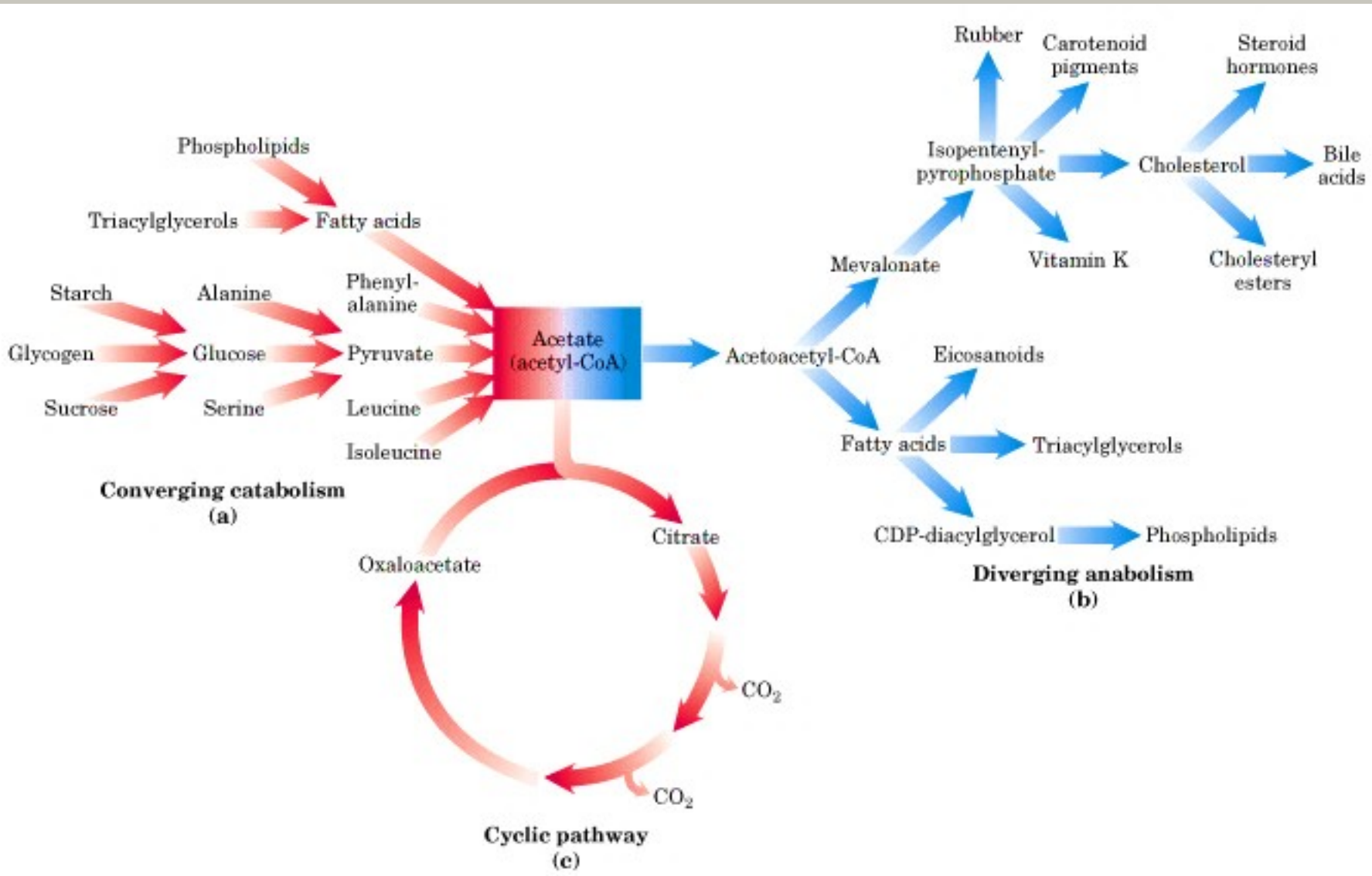

(a) Mitochondrion
1 Reduced substrate (fuel) donates e⁻.
2 Electron carriers pump H⁺ out as electrons flow to O2.
Inner mitochondrial membrane
Electron carriers (respiratory chain)
e⁻
Reduced e⁻ donor
O2 (e⁻ acceptor)
3 Energy of e⁻ flow stored as electrochemical potential.
ATP synthase
ADP + Pi
ATP
4 ATP synthase uses electrochemical potential to synthesize ATP.


Stage 1
Acetyl-CoA
production
Amino
acids
Fatty
acids
Glucose
Glycolysis
Pyruvate
pyruvate
dehydrogenase
complex
e^-
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e^-
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CO_2
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Stage 2
Acetyl-CoA
oxidation
Citrate
Oxaloacetate
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acid cycle
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Stage 3
Electron transfer
and oxidative
phosphorylation
e^-
$2H^+ + \frac{1}{2}O_2$
Respiratory
(electron transfer)
chain
H_2O
ADP + P_i
ATP

Biochemist

metabolism


Energy-containing nutrients
Carbohydrates
Fats
Proteins
Catabolism
Energy-depleted end products
CO_2
H_2O
NH_3
$ADP + HPO_4^{2-}$
NAD^+
$NADP^+$
FAD
ATP
NADH
NADPH
$FADH_2$
Chemical energy
Cell macromolecules
Proteins
Polysaccharides
Lipids
Nucleic acids
Anabolism
Precursor molecules
Amino acids
Sugars
Fatty acids
Nitrogenous bases


Phospholipids
Triacylglycerols
Fatty acids
Starch
Alanine
Phenyl-alanine
Glycogen
Glucose
Pyruvate
Sucrose
Serine
Leucine
Isoleucine
Converging catabolism
(a)
Acetate (acetyl-CoA)
Acetoacetyl-CoA
Mevalonate
Isopentenyl-pyrophosphate
Rubber
Carotenoid pigments
Steroid hormones
Cholesterol
Bile acids
Vitamin K
Cholesteryl esters
Eicosanoids
Fatty acids
Triacylglycerols
CDP-diacylglycerol
Phospholipids
Diverging anabolism
(b)
Citrate
Oxaloacetate
CO_2
CO_2
Cyclic pathway
(c)

- **Bioenergetics (生物能学)** : the quantitative study of energy transductions in living cells and the chemical nature underlying these processes.



2. Cells have to use chemical energy to do all their work

- Antoine Lavoisier`s （1743－1794） insight on animal respiration in the 18th century: it is nothing but a slow combustion of carbon and hydrogen (the same nature as a lighting candle).
- Living cells are generally held at constant temperature and pressure: **chemical energy** (free energy) has to be used by living organisms, no thermal energy, neither mechanical energy is available to do work in cells.



- Biological energy transformation obey the two basic laws of thermodynamics revealed by physicists and chemists in the 19th century: energy can neither be created nor be destroyed (but conserved); energy conversion is never 100% efficient (some will always be wasted in increasing the disorder or "entropy" of the universe).
- **The free energy** concept of thermodynamic is more important to biochemists than to chemists (who can always increase the temperature and pressure to make a reaction to occur!).



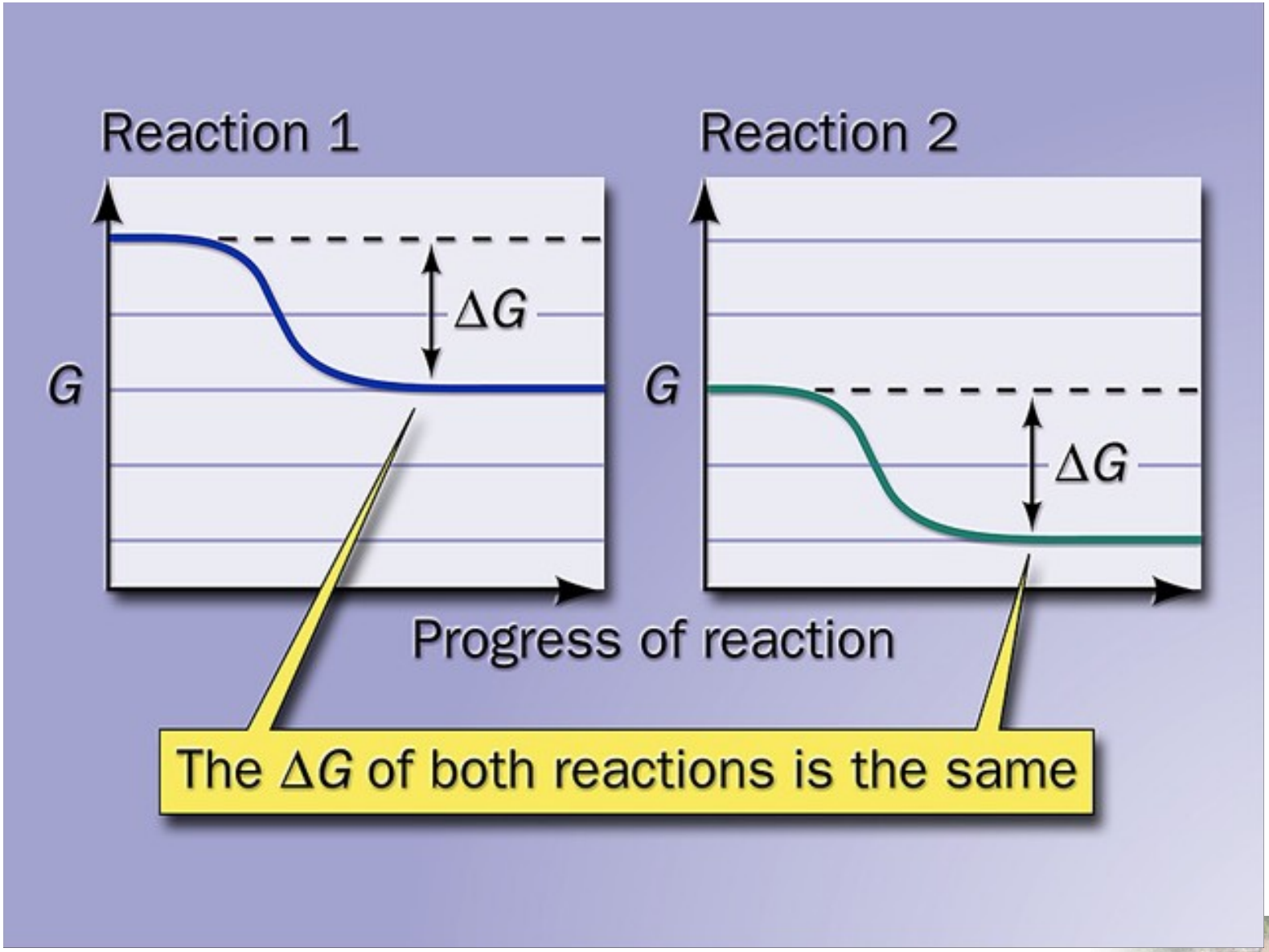
3. Application of the free energy (*G*) concept to biochemical reactions

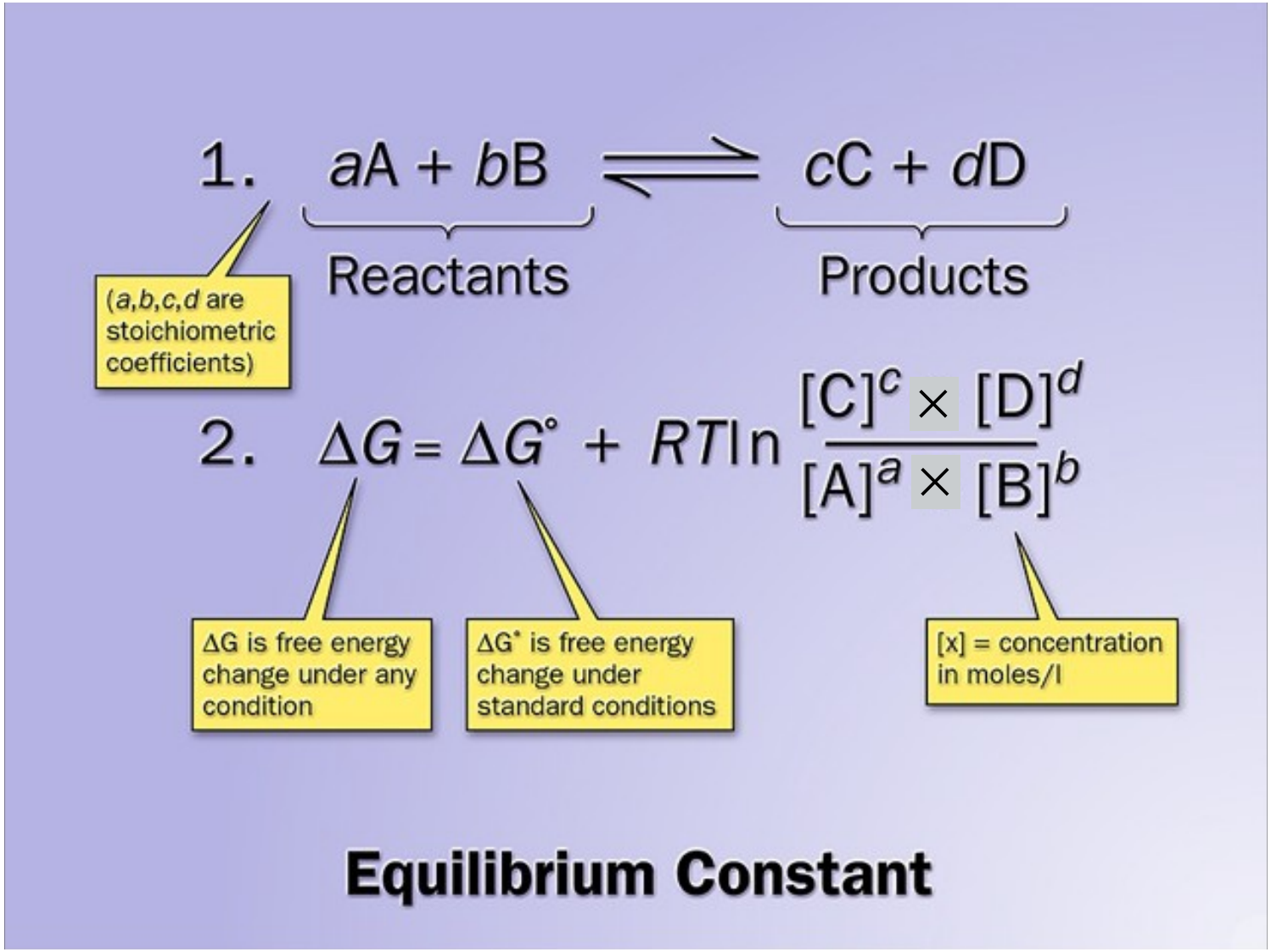
- Free energy (*G*): the amount of energy available to do work during a reaction at a constant temperature and pressure; change, not absolute value can be measured.
- Free energy change (ΔG): The free energy difference between the products and the reactants.



- Gibbs (Josiah W. Gibbs)observation: under constant temperature and pressure, all systems change in such a way that free energy is minimized (products should have less free energy than reactants for a reaction to occur spontaneously, i.e., ΔG has a negative value).




Reaction 1
Reaction 2
G
ΔG
G
ΔG
Progress of reaction
The ΔG of both reactions is the same

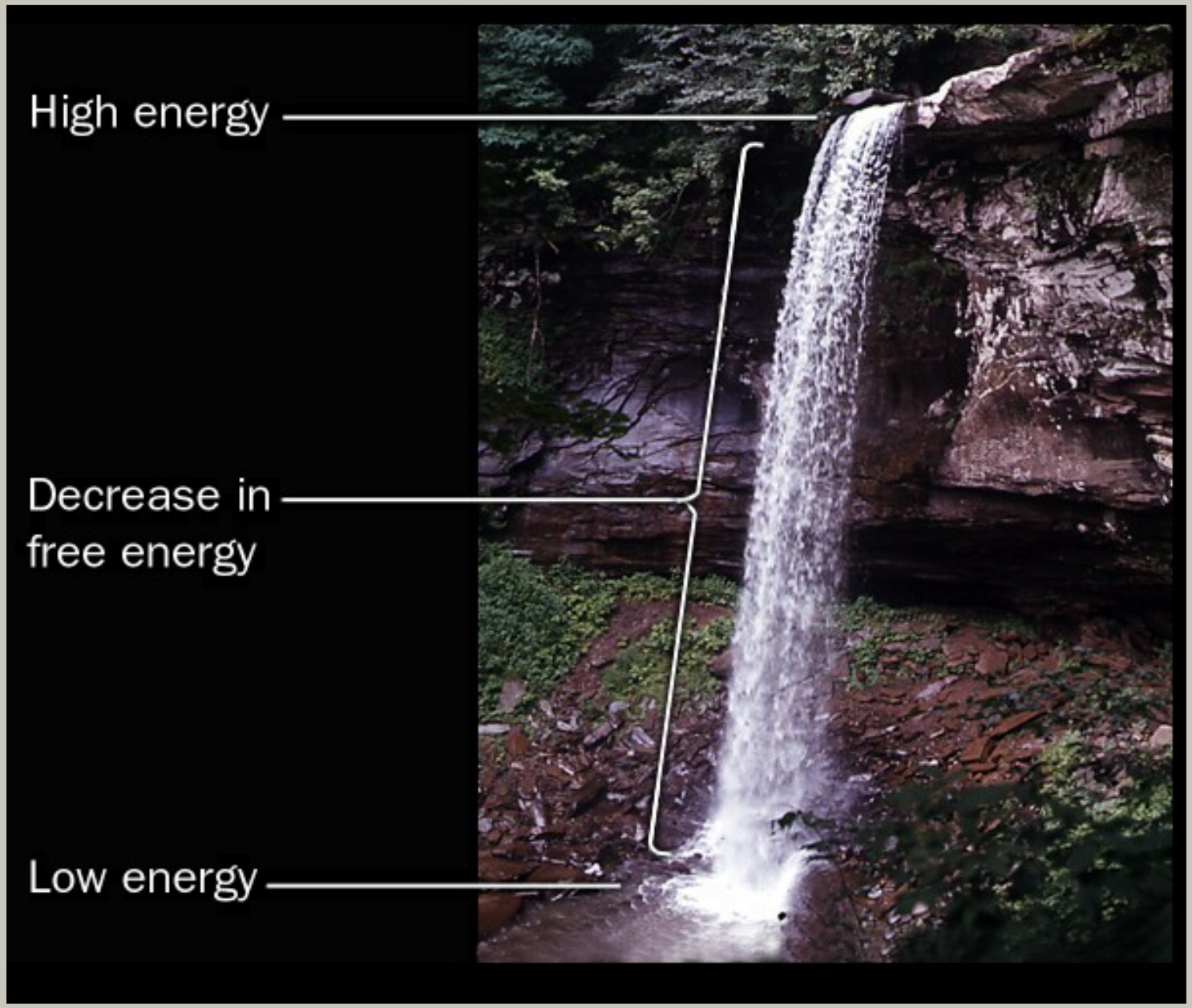

1. aA + bB ⇌ cC + dD
Reactants
Products
(a,b,c,d are stoichiometric coefficients)
2. ΔG = ΔG° + RT ln [C]^c × [D]^d / [A]^a × [B]^b
ΔG is free energy change under any condition
ΔG° is free energy change under standard conditions
[x] = concentration in moles/l
Equilibrium Constant

- **Spontaneity** has nothing to do with rate!
- Standard free energy change in biochemistry ($\Delta G'^{o}$): value of the change in free energy under conditions of 298 K (25°C), 1 atm pressure, pH 7.0 (chemists use pH 0, i.e., the concentration of H^+ they use is 1M, not 10^{-7} M as biochemists use here) and initial concentrations of 1 M for all reactants and products (except H^+).



- The actual free energy chang *(ΔG)*depends on *ΔG′°*, temperature, ratio of product and reactant concentrations (Q):
- $$\Delta G = \Delta G'^{o} + RT \ln Q$$
- Enzymes only speed up thermodynamically favorable reactions (having a negative *ΔG)* !




High energy
Decrease in free energy
Low energy

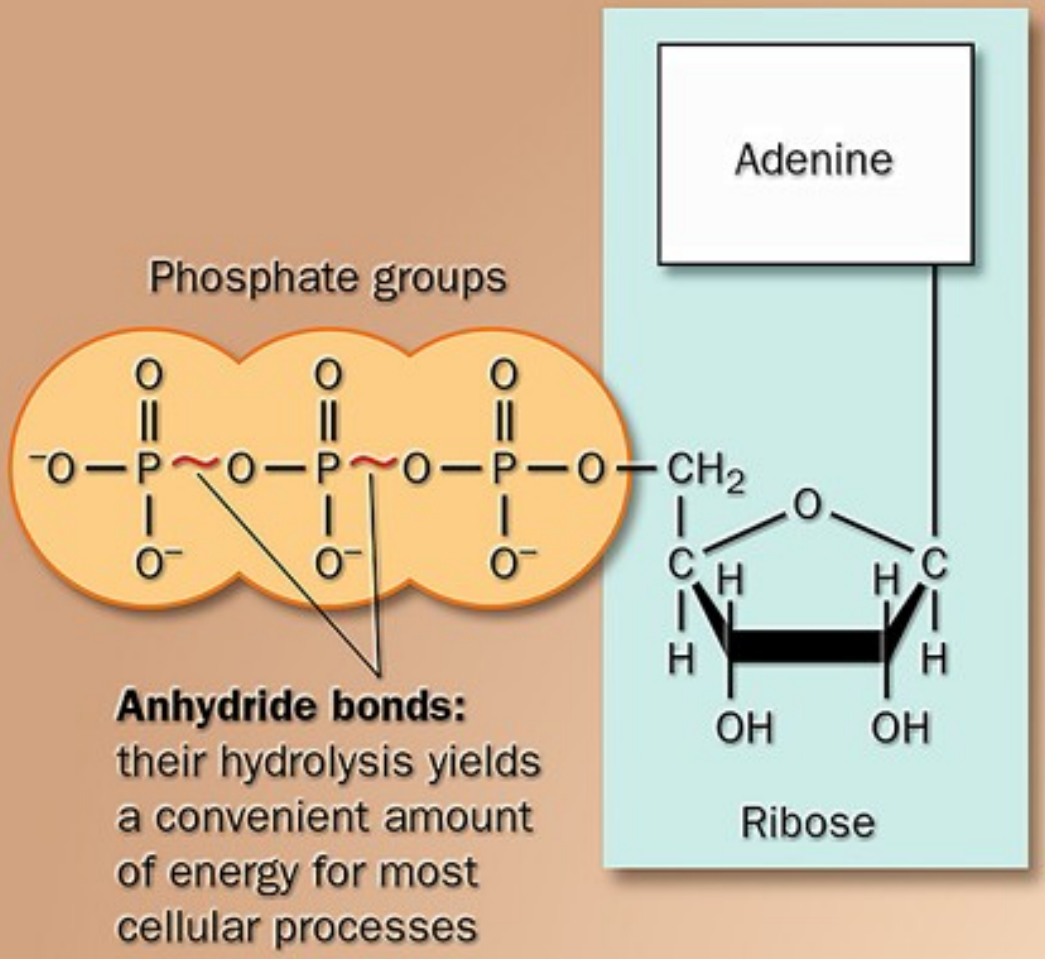
4. ATP is the universal currency for biological energy

- This was first perceived by Fritz Lipmann and Herman Kalckar in 1941 when studying glycolysis.
- Hydrolysis of the two phosphoanhydride (磷酸酐键) bonds in ATP generate more stable products releasing large amount of free energy ($\Delta G'^{o}$ is -30.5 kJ/mol; ΔG in cells is -50 to -65 kJ/mol).



- The ATP molecule is kinetically stable at pH 7 (i.e., it has a high activation energy, $\Delta G^{\ddagger}$ for hydrolysis) and enzyme catalysis is needed for its hydrolysis.
- ATP is not a long-term storage form of free energy in living cells, being consumed within a minute following its formation (phosphocreatine, 磷酸肌酸, act as a energy storage form for longer term).
- A resting human consumes about 40 kg of ATP in 24 hours!





ATP as Energy Currency

ATP has an intermediate phosphoryl group transfer potential.

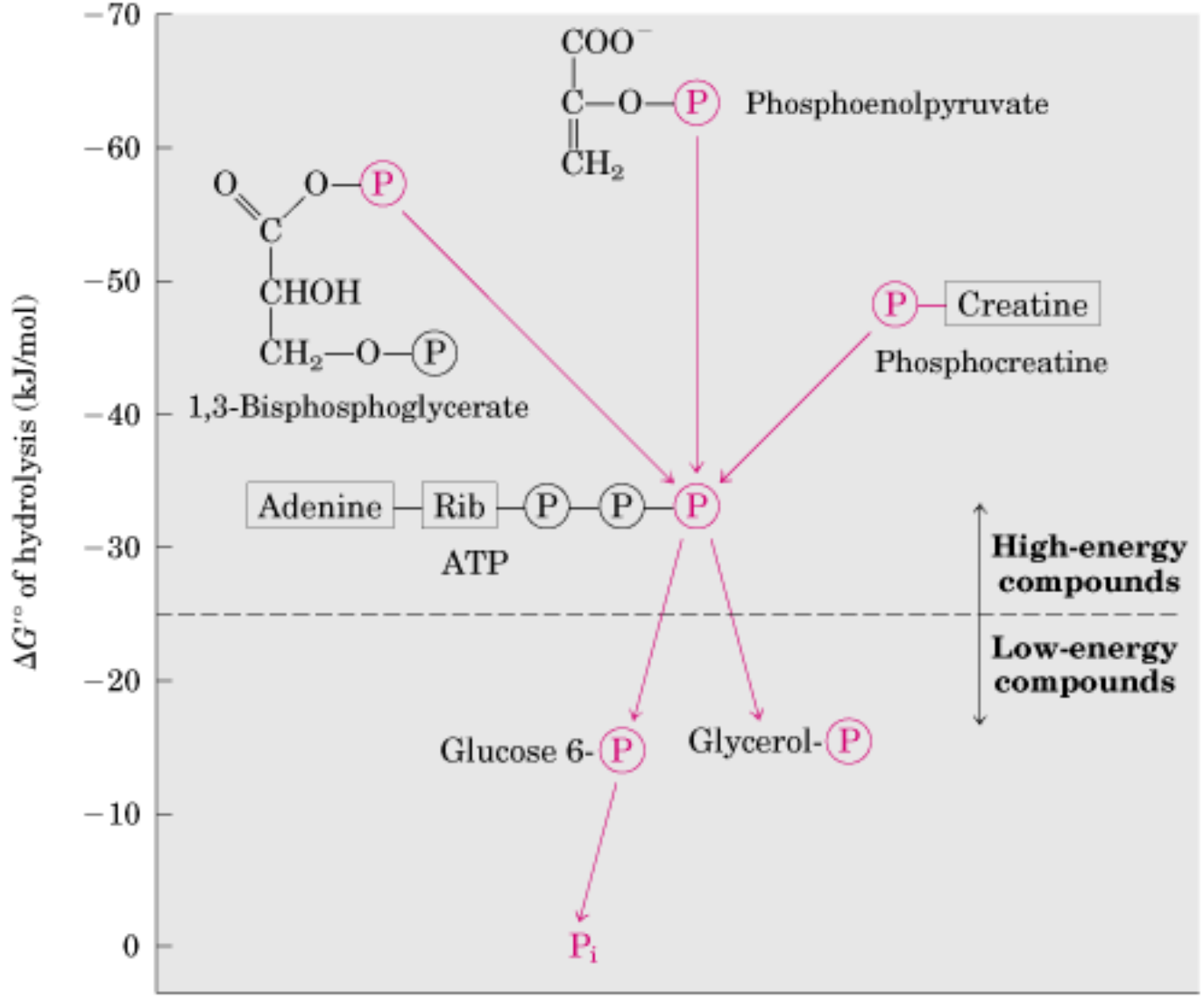
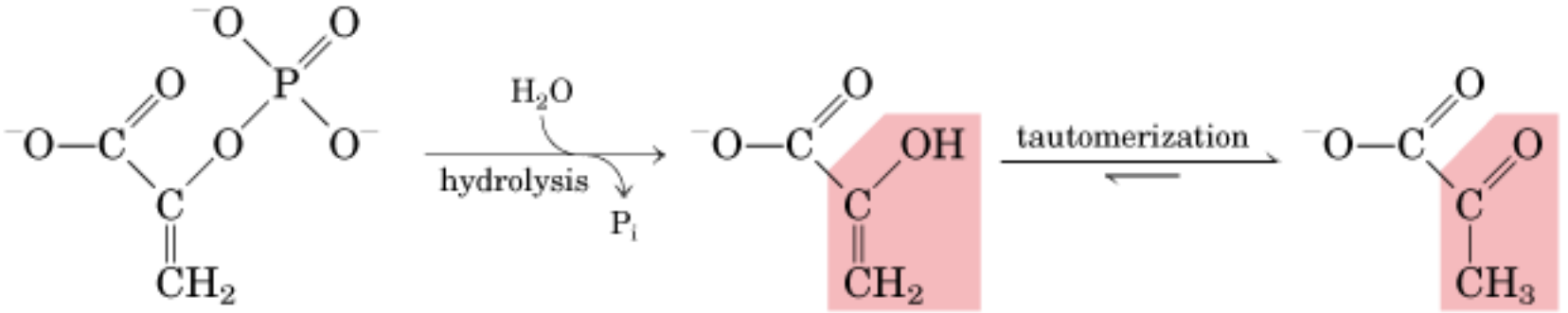


table 14–6

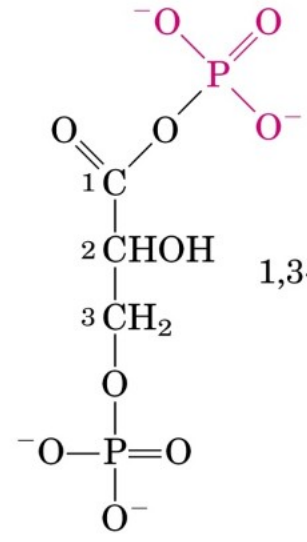
Standard Free Energies of Hydrolysis of Some Phosphorylated Compounds and Acetyl-CoA (a Thioester)

	$\Delta G'^{\circ}$	
	(kJ/mol)	(kcal/mol)
Phosphoenolpyruvate	−61.9	−14.8
1,3-bisphosphoglycerate (⟶ 3-phosphoglycerate + P_i)	−49.3	−11.8
Phosphocreatine	−43.0	−10.3
ADP (⟶ AMP + P_i)	−32.8	−7.8
ATP (⟶ ADP + P_i)	−30.5	−7.3
ATP (⟶ AMP + PP_i)	−45.6	−10.9
AMP (⟶ adenosine + P_i)	−14.2	−3.4
PP_i (⟶ $2P_i$)	−19	−4.0
Glucose 1-phosphate	−20.9	−5.0
Fructose 6-phosphate	−15.9	−3.8
Glucose 6-phosphate	−13.8	−3.3
Glycerol 1-phosphate	−9.2	−2.2
Acetyl-CoA	−31.4	−7.5

Source: Data mostly from Jencks, W.P. (1976) in *Handbook of Biochemistry and Molecular Biology,* 3rd edn (Fasman, G.D., ed.), *Physical and Chemical Data,* Vol. I, pp. 296–304, CRC Press, Boca Raton, FL.

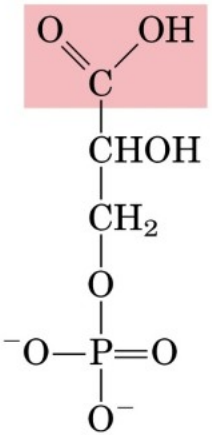


$$PEP^{3-} + H_2O \longrightarrow pyruvate^{-} + P_i^{2-}$$
$$\Delta G'^{\circ} = -61.9 \text{ kJ/mol}$$



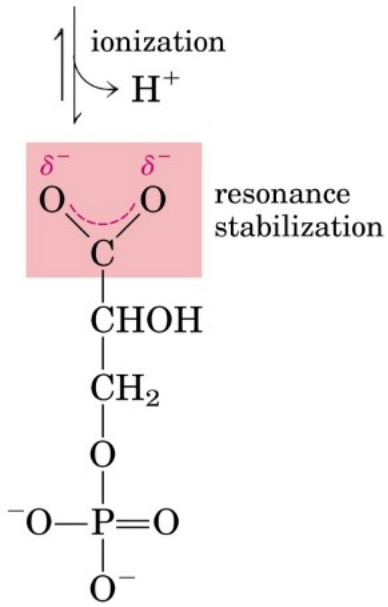
1,3-Bisphosphoglycerate

H_2O hydrolysis P_i



3-Phosphoglyceric acid

ionization H^+



3-Phosphoglycerate

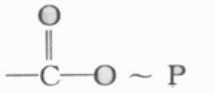
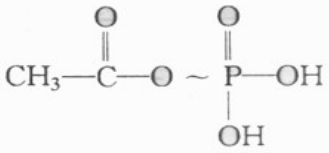
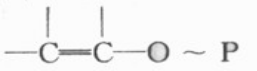
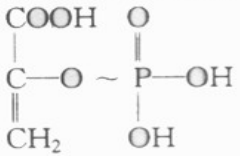
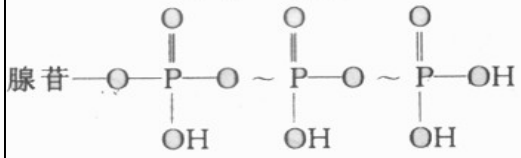
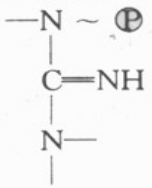
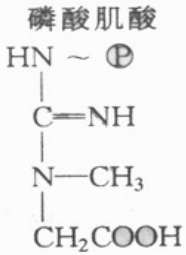
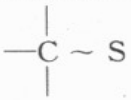
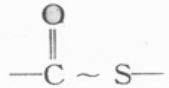
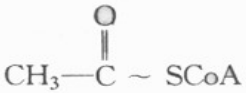
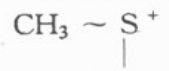
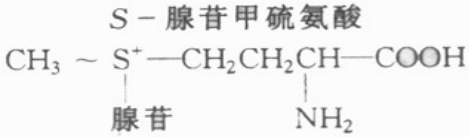
resonance stabilization

ionization H^+

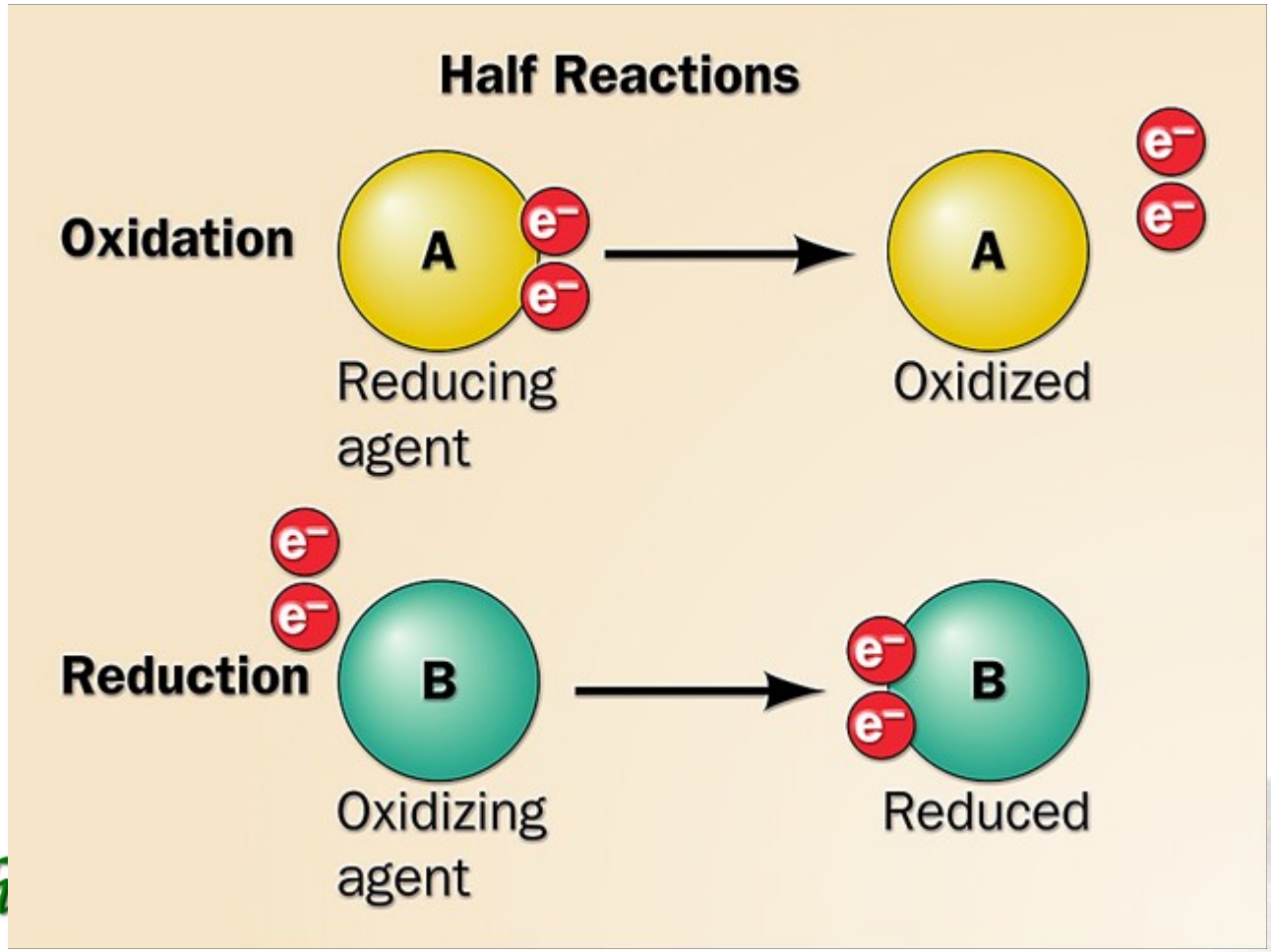
$$\text{1,3-Bisphosphoglycerate}^{4-} + H_2O \longrightarrow \text{3-phosphoglycerate}^{3-} + P_i^{2-} + H^+$$

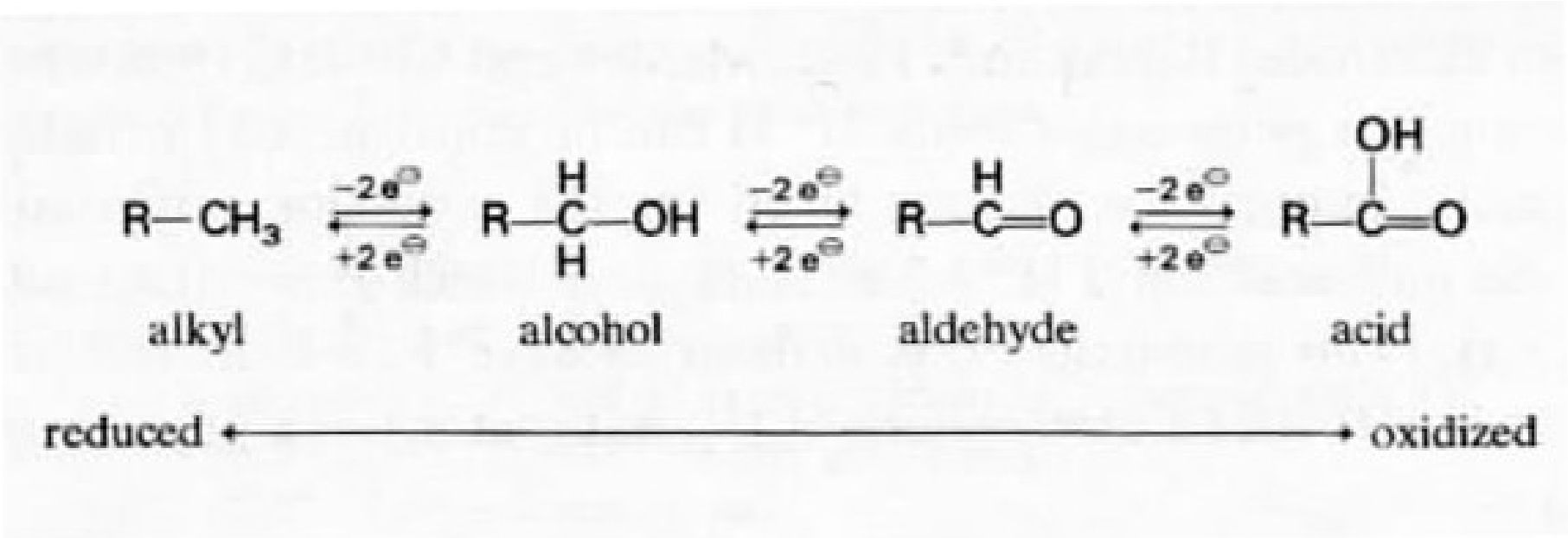
$$\Delta G'^\circ = -49.3 \text{ kJ/mol}$$

高能化合物类型

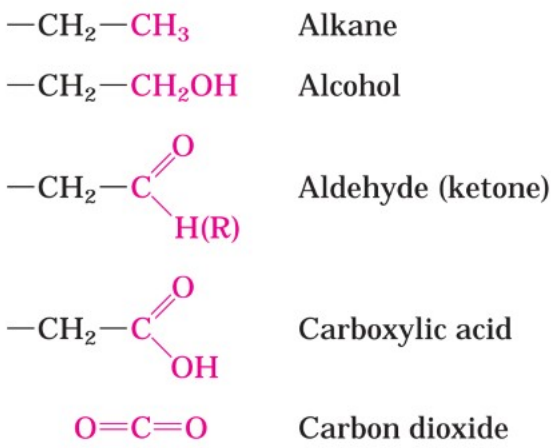
	高能键型	高能化合物举例	水解时释放的标准自由能 $\Delta G^{\ominus\prime}$
磷氧键型 $-O\sim P$	酰基磷酸化合物 	乙酰磷酸 	$-42.3\ kJ\cdot mol^{-1}$ $(-10.1\ kcal\cdot mol^{-1})$
	烯醇式磷酸化合物 	磷酸烯醇式丙酮酸 	$-61.9\ kJ\cdot mol^{-1}$ $(-14.8\ kcal\cdot mol^{-1})$
	焦磷酸化合物 	腺苷三磷酸 	$-30.5\ kJ\cdot mol^{-1}$ $(-7.3\ kcal\cdot mol^{-1})$
磷氮键型 $-N\sim P$	胍基磷酸化合物 	磷酸肌酸 	$-43.1\ kJ\cdot mol^{-1}$ $(-10.3\ kcal\cdot mol^{-1})$
硫碳键型 	硫酯键化合物 	乙酰辅酶 A 	$-31.4\ kJ\cdot mol^{-1}$ $(-7.5\ kcal\cdot mol^{-1})$
	甲硫键化合物 	S－腺苷甲硫氨酸 	$-41.8\ kJ\cdot mol^{-1}$ $(-10.0\ kcal\cdot mol^{-1})$

5. Electron transfer via redox reactions generates biological energy





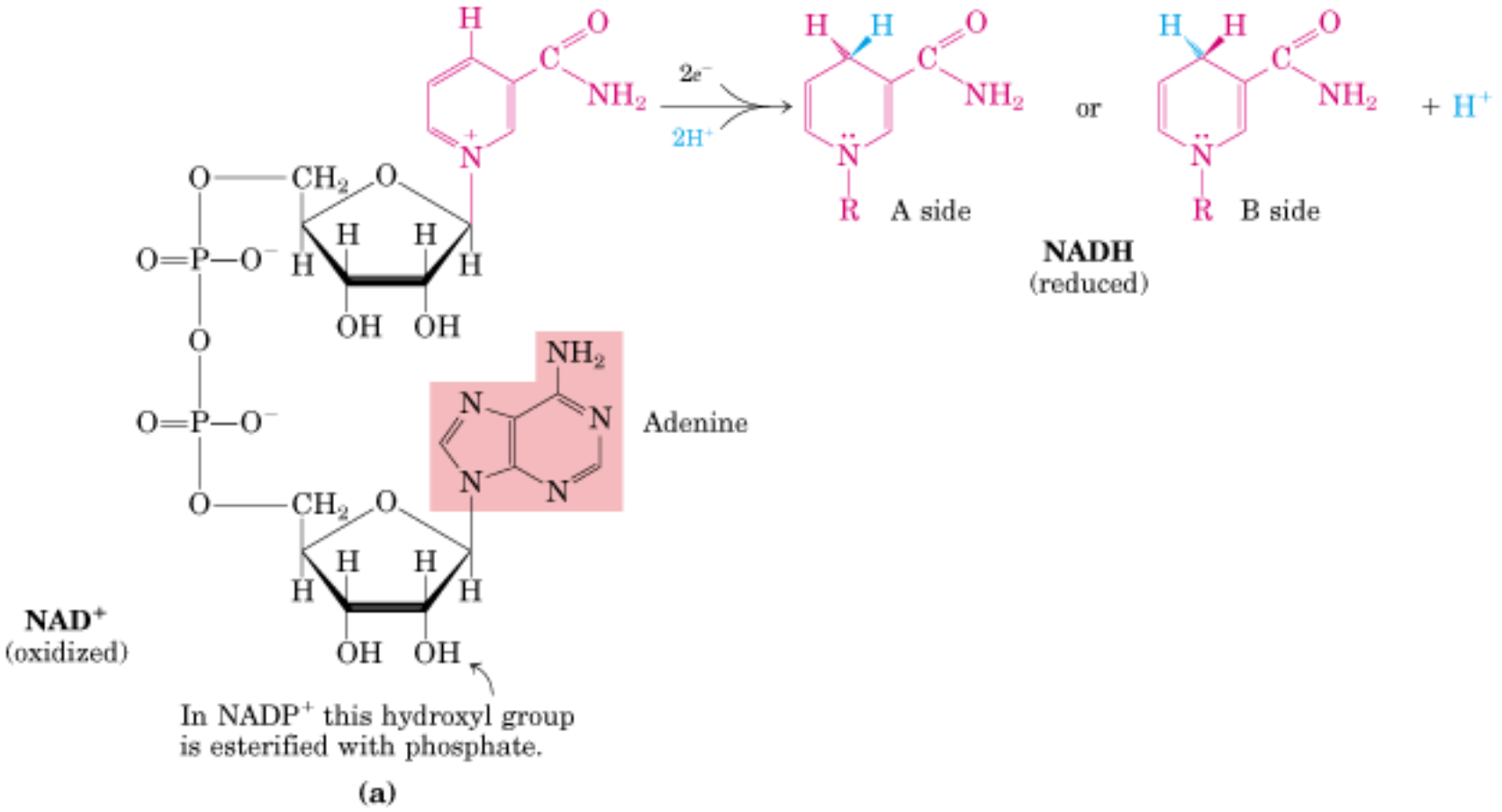
Carbons have various oxidation states, with hydrocarbon being the most reduced and CO_2 being the most oxidized.



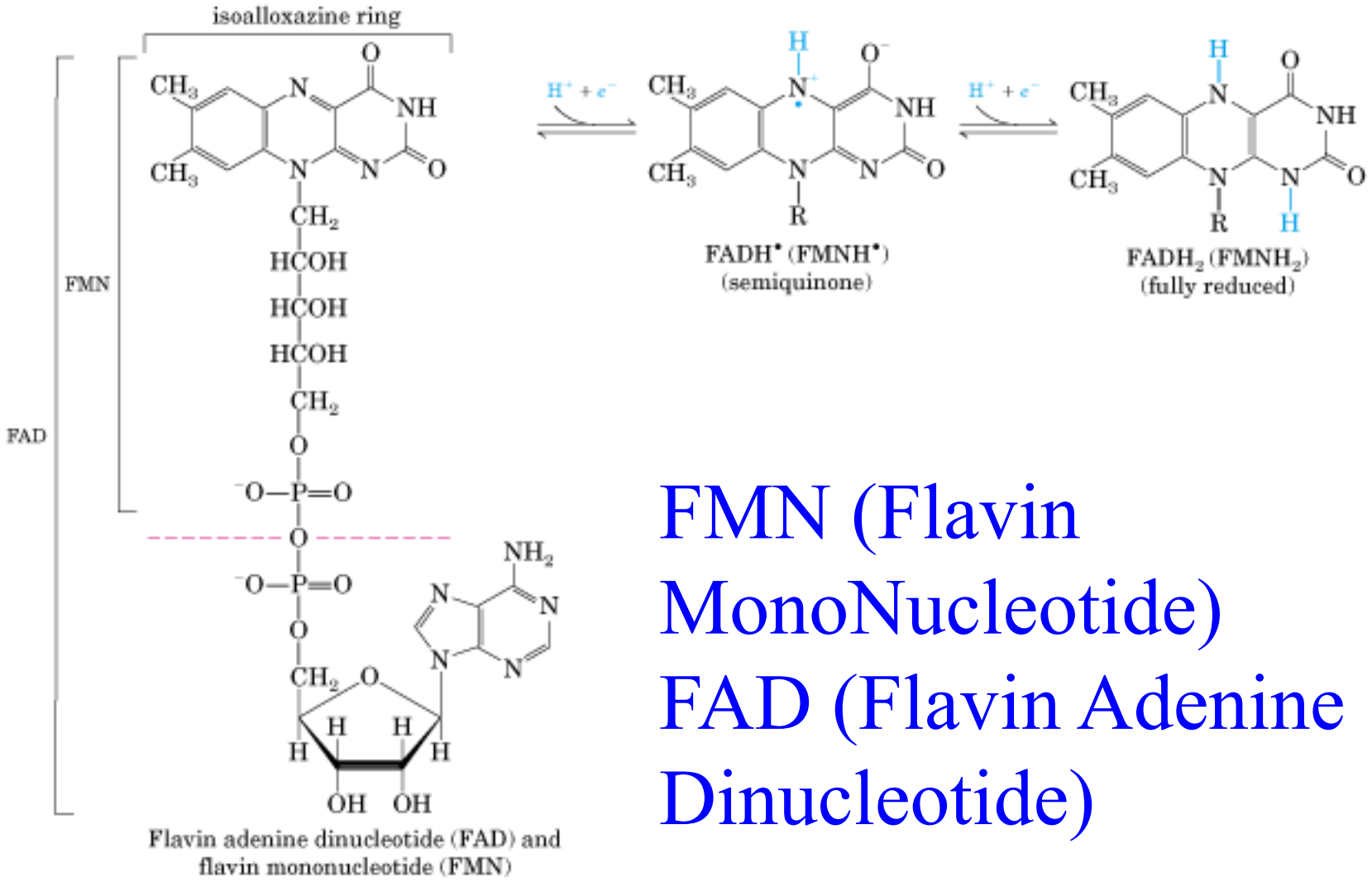


- When electrons flow from a low affinity carrier (reductant) to a high affinity carrier (oxidant), either in an electric battery or in a living cell, energy is released and work can be done.
- Oxidation of energy-rich biological fuels often means dehydrogenation (catalyzed by dehydrogenases, 脱氢酶) from carbons having various oxidation states.
- In the living cells, electrons are transferred directly as electrons (between metal ions), as hydrogen atoms ($H^+ + e^-$), or as a hydride ion ($:H^-$ or $H^+ + 2e^-$).





Nicotinamide is derived From niacin(B5)

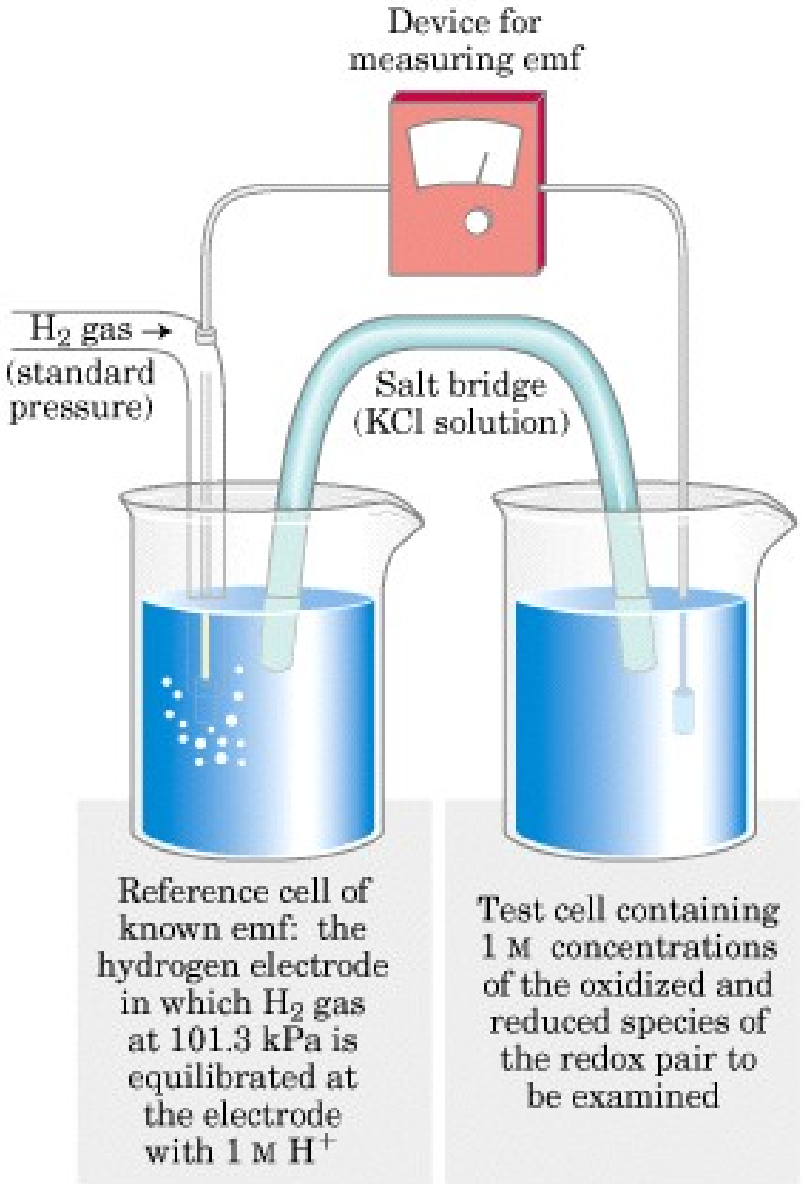


Flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN)

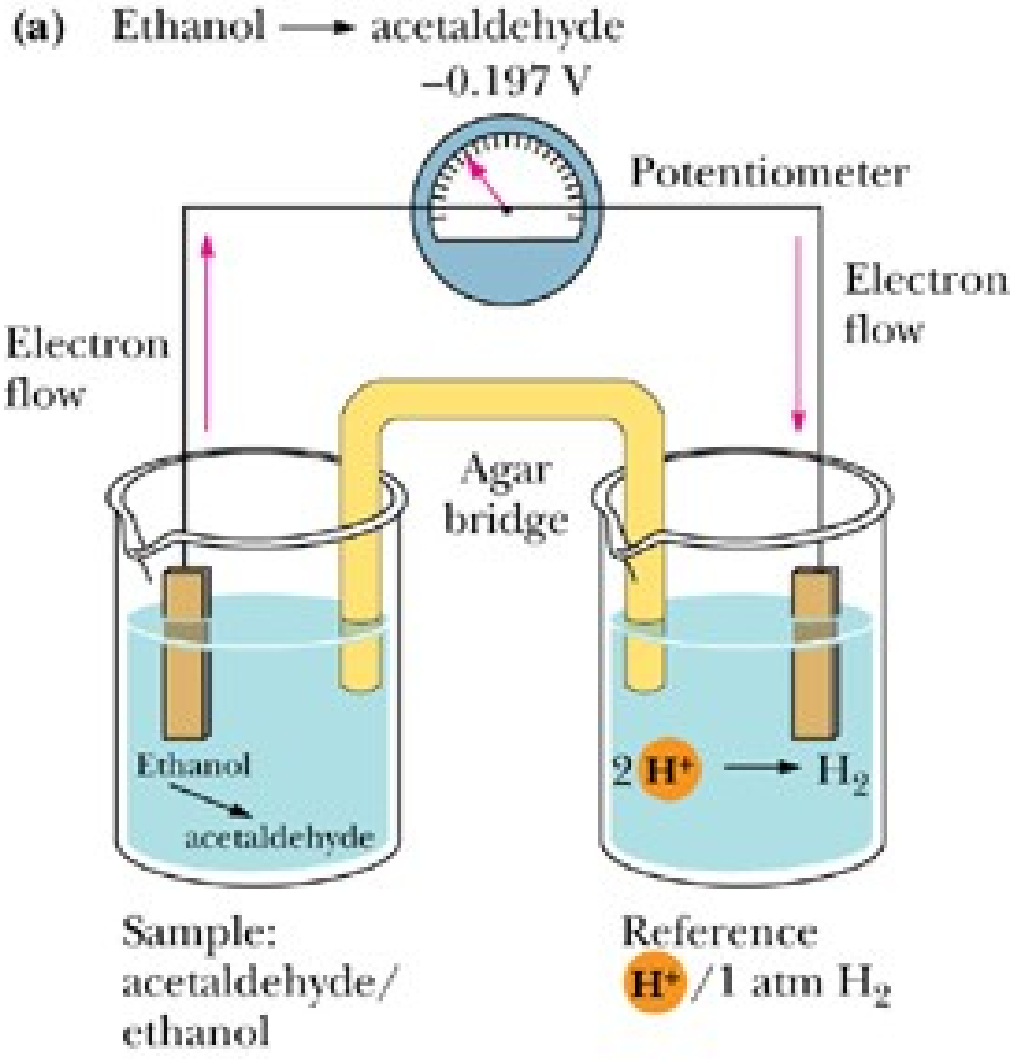
FMN (Flavin MonoNucleotide)
FAD (Flavin Adenine Dinucleotide)

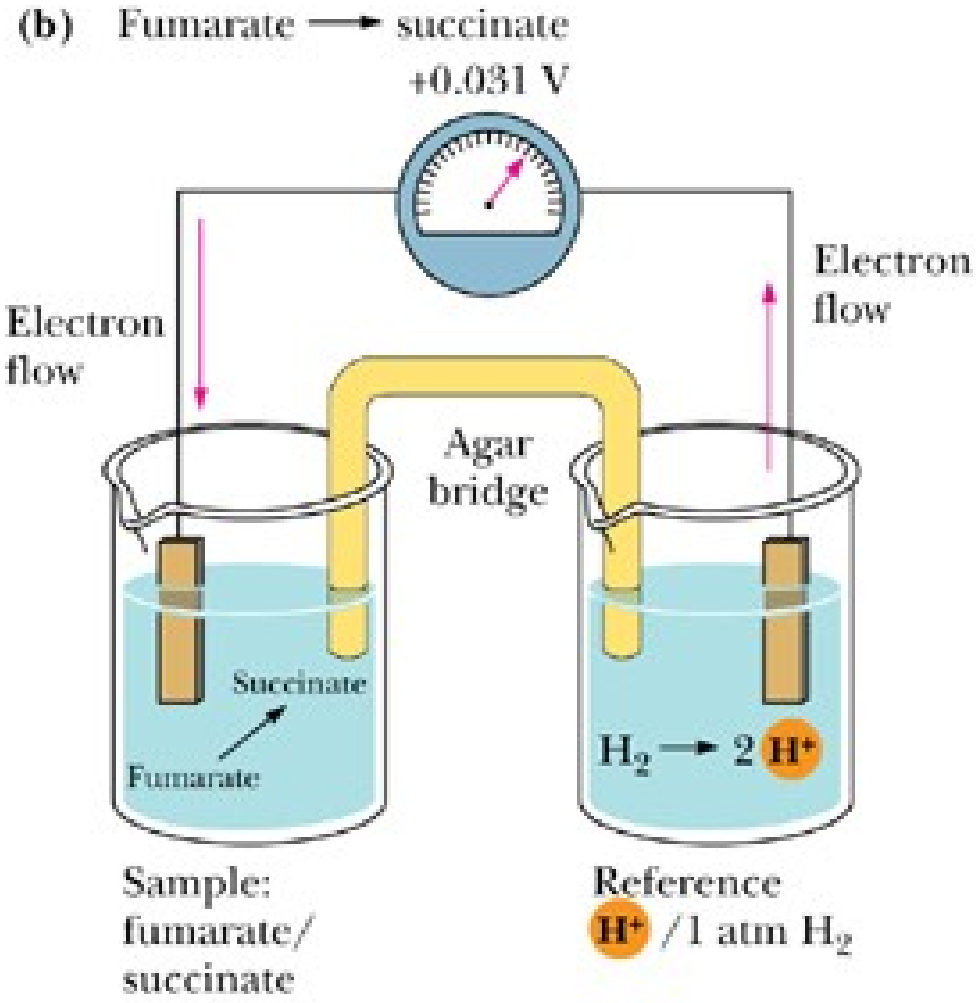
- The affinity for electrons of a compound (in its oxidized form) is indicated by its reduction potential (E).
- Standard reduction potential (E'^{o}) of each oxidant (a constant) is measured by connecting a half-cell having the oxidized and reduced species of the redox pair each at 1 M, or 1 atm for gases, pH 7 to a reference half-cell having 1 M H^+ and 1 atm H_2, whose E'^{o} is arbitrarily assigned as 0.00 V.
- A positive value of E'^{o} indicates a tendency to acquire electron from the reference half cell (with 1M H^+/1atm H_2).





The standard reduction potential (E'^o) of a conjugate redox pair is measured by connecting the sample half cell to the H^+/H_2 reference half cell.


(a) Ethanol → acetaldehyde
−0.197 V
Potentiometer
Electron flow
Electron flow
Agar bridge
Ethanol
acetaldehyde
2 H+ → H2
Sample: acetaldehyde/ ethanol
Reference H+ /1 atm H2


(b) Fumarate → succinate
+0.031 V
Electron flow
Electron flow
Agar bridge
Succinate
Fumarate
H_2 → 2 H^+
Sample: fumarate/ succinate
Reference H^+ / 1 atm H_2

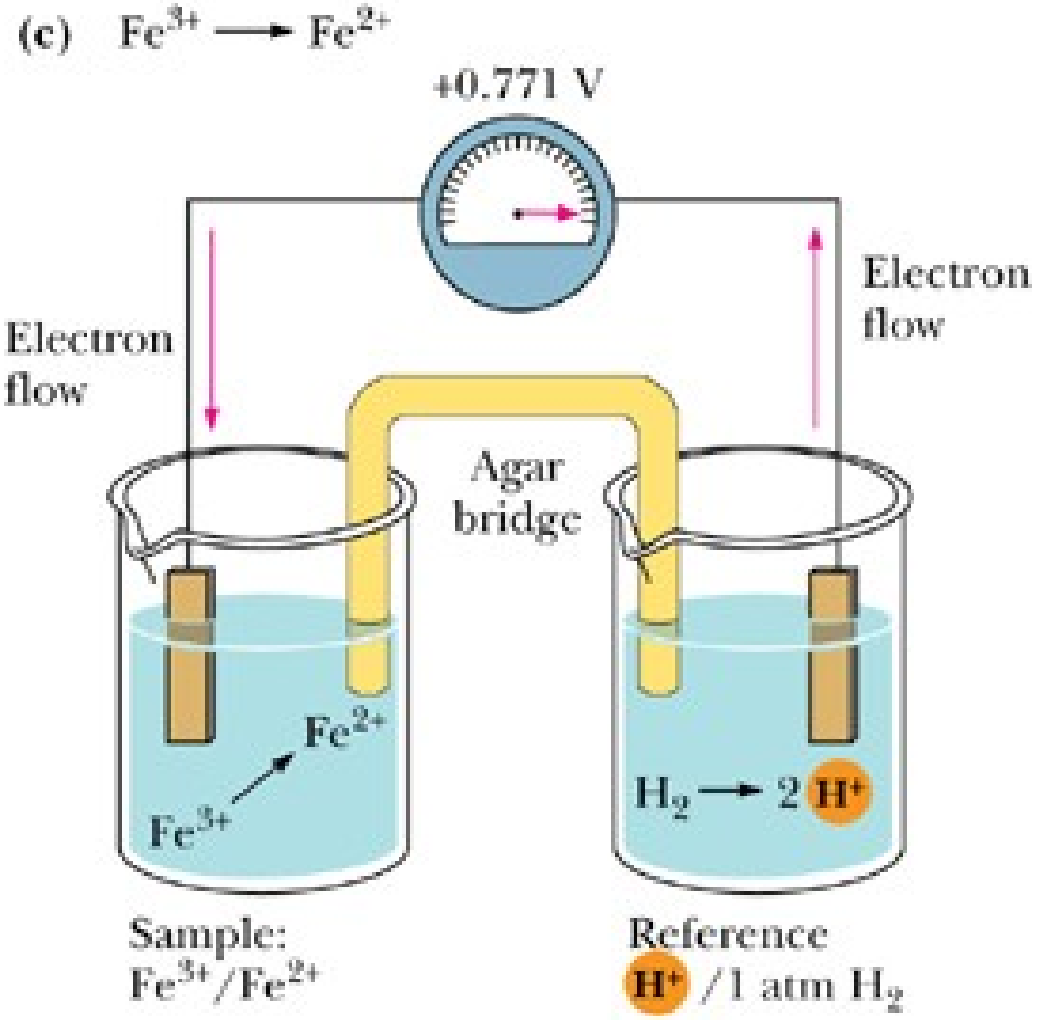

(c) $Fe^{3+} \longrightarrow Fe^{2+}$
+0.771 V
Electron flow
Electron flow
Agar bridge
Fe^{2+}
Fe^{3+}
$H_2 \longrightarrow 2\ H^+$
Sample: Fe^{3+}/Fe^{2+}
Reference H^+ / 1 atm H_2

table 14–7

Standard Reduction Potentials of Some Biologically Important Half-Reactions, at 25 °C and pH 7

Half-reaction	E'° (V)
$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$	0.816
$Fe^{3+} + e^- \longrightarrow Fe^{2+}$	0.771
$NO_3^- + 2H^+ + 2e^- \longrightarrow NO_2^- + H_2O$	0.421
Cytochrome f (Fe^{3+}) + e^- ⟶ cytochrome f (Fe^{2+})	0.365
$Fe(CN)_6^{3-}$ (ferricyanide) + e^- ⟶ $Fe(CN)_6^{4-}$	0.36
Cytochrome a_3 (Fe^{3+}) + e^- ⟶ cytochrome a_3 (Fe^{2+})	0.35
$O_2 + 2H^+ + 2e^- \longrightarrow H_2O_2$	0.295
Cytochrome a (Fe^{3+}) + e^- ⟶ cytochrome a (Fe^{2+})	0.29
Cytochrome c (Fe^{3+}) + e^- ⟶ cytochrome c (Fe^{2+})	0.254
Cytochrome c_1 (Fe^{3+}) + e^- ⟶ cytochrome c_1 (Fe^{2+})	0.22
Cytochrome b (Fe^{3+}) + e^- ⟶ cytochrome b (Fe^{2+})	0.077
Ubiquinone + $2H^+ + 2e^-$ ⟶ ubiquinol + H_2	0.045
Fumarate^{2-} + $2H^+ + 2e^-$ ⟶ succinate^{2-}	0.031
$2H^+ + 2e^- \longrightarrow H_2$ (at standard conditions, pH 0)	0.000
Crotonyl-CoA + $2H^+ + 2e^-$ ⟶ butyryl-CoA	−0.015
Oxaloacetate^{2-} + $2H^+ + 2e^-$ ⟶ malate^{2-}	−0.166
Pyruvate$^-$ + $2H^+ + 2e^-$ ⟶ lactate$^-$	−0.185
Acetaldehyde + $2H^+ + 2e^-$ ⟶ ethanol	−0.197
FAD + $2H^+ + 2e^- \longrightarrow FADH_2$	−0.219*
Glutathione + $2H^+ + 2e^-$ ⟶ 2 reduced glutathione	−0.23
$S + 2H^+ + 2e^- \longrightarrow H_2S$	−0.243
Lipoic acid + $2H^+ + 2e^-$ ⟶ dihydrolipoic acid	−0.29
$NAD^+ + H^+ + 2e^- \longrightarrow NADH$	−0.320
$NADP^+ + H^+ + 2e^- \longrightarrow NADPH$	−0.324
Acetoacetate + $2H^+ + 2e^-$ ⟶ β-hydroxybutyrate	−0.346
α-Ketoglutarate + $CO_2 + 2H^+ + 2e^-$ ⟶ isocitrate	−0.38
$2H^+ + 2e^- \longrightarrow H_2$ (at pH 7)	−0.414
Ferredoxin (Fe^{3+}) + e^- ⟶ ferredoxin (Fe^{2+})	−0.432

Data mostly from Loach, P.A. (1976) In *Handbook of Biochemistry and Molecular Biology*, 3rd edn (Fasman, G.D., ed.), *Physical and Chemical Data*, Vol. I, pp. 122–130, CRC Press, Boca Raton, FL.

*This is the value for free FAD; FAD bound to a specific flavoprotein (for example succinate dehydrogenase) has a different E'°.



- The actual reduction potential (E) depends on , electrons transferred per molecule, temperature, ratio of [electron acceptor]/[electron donor]:

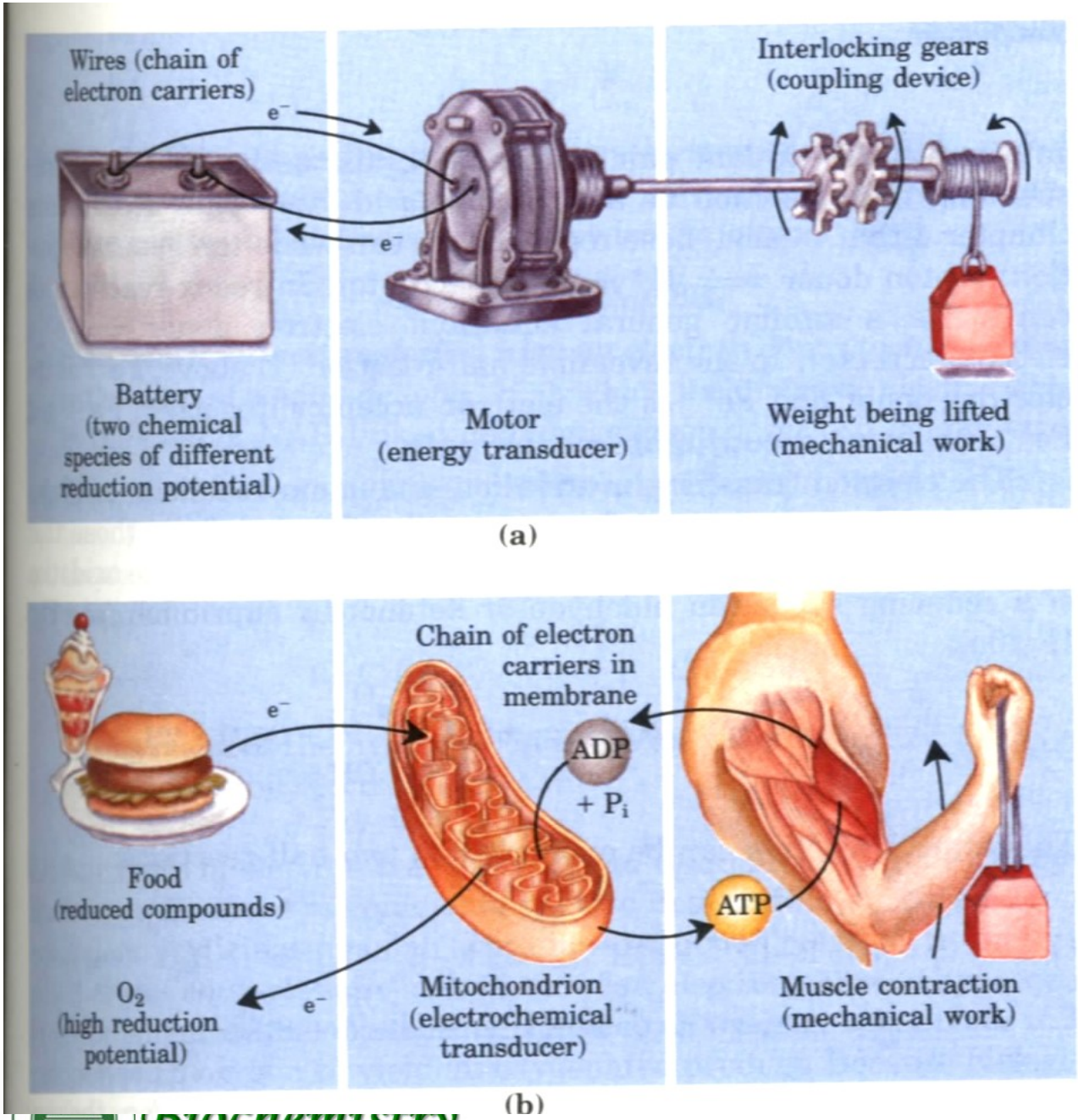
$$E = E^\circ + \frac{RT}{n\mathcal{F}} \ln \frac{[\text{electron acceptor}]}{[\text{electron donor}]}$$

- l

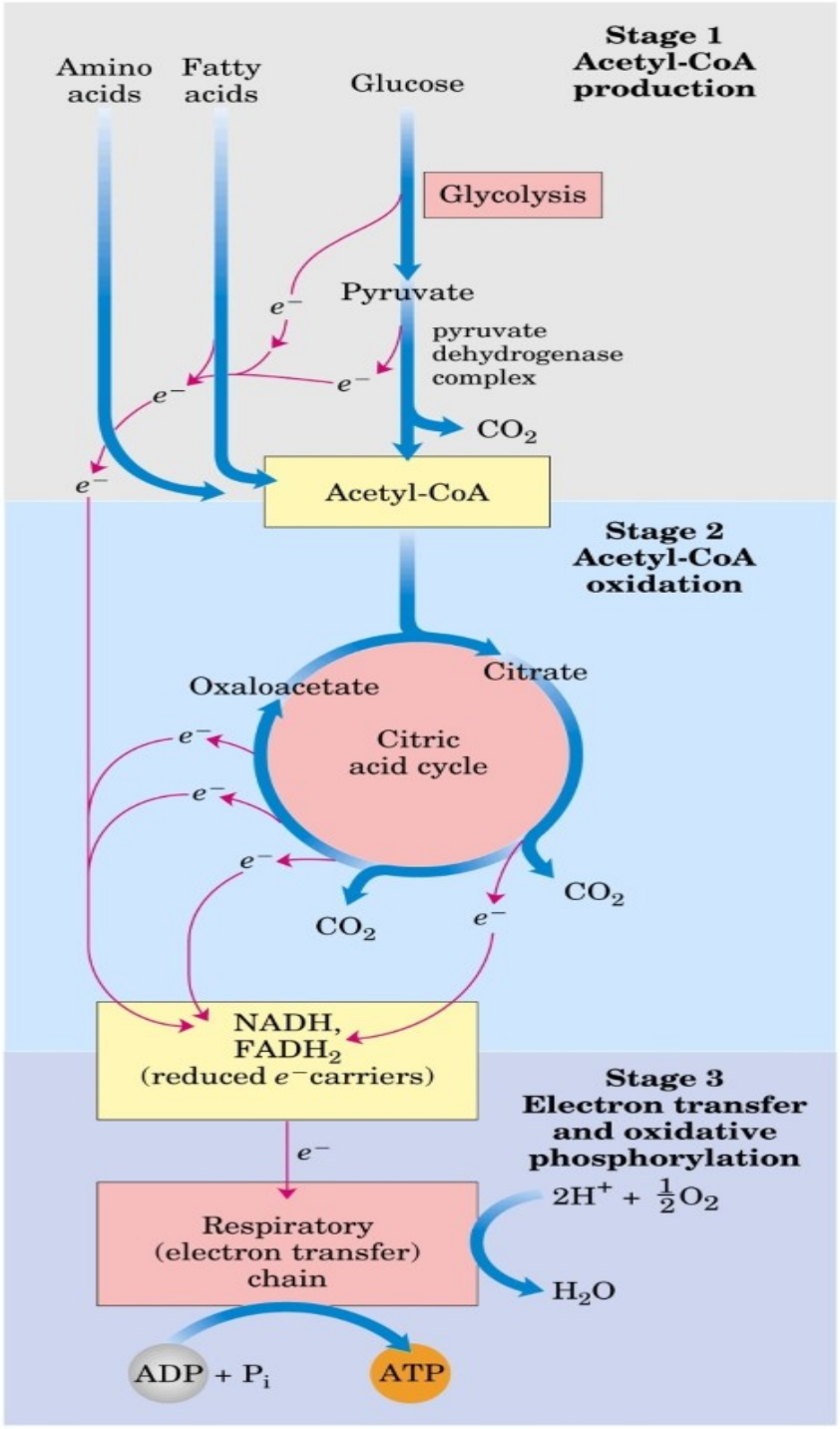
the $\Delta E'^{o}$ of the two redox pairs:

$$\Delta G = -n\mathcal{F}\,\Delta E, \quad \text{or} \quad \Delta G'^\circ = -n\mathcal{F}\,\Delta E'^\circ$$





Energy is "generated" via electron flow both in a battery and in a cell!


Stage 1
Acetyl-CoA
production
Amino
acids
Fatty
acids
Glucose
Glycolysis
Pyruvate
pyruvate
dehydrogenase
complex
e^-
e^-
e^-
e^-
CO_2
Acetyl-CoA
Stage 2
Acetyl-CoA
oxidation
Citrate
Oxaloacetate
Citric
acid cycle
e^-
e^-
e^-
e^-
CO_2
CO_2
NADH,
$FADH_2$
(reduced e^- carriers)
Stage 3
Electron transfer
and oxidative
phosphorylation
e^-
$2H^+ + \frac{1}{2}O_2$
Respiratory
(electron transfer)
chain
H_2O
ADP + P_i
ATP

Biochemist

6. A few universal carriers (as coenzymes) collect electrons from the oxidation of various substrates



- NAD^+, $NADP^+$, FAD are the few commonly used such reversible electron carriers.
- NAD and NADP are dinucleotides able to accept/donate a hydride ion ($:H^-$, $=H^+ + 2e^-$) for each round of reduction/oxidation.

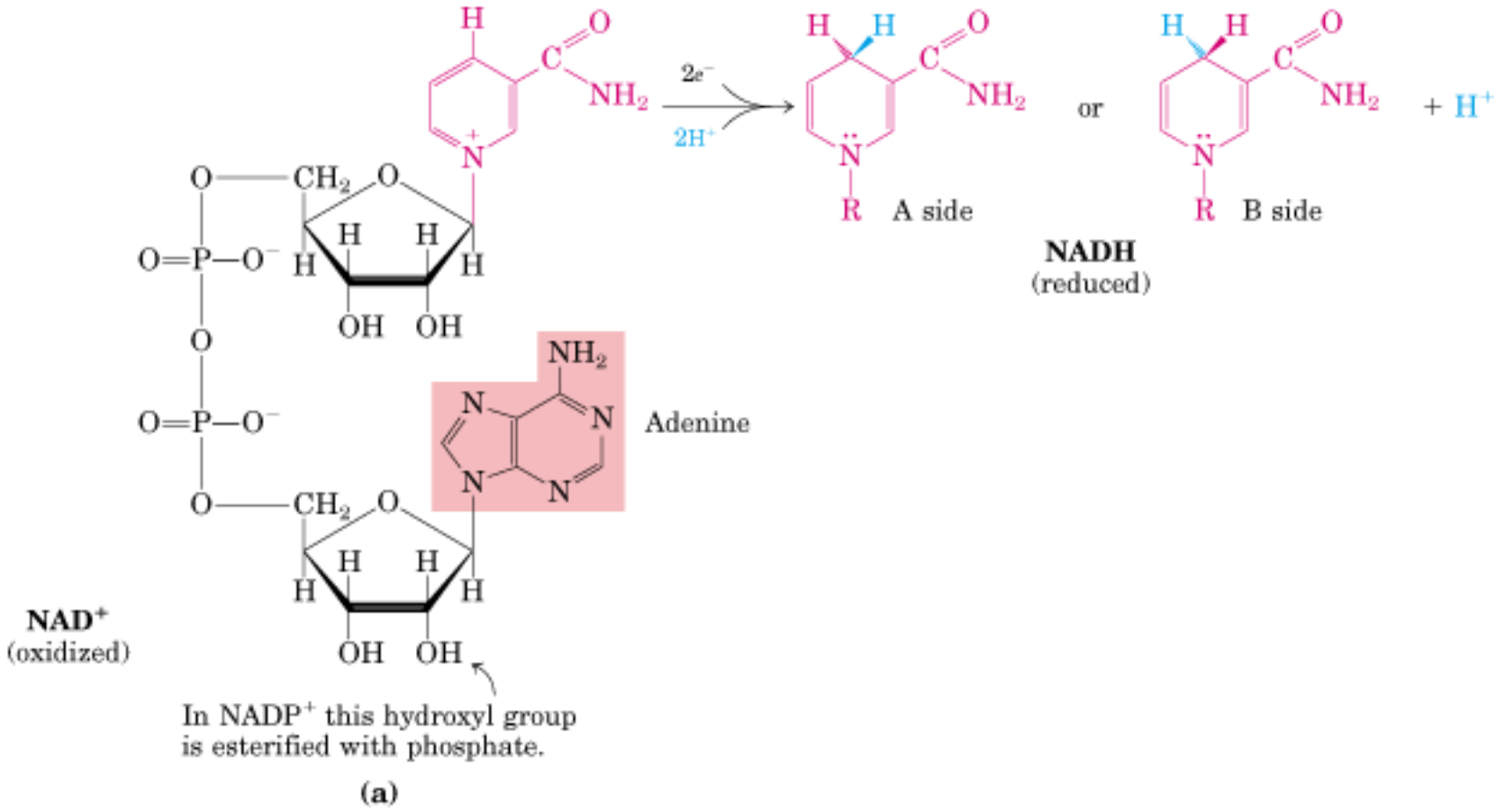


- In each specific NAD- or NADP-containing dehydrogenase, the hydride ion is added/taken stereospecifically from one side (A or B) of the nicotinamide (烟碱) ring.
- FAD is able to accept/donate one or two electrons (as hydrogen atom), with absorption maximum shifts from 570 nm to 450nm.
- NAD and NADP can easily diffuse out of the enzymes, but FMN and FAD are tightly bound to the enzymes.

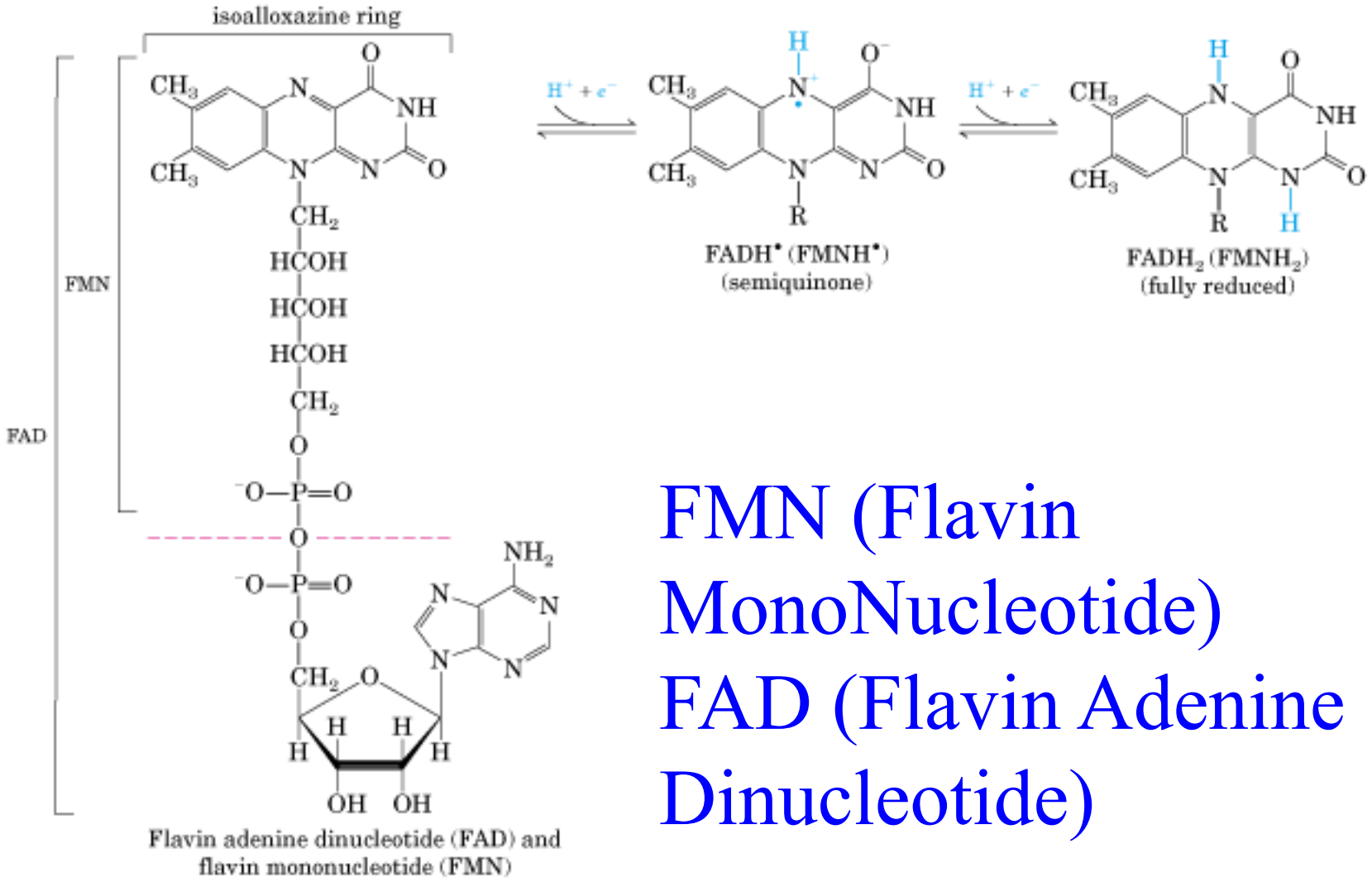


- They serve as cofactors of various enzymes catalyzing the oxidation of nutrients.
- NADH and $FADH_2$ will be further oxidized via the respiratory chain on the inner membrane of mitochondria or plasma membrane of bacteria for energy generation (transduction).





Nicotinamide is derived From niacin(B5)



Flavin adenine dinucleotide (FAD) and flavin mononucleotide (FMN)

FMN (Flavin MonoNucleotide)
FAD (Flavin Adenine Dinucleotide)

table 14–8

Stereospecificity of Dehydrogenases That Employ NAD^+ or $NADP^+$ as Coenzymes

Enzyme	Coenzyme	Stereochemical specificity for nicotinamide ring (A or B)
Isocitrate dehydrogenase	NAD^+	A
α-Ketoglutarate dehydrogenase	NAD^+	B
Glucose 6-phosphate dehydrogenase	$NADP^+$	B
Malate dehydrogenase	NAD^+	A
Glutamate dehydrogenase	NAD^+ or $NADP^+$	B
Glyceraldehyde 3-phosphate dehydrogenase	NAD^+	B
Lactate dehydrogenase	NAD^+	A
Alcohol dehydrogenase	NAD^+	A



table 14–9

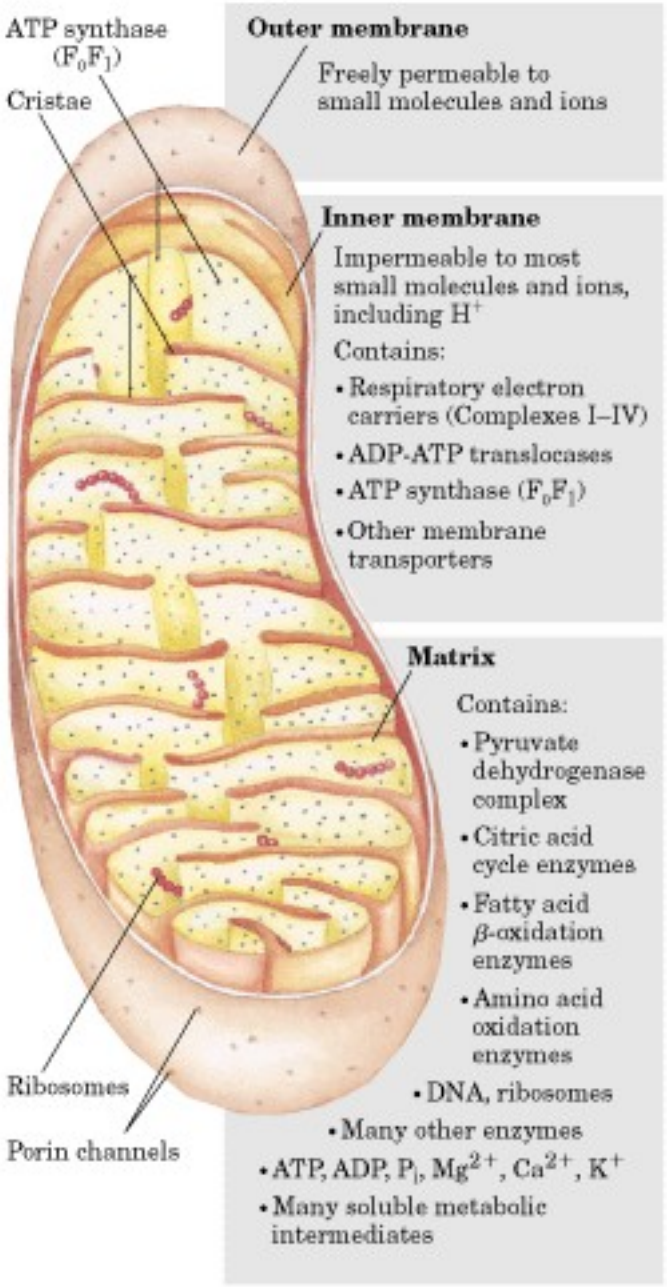
Some Enzymes (Flavoproteins) That Employ Flavin Nucleotide Coenzymes

Enzyme	Flavin nucleotide
Fatty acyl–CoA dehydrogenase	FAD
Dihydrolipoyl dehydrogenase	FAD
Succinate dehydrogenase	FAD
Glycerol 3-phosphate dehydrogenase	FAD
Thioredoxin reductase	FAD
NADH dehydrogenase (Complex I)	FMN
Glycolate dehydrogenase	FMN



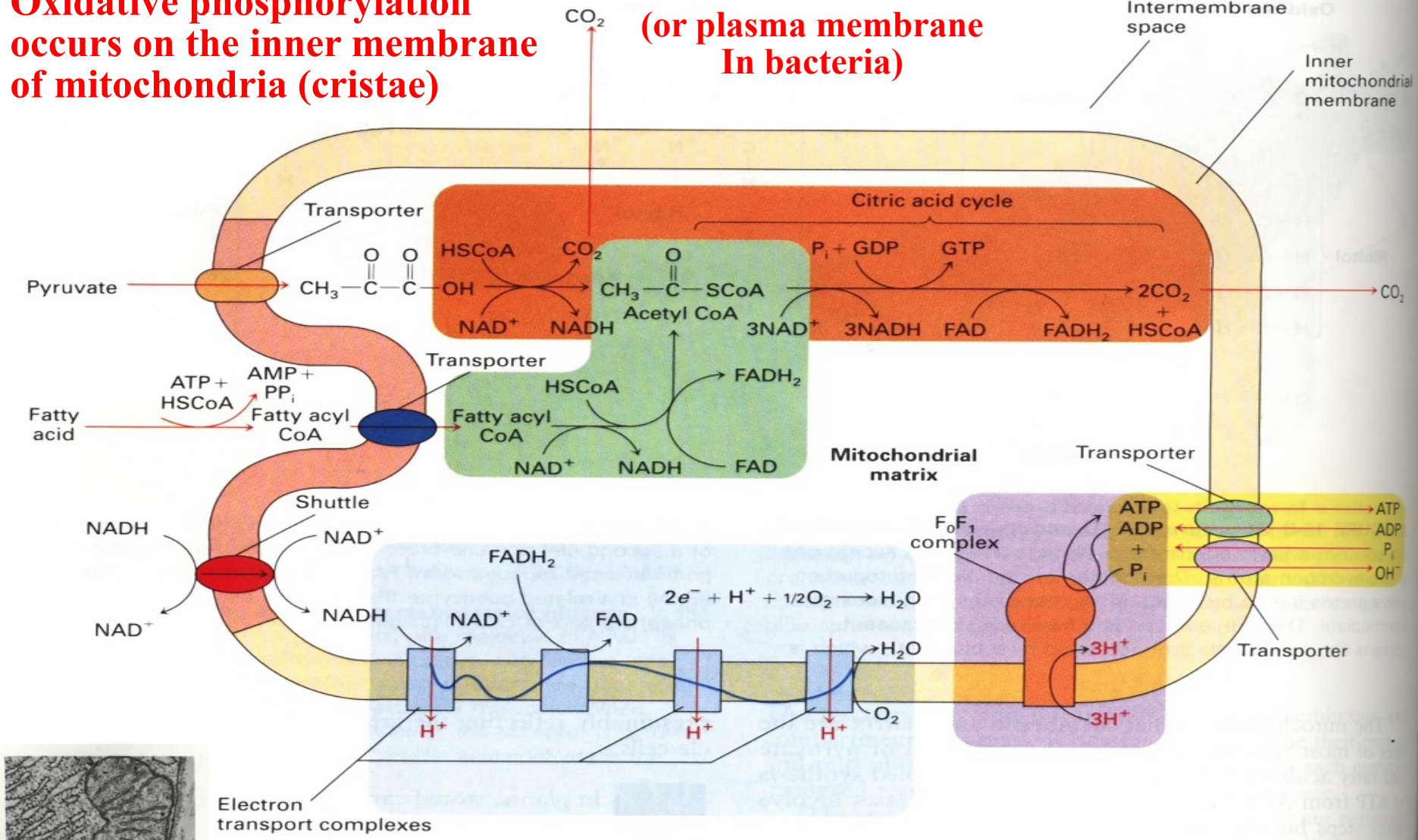
7. Electrons of NADH and FADH2 are transferred to O_2 via many intermediate electron carriers on the way.




ATP synthase (F_0F_1)
Cristae
Outer membrane
Freely permeable to small molecules and ions
Inner membrane
Impermeable to most small molecules and ions, including H^+
Contains:
• Respiratory electron carriers (Complexes I–IV)
• ADP-ATP translocases
• ATP synthase (F_0F_1)
• Other membrane transporters
Matrix
Contains:
• Pyruvate dehydrogenase complex
• Citric acid cycle enzymes
• Fatty acid β-oxidation enzymes
• Amino acid oxidation enzymes
• DNA, ribosomes
• Many other enzymes
• ATP, ADP, P_i, Mg^{2+}, Ca^{2+}, K^+
• Many soluble metabolic intermediates
Ribosomes
Porin channels

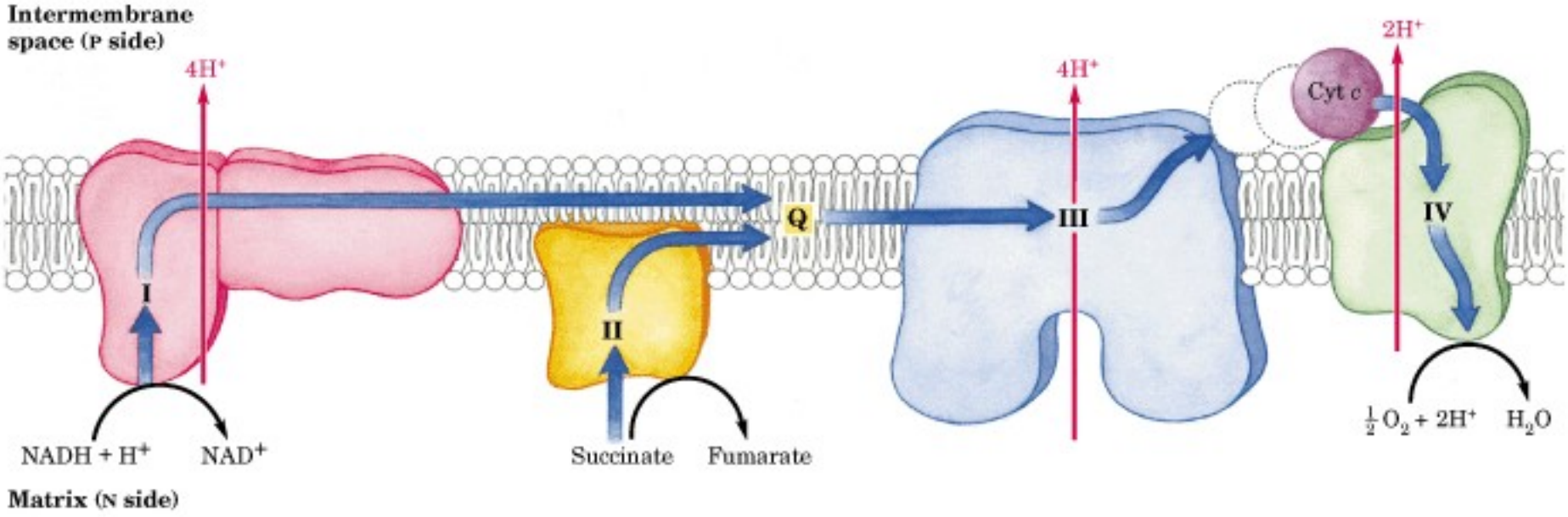
Oxidative phosphorylation occurs on the inner membrane of mitochondria (cristae)

(or plasma membrane In bacteria)





- Transport of metabolites into the mitochondrion
- Fatty acid metabolism
- Electron transport from NADH and $FADH_2$ to oxygen; generation of proton-motive force
- Pyruvate dehydrogenase and citric acid cycle
- ATP synthesis by F_0F_1 using proton-motive force
- ATP export, ADP and P_i import


Intermembrane space (P side)
4H+
4H+
2H+
Cyt c
I
II
Q
III
IV
NADH + H+
NAD+
Succinate
Fumarate
½ O2 + 2H+
H2O
Matrix (N side)

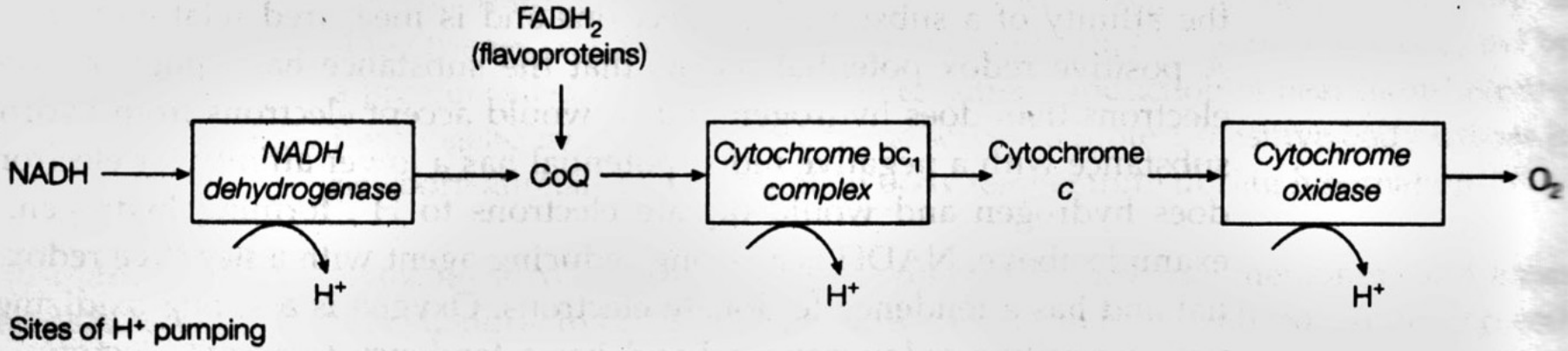


Fig. 1. Overview of the electron transport chain (respiratory chain).

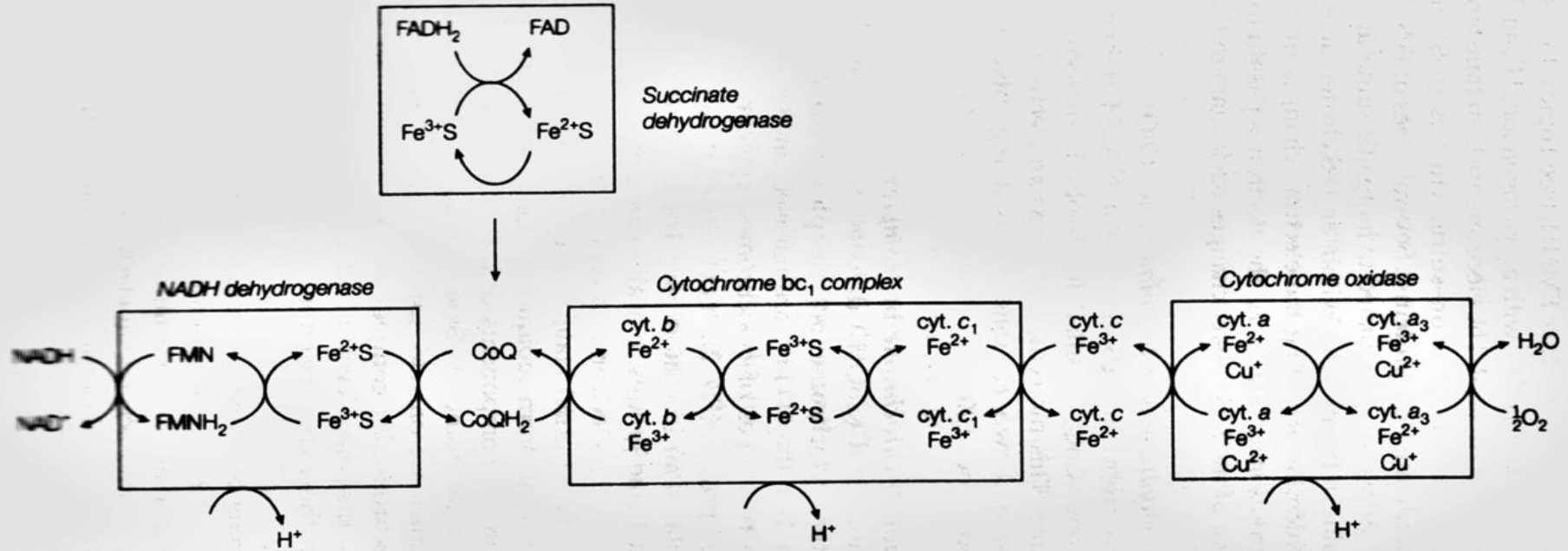
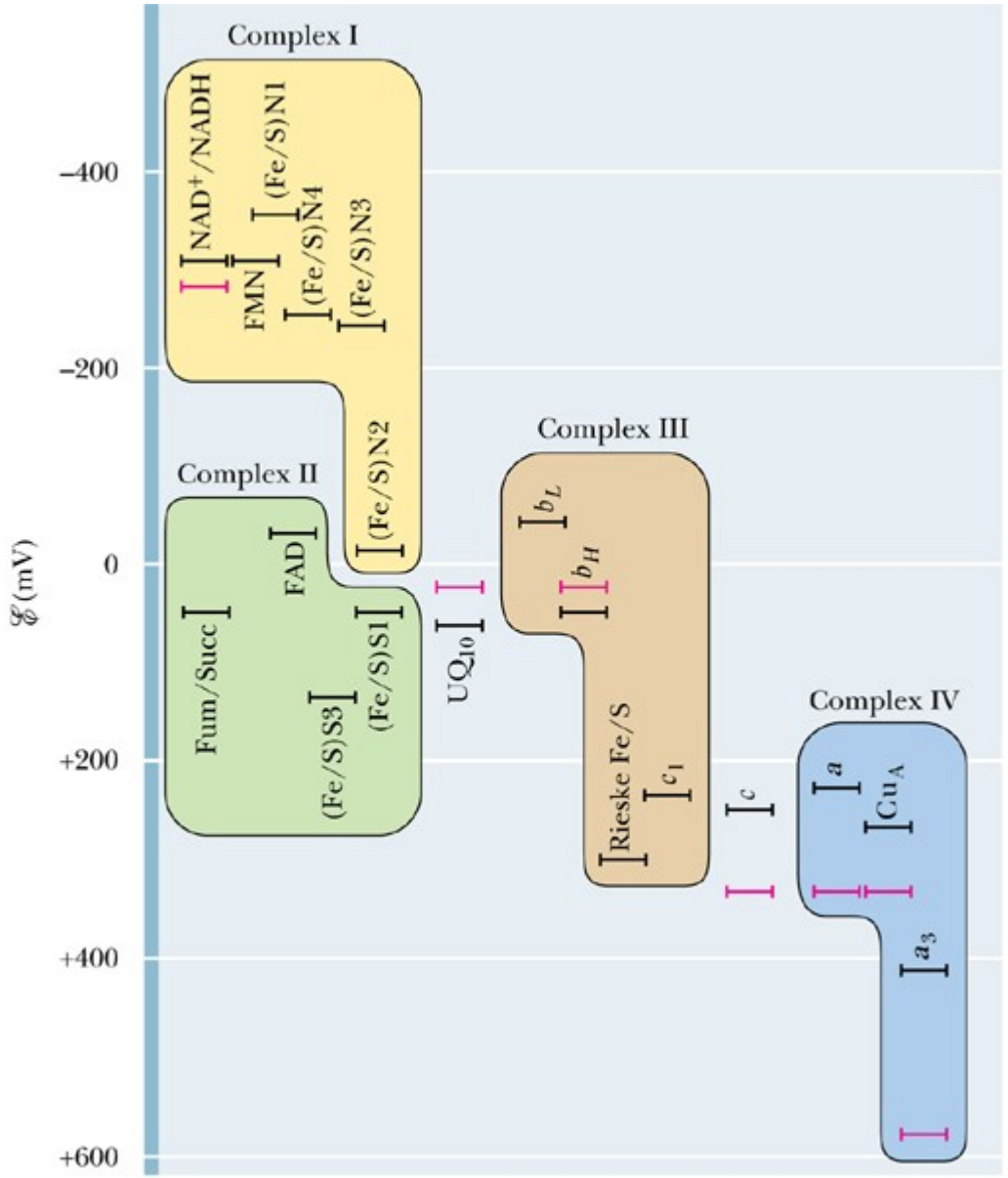
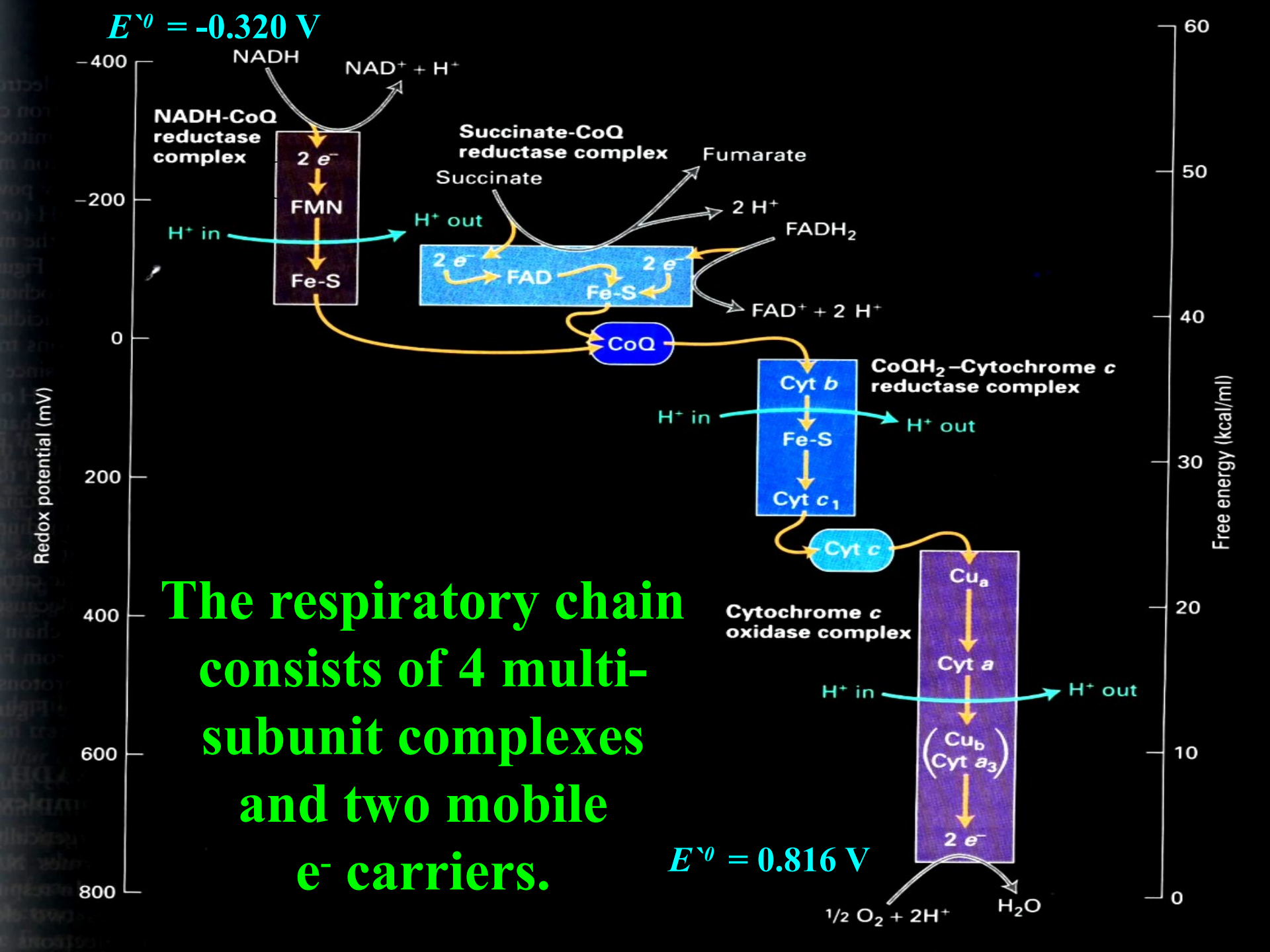


Fig. 2. Details of electron transport.



Biochemistry

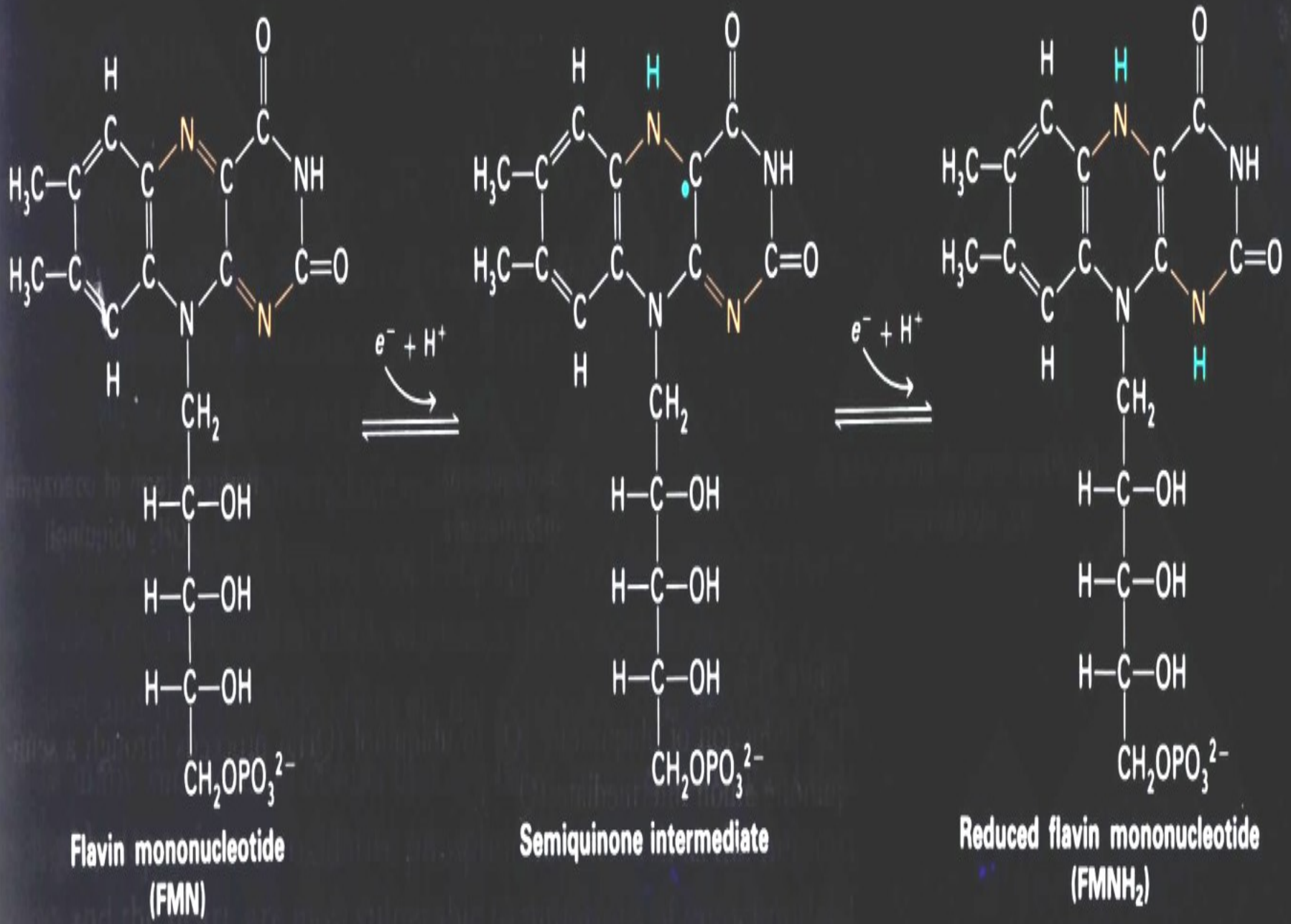


The respiratory chain consists of 4 multi-subunit complexes and two mobile e^- carriers.

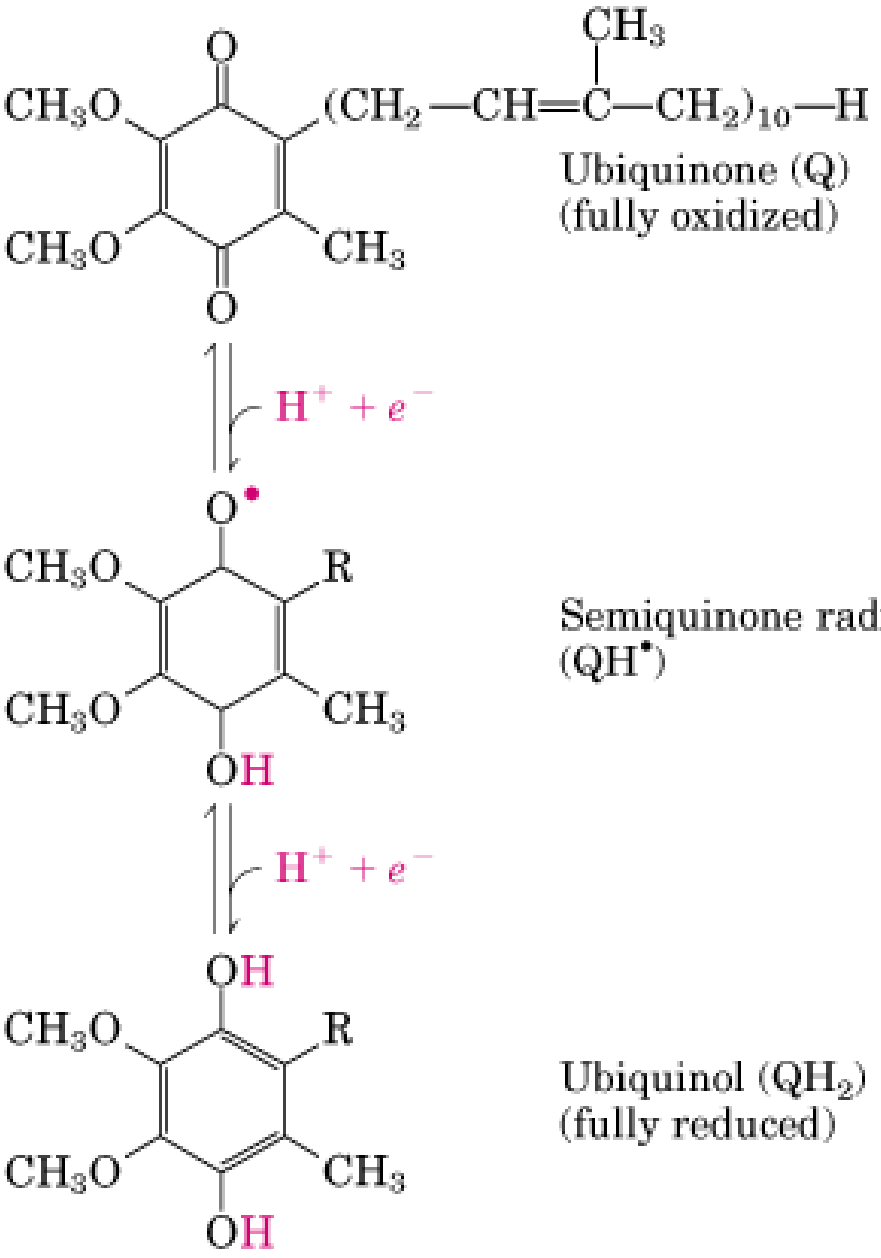
8. Various prosthetic groups act as intermediate electron carriers in the respiratory chain



FMN and FAD can accept/donate one/two e-



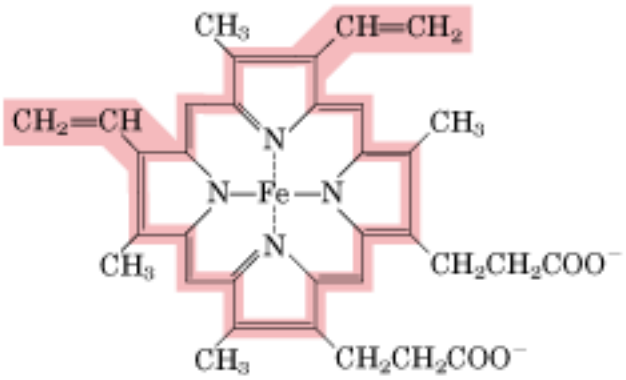
Q is a mobile one/two e^-/H^+ carrier



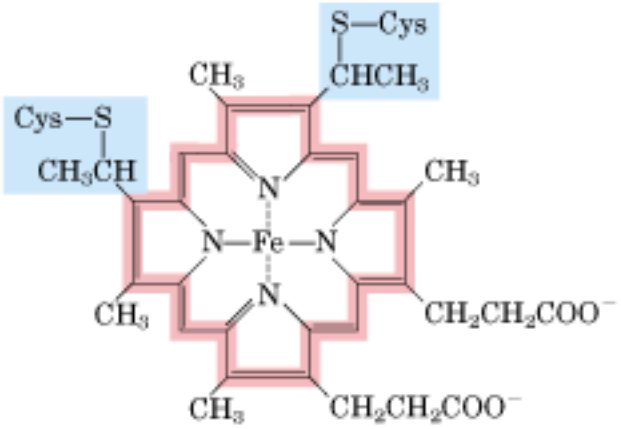
The only e^- carrier not bound to a protein

Q/QH2 diffuses freely in the lipid bilayer

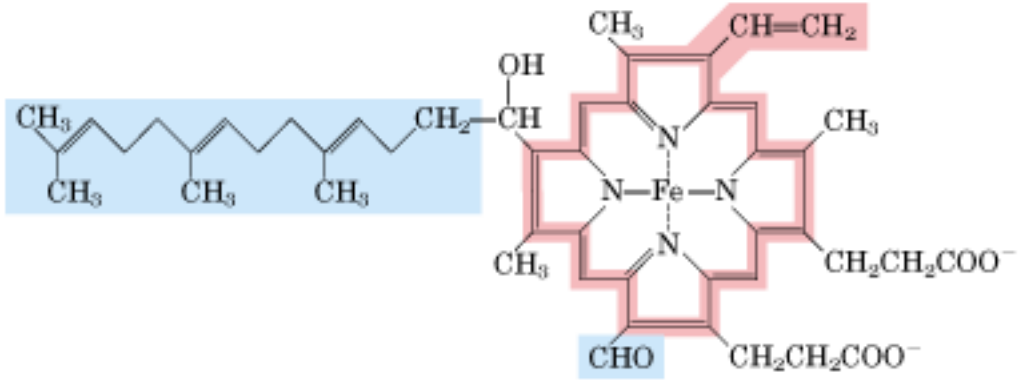




Iron protoporphyrin IX
(in *b*-type cytochromes)

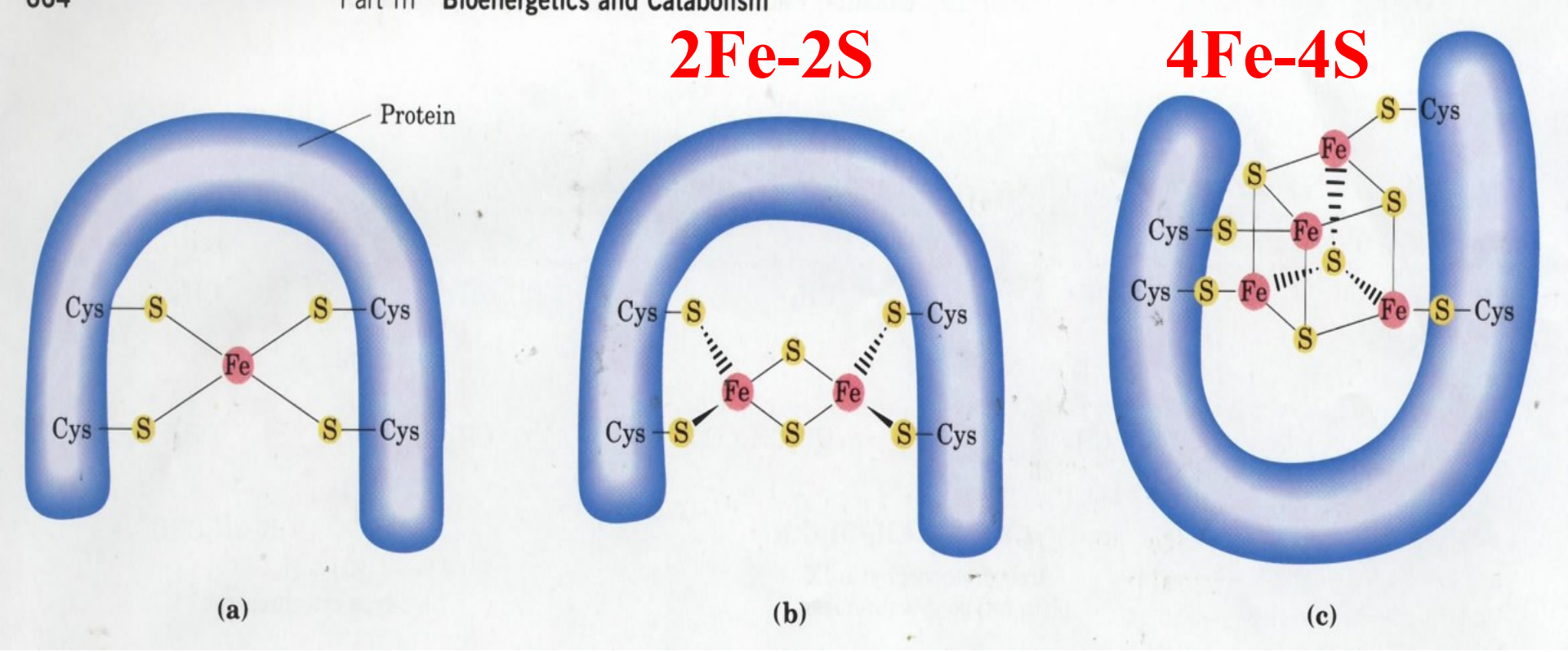


Heme C
(in *c*-type cytochromes)



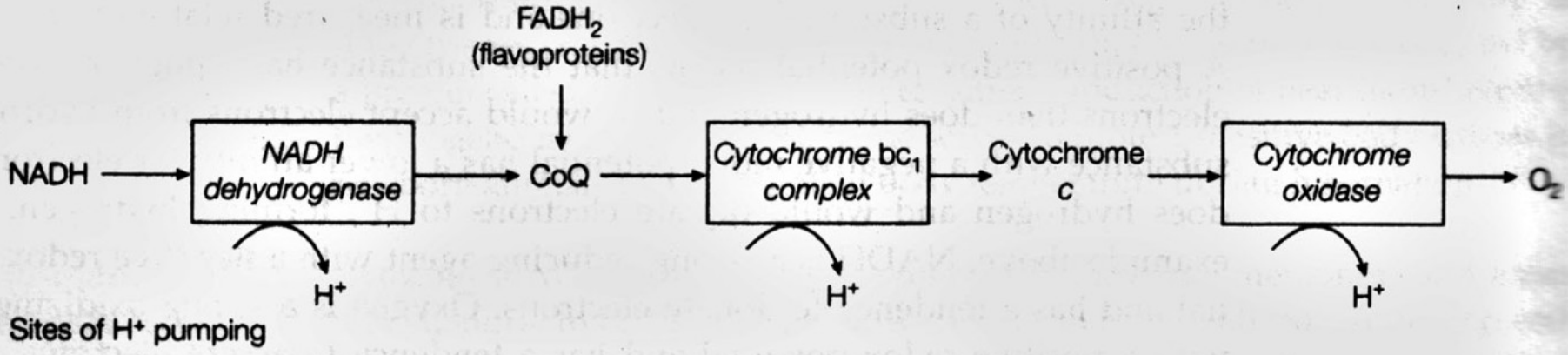
Heme A
(in *a*-type cytochromes)

Three types of heme groups are found in cytochromes.



At least eight different types of iron-sulfur centers act in the respiratory chain: iron atoms cycle between Fe^{2+} (reduced) and Fe^{3+} (oxidized).





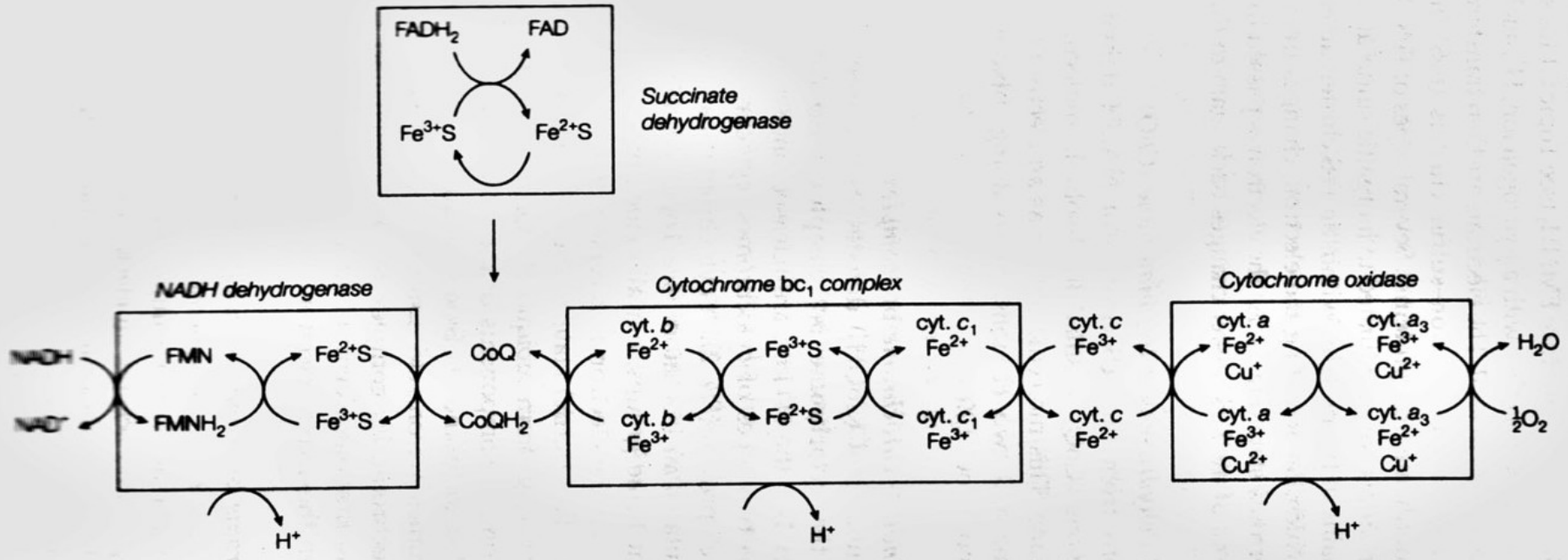
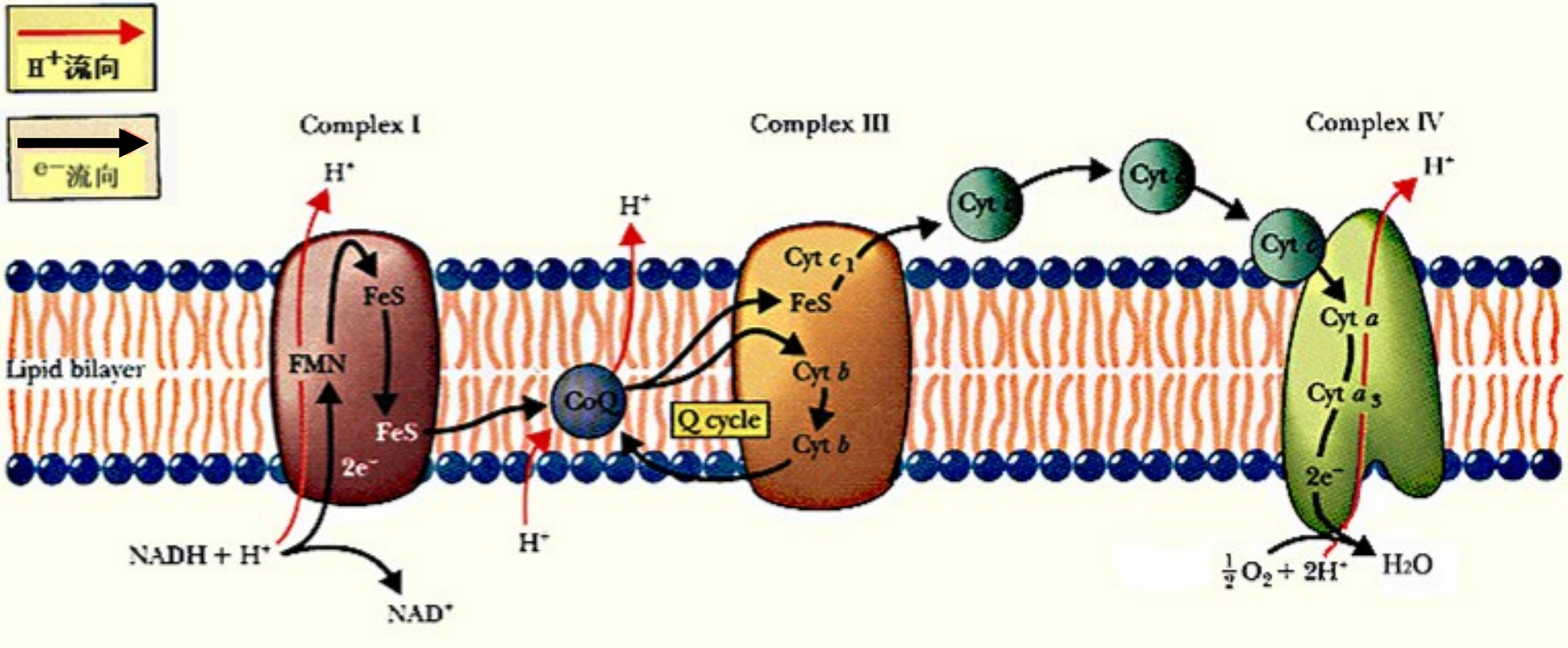


Fig. 2 *Details of electron transport.*

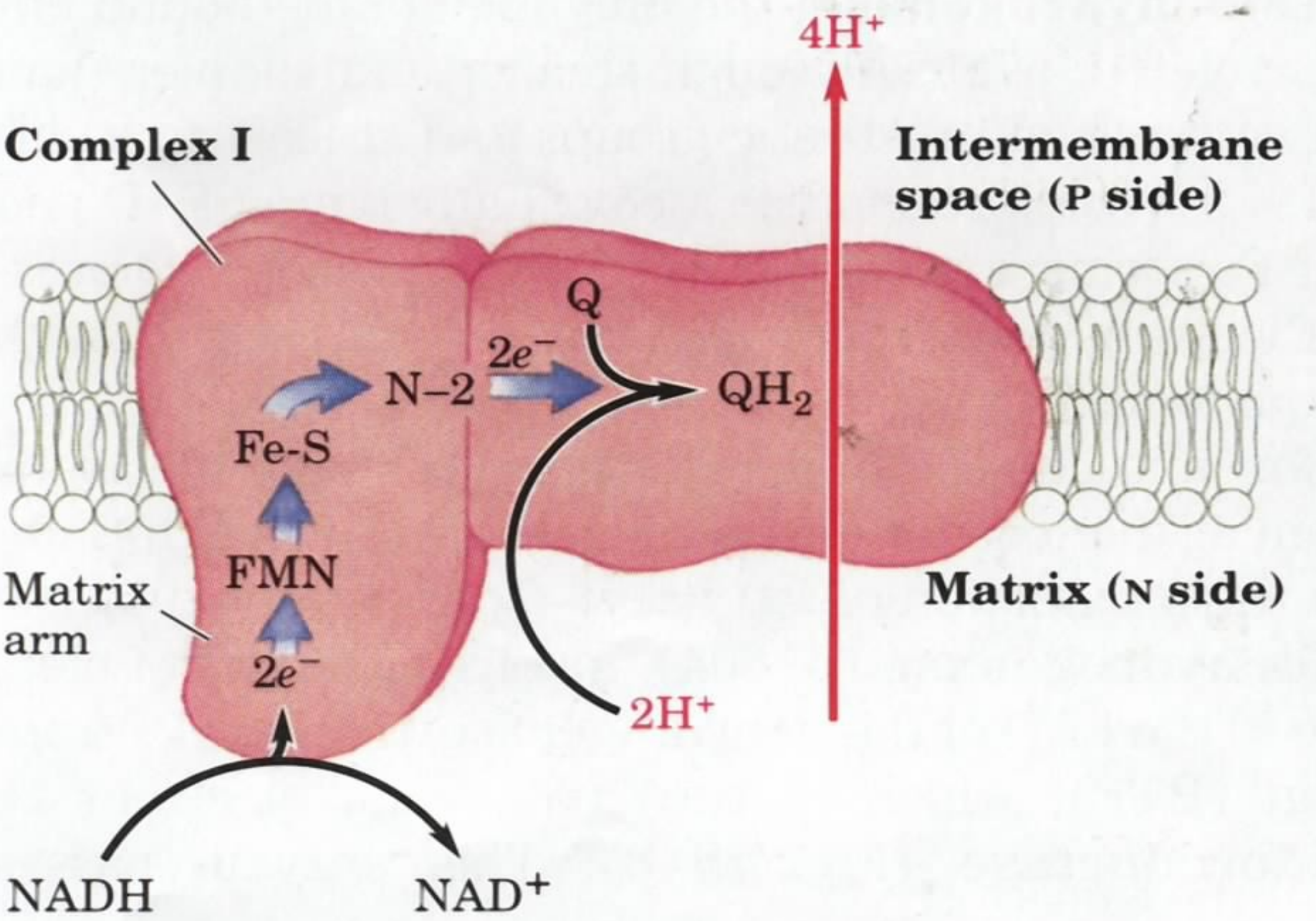


递氢体和递电子体交替排列

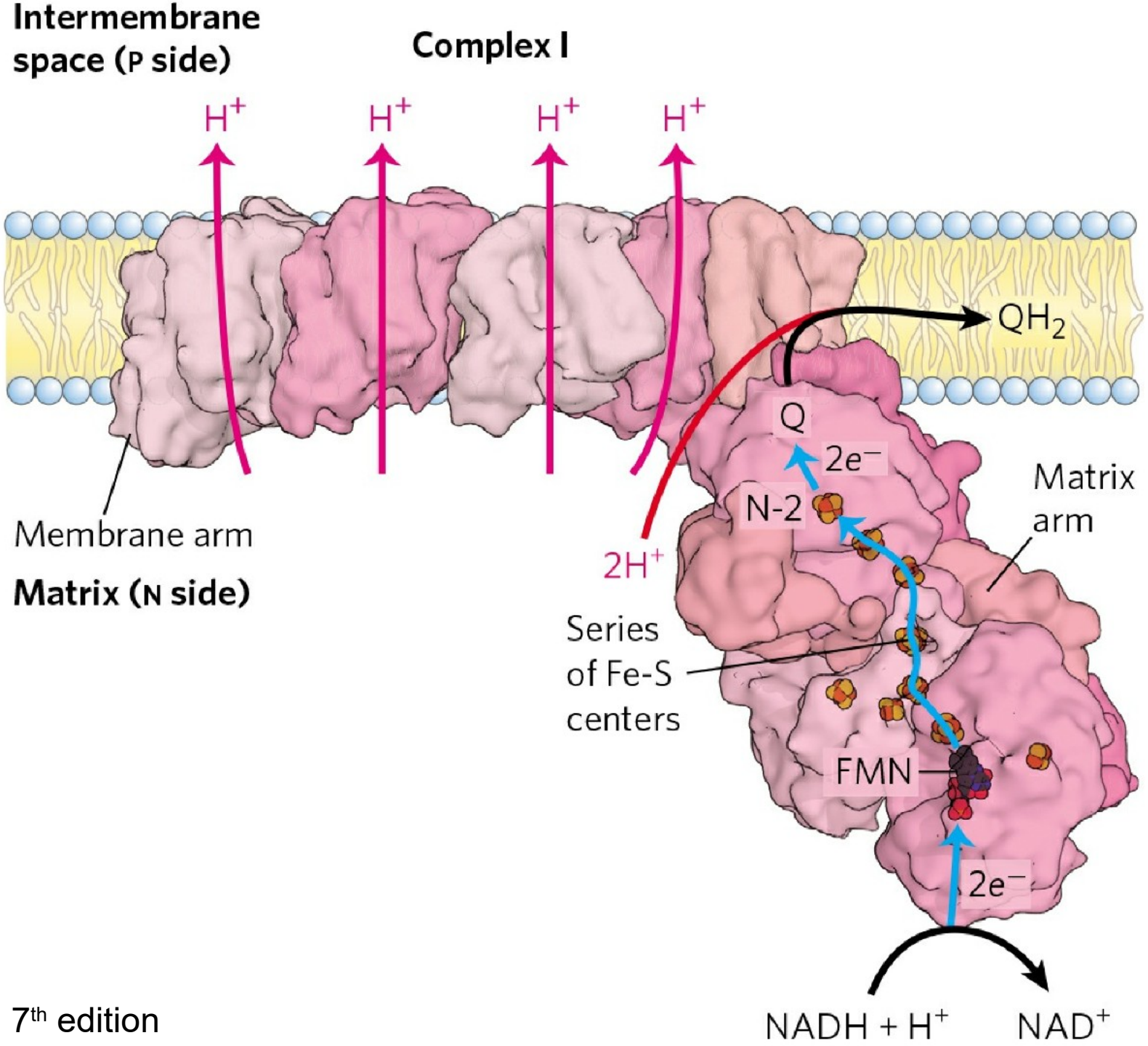
Complex I:

NADH enters at NADH dehydrogenase (also NADH:ubiquinone oxidoreductase or NADH-Q oxidoreductase).




4H+
Complex I
Intermembrane space (P side)
Q
2e−
N–2
QH2
Fe-S
FMN
Matrix arm
Matrix (N side)
2e−
2H+
NADH
NAD+

4th edition

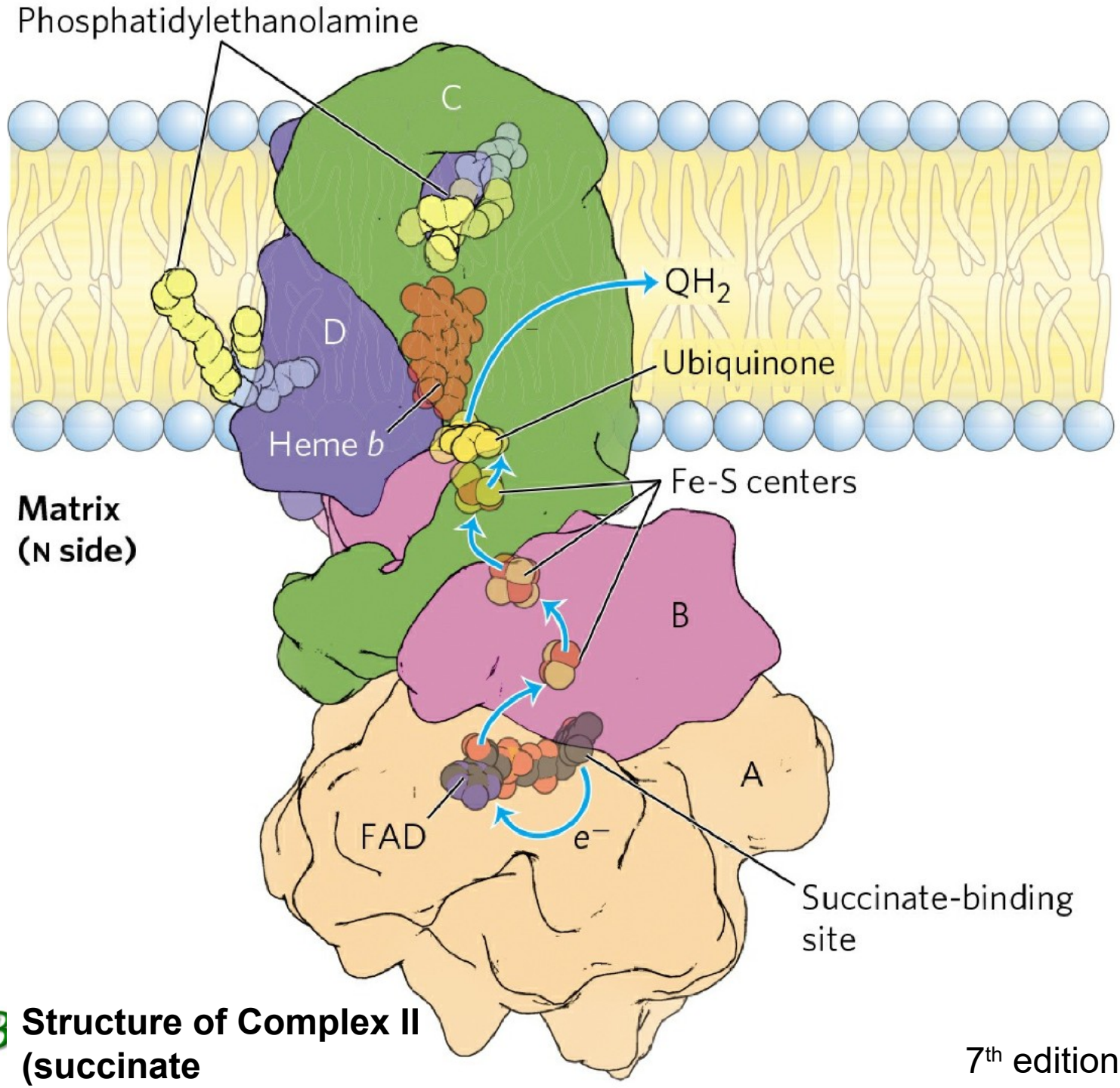

Intermembrane space (P side)
Complex I
H+
H+
H+
H+
QH2
Q
2e−
N-2
Matrix arm
Membrane arm
2H+
Matrix (N side)
Series of Fe-S centers
FMN
2e−
NADH + H+
NAD+

7th edition

(complex II) :
succinate Dehydrogenase

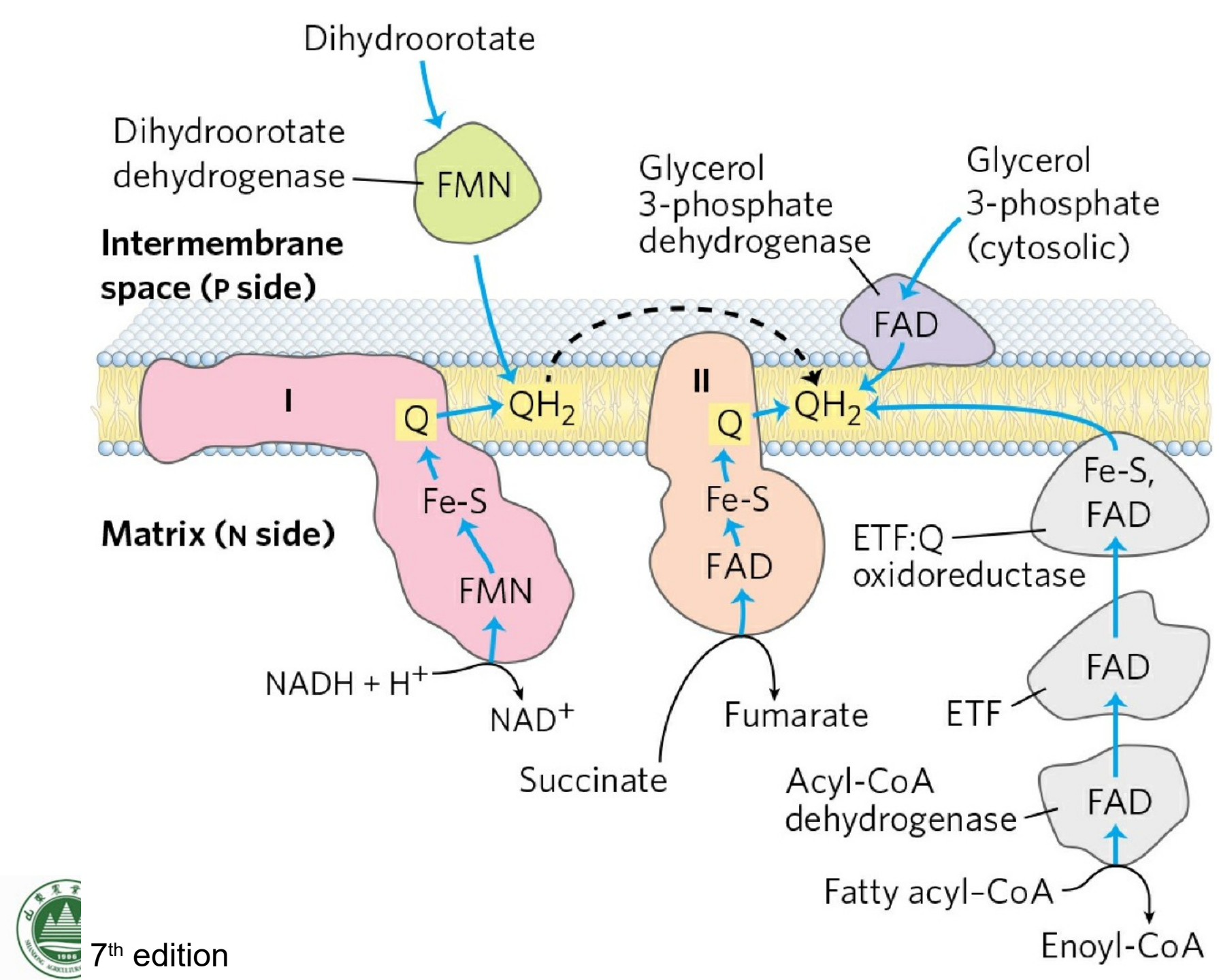
$FADH_2$ of flavoproteins also transfer their electrons to ubiquinone (Q), with no H^+ pumped.



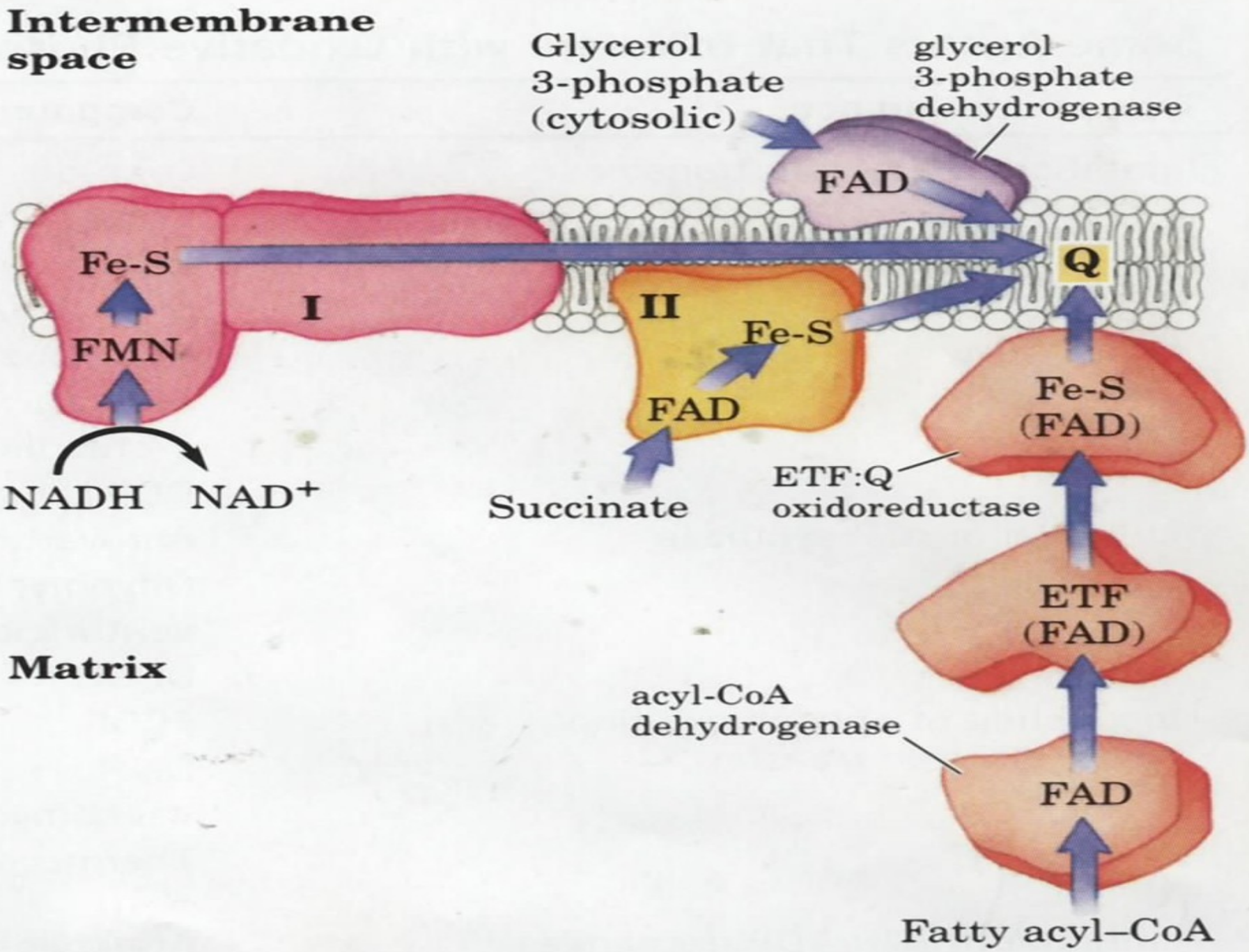


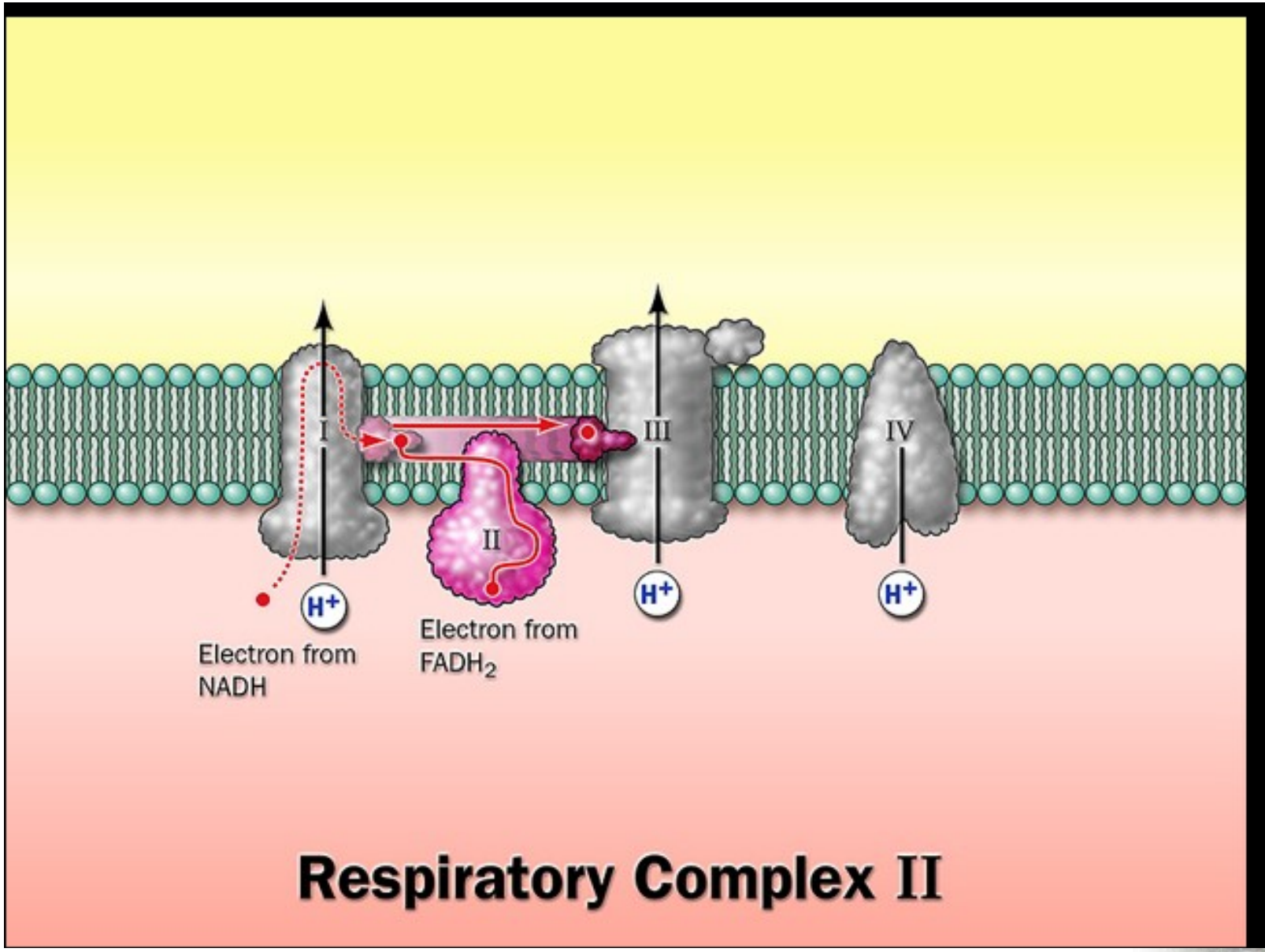
Structure of Complex II (succinate

7th edition


Dihydroorotate
Dihydroorotate dehydrogenase
FMN
Intermembrane space (P side)
Glycerol 3-phosphate dehydrogenase
Glycerol 3-phosphate (cytosolic)
FAD
I
Q
QH_2
II
Q
QH_2
Fe-S
FMN
Fe-S
FAD
Fe-S, FAD
ETF:Q oxidoreductase
Matrix (N side)
$NADH + H^+$
NAD^+
Succinate
Fumarate
FAD
ETF
Acyl-CoA dehydrogenase
FAD
Fatty acyl-CoA
Enoyl-CoA

7th edition


Intermembrane space
Glycerol 3-phosphate (cytosolic)
glycerol 3-phosphate dehydrogenase
FAD
Fe-S
I
FMN
II
Fe-S
FAD
Q
Fe-S (FAD)
ETF:Q oxidoreductase
NADH
NAD+
Succinate
ETF (FAD)
Matrix
acyl-CoA dehydrogenase
FAD
Fatty acyl–CoA

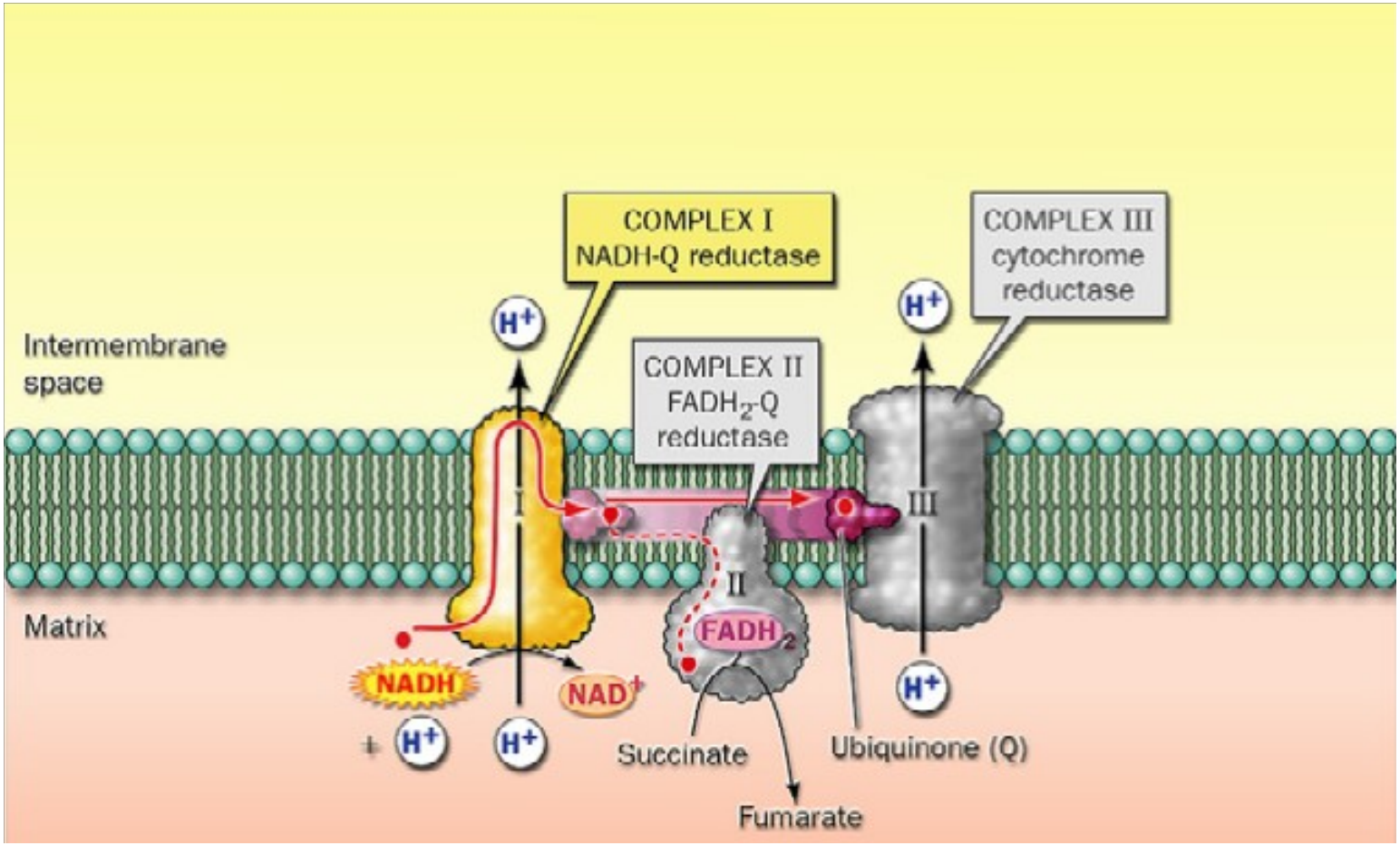

I
II
III
IV
H+
H+
H+
Electron from NADH
Electron from FADH2
Respiratory Complex II

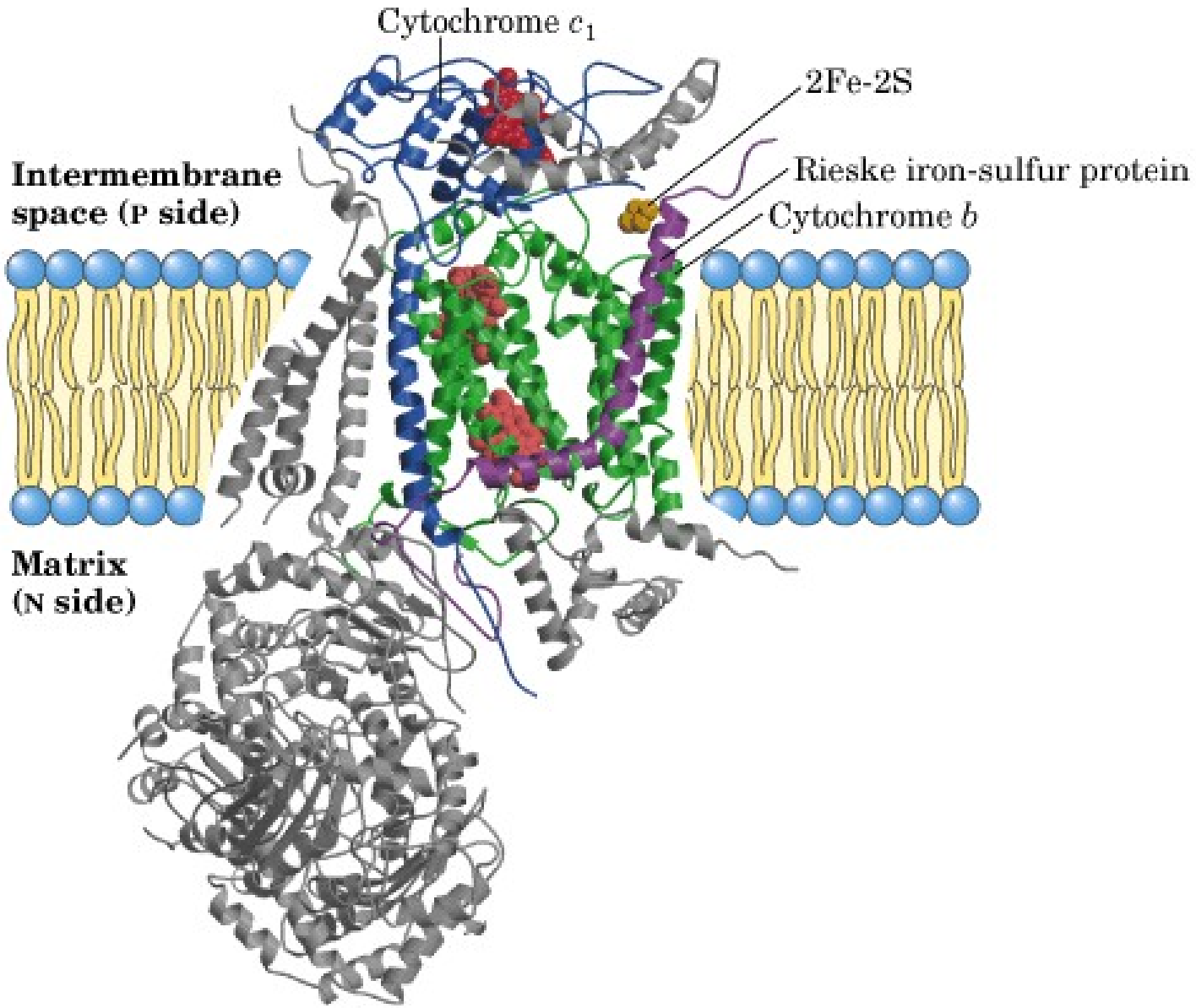
(Complex III)

Cytochrome bc_1 complex passes $2e^-$ from 1 QH_2 to 2 cytochrome *c*

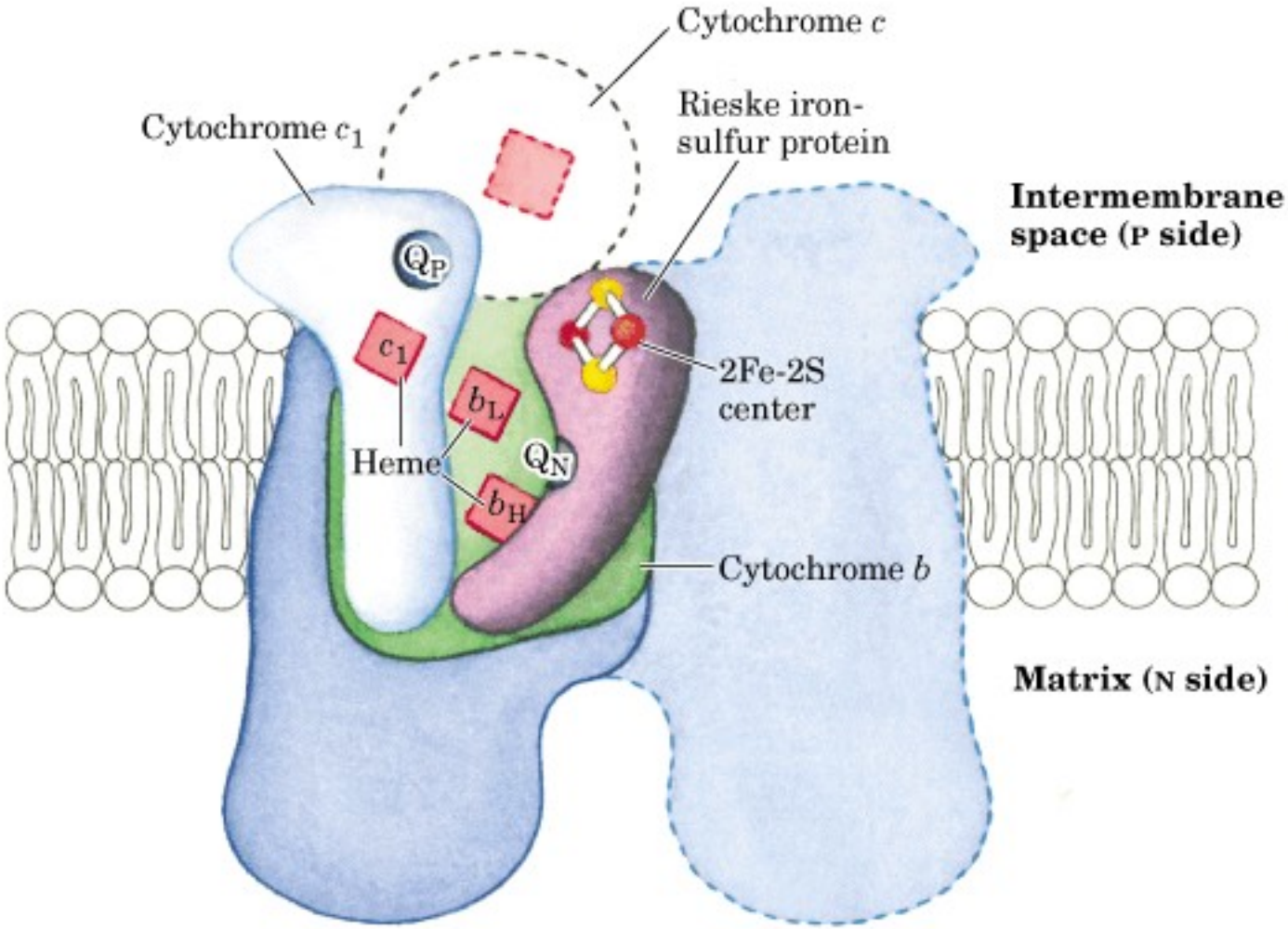
e^- transferring & H^+ pumping in Complex III via the Q cycle (for each 2 e^- transferred, $4H^+$ are released at the P side and $2H^+$ taken from the N side)



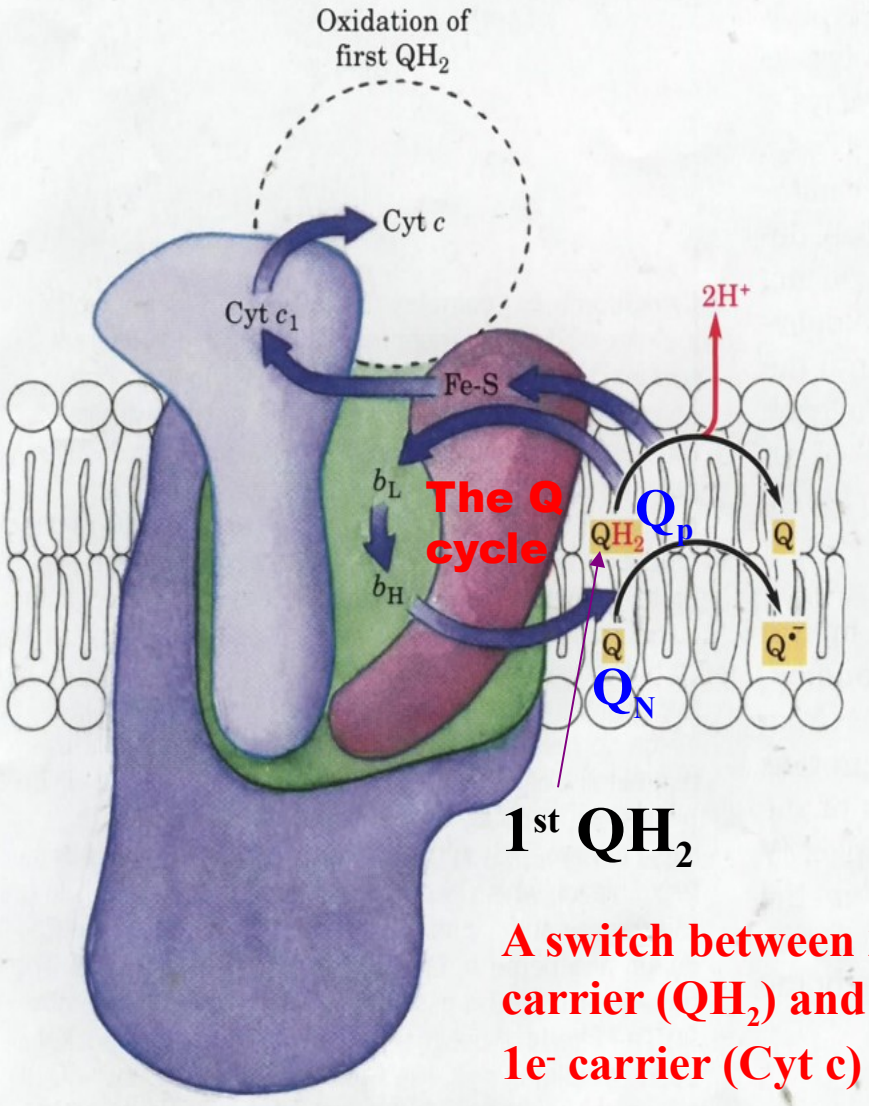

COMPLEX I
NADH-Q reductase
COMPLEX II
FADH2-Q
reductase
COMPLEX III
cytochrome
reductase
Intermembrane
space
Matrix
H+
NADH
NAD+
FADH2
Succinate
Fumarate
Ubiquinone (Q)
I
II
III

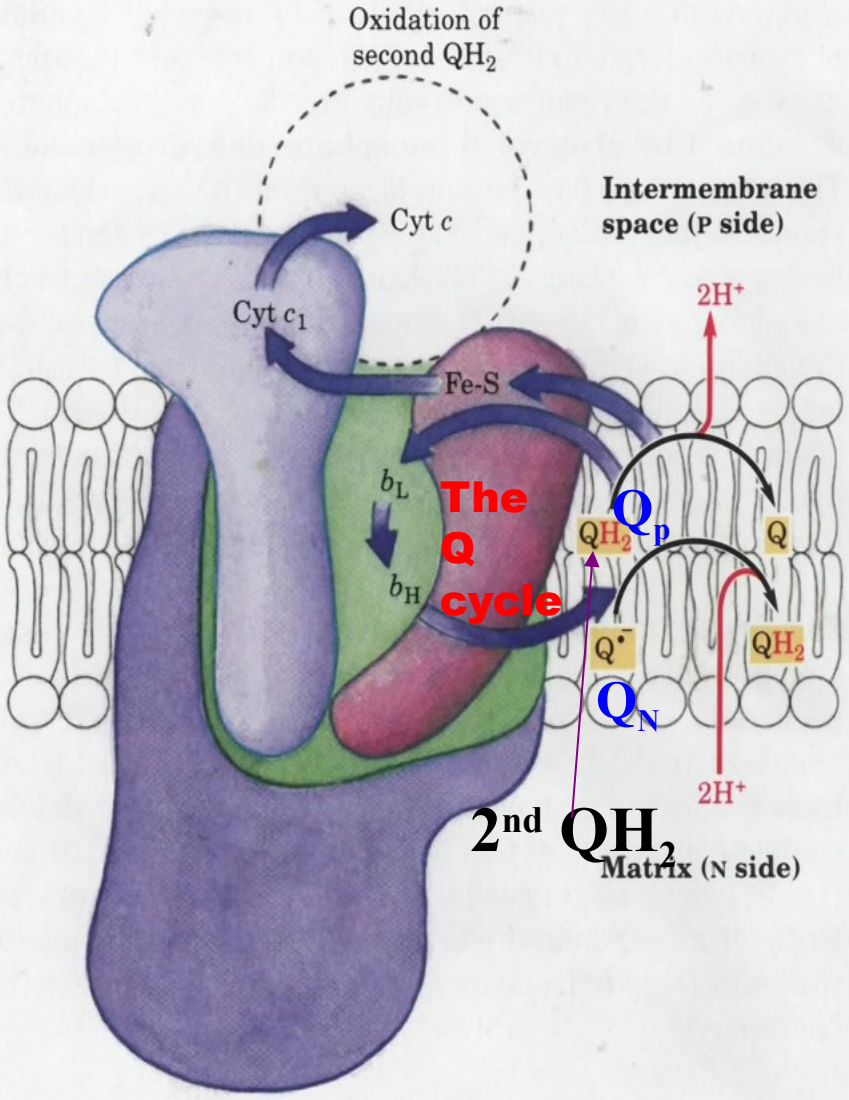

Cytochrome c_1
2Fe-2S
Rieske iron-sulfur protein
Cytochrome b
Intermembrane space (P side)
Matrix (N side)

(a)


Cytochrome c
Rieske iron-sulfur protein
Cytochrome c_1
Intermembrane space (P side)
Q_P
c_1
b_L
2Fe-2S center
Heme
Q_N
b_H
Cytochrome b
Matrix (N side)

(b)





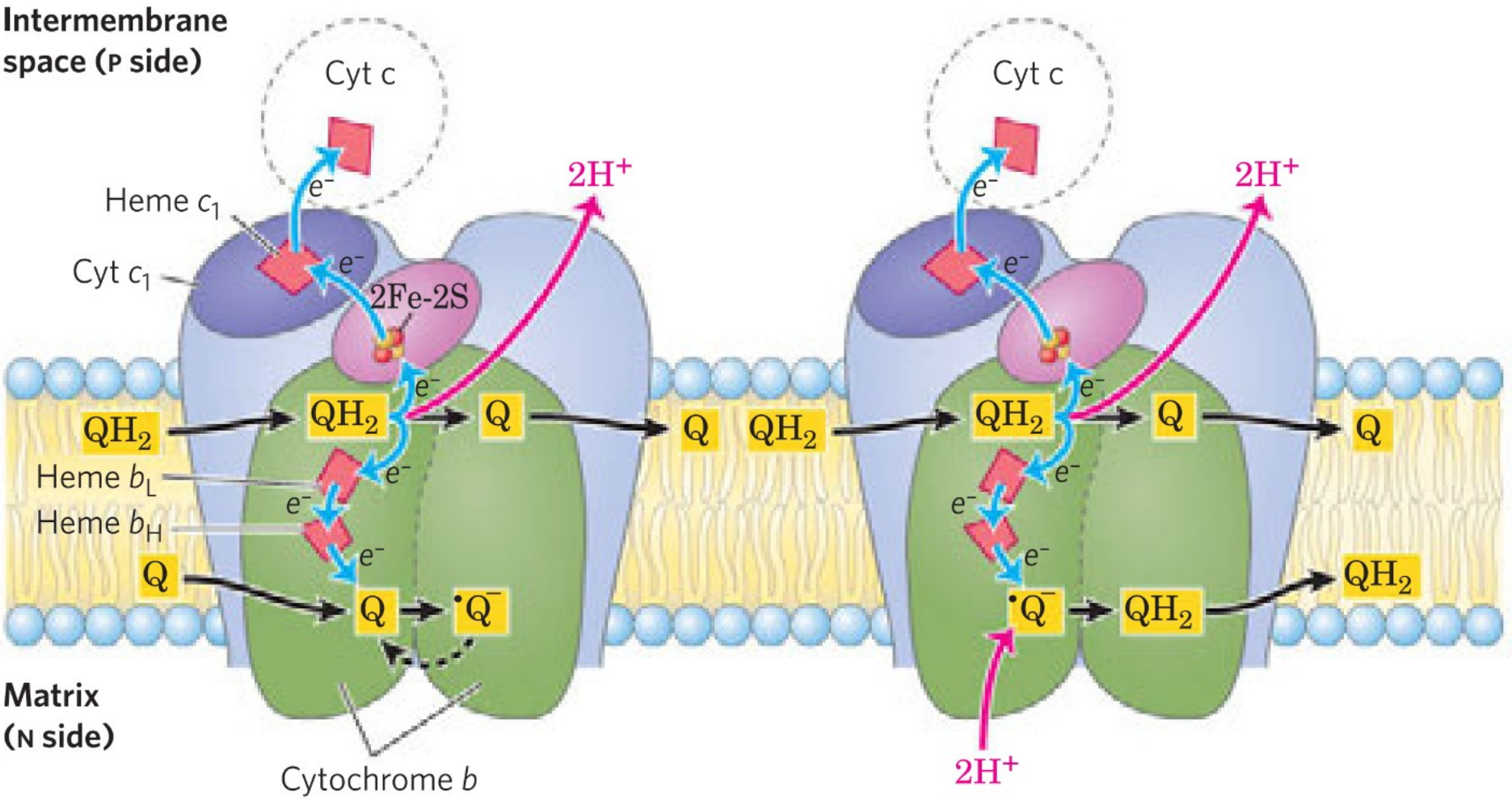
A switch between $2e^-$ carrier (QH_2) and $1e^-$ carrier (Cyt c)

$$QH_2 + \text{Cyt } c_1 \text{ (oxidized)} \longrightarrow Q^{\bullet-} + 2H_P^+ + \text{Cyt } c_1 \text{ (reduced)}$$

$$QH_2 + Q^{\bullet-} + 2H_N^+ + \text{Cyt } c_1 \text{ (oxidized)} \longrightarrow QH_2 + 2H_P^+ + Q + \text{Cyt } c_1 \text{ (reduced)}$$

Net equation:

$$QH_2 + 2 \text{ Cyt } c_1 \text{ (oxidized)} + 2H_N^+ \longrightarrow Q + 2 \text{ Cyt } c_1 \text{ (reduced)} + 4H_P^+$$



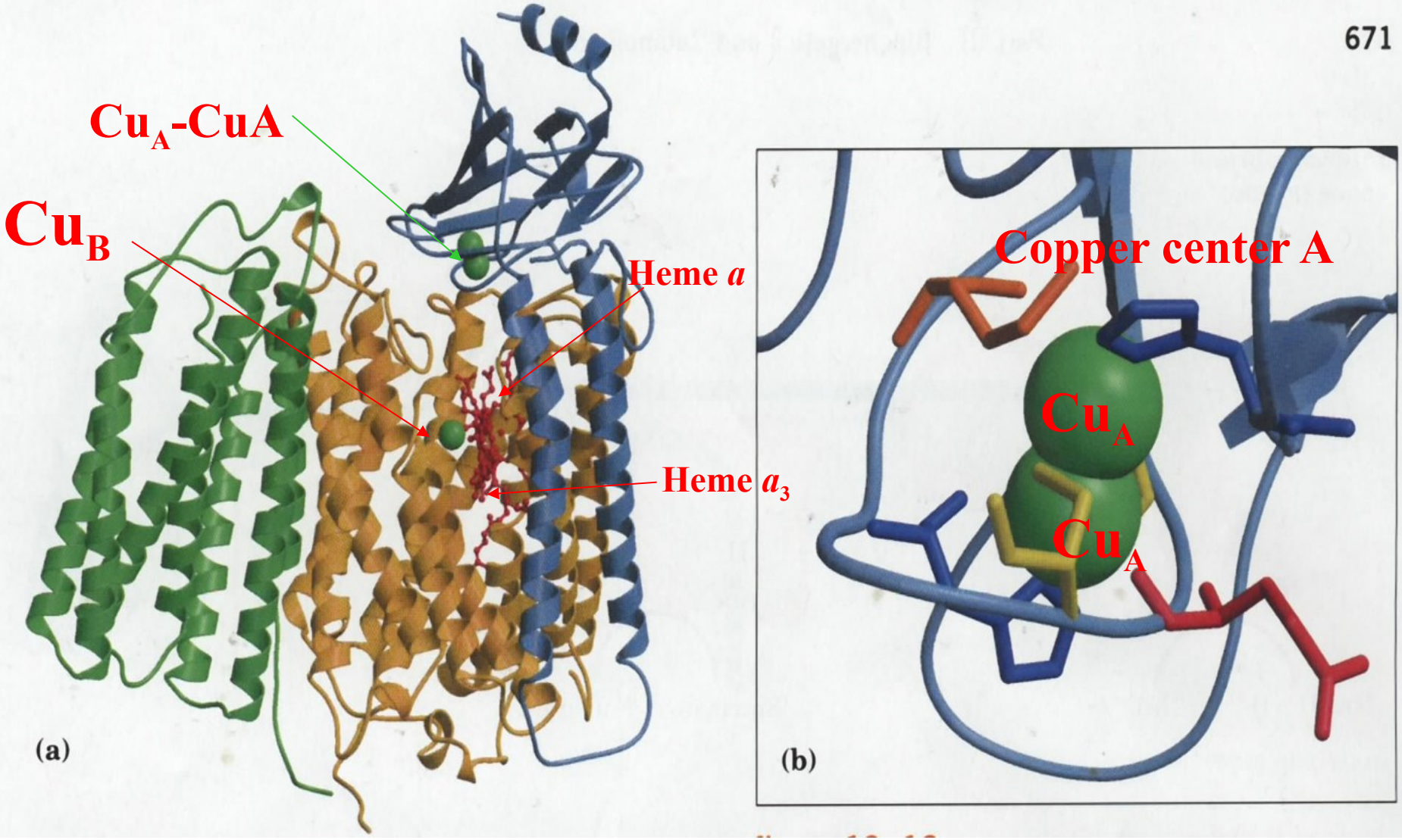
$$QH_2 + Q + \text{cyt } c_1\,(\text{oxidized}) \longrightarrow Q + {}^{\bullet}Q^- + 2H_P^+ + \text{cyt } c_1\,(\text{reduced})$$

$$QH_2 + {}^{\bullet}Q^- + 2H_N^+ + \text{cyt } c_1\,(\text{oxidized}) \longrightarrow Q + 2H_P^+ + QH_2 + \text{cyt } c_1\,(\text{reduced})$$

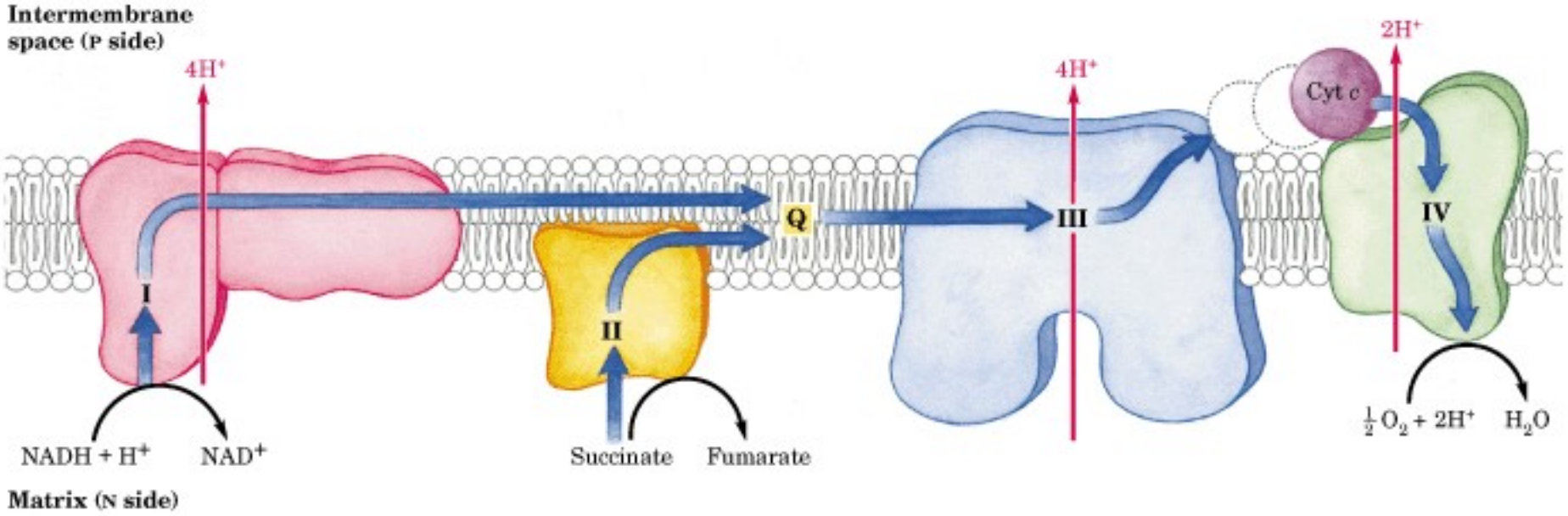
Net equation: $QH_2 + 2 \text{ cyt } c_1\,(\text{oxidized}) + 2H_N^+ \longrightarrow Q + 2 \text{ cyt } c_1\,(\text{reduced}) + 4H_P^+$

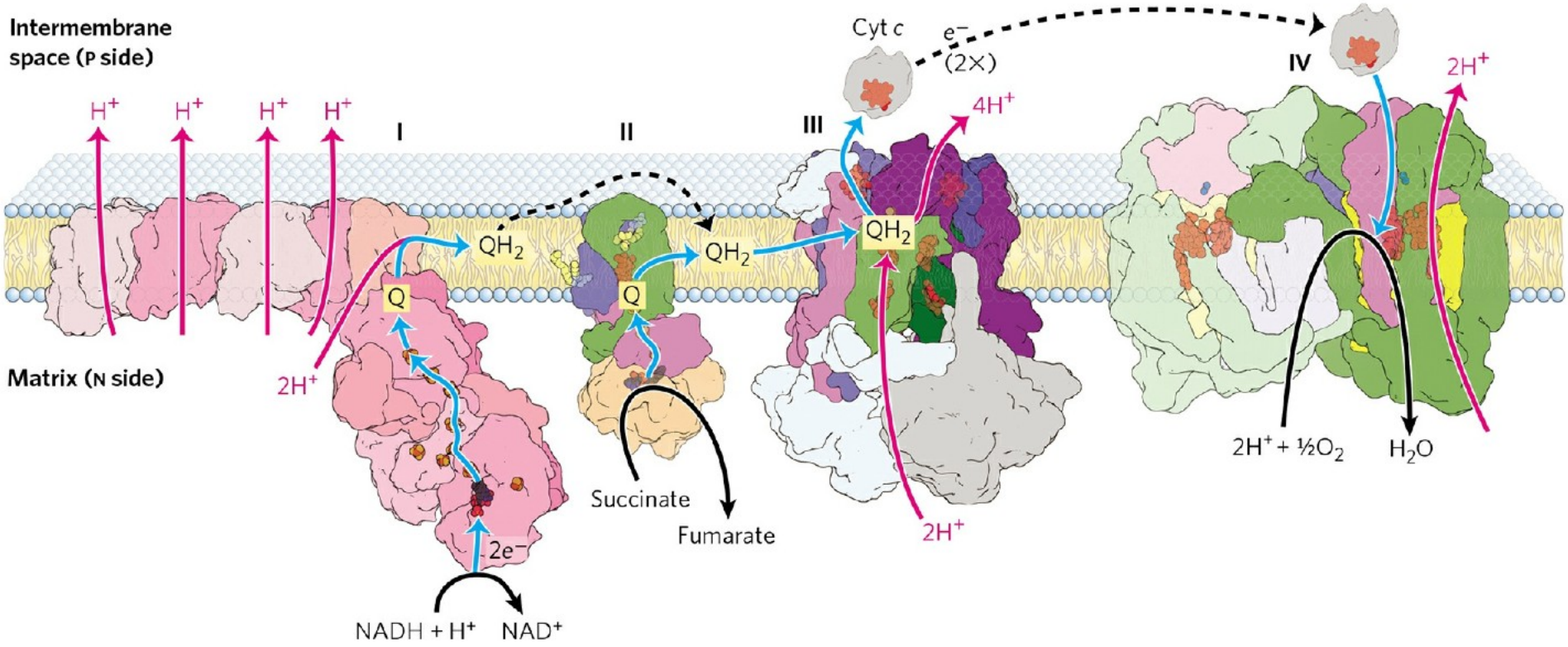
Complex IV :
Cytochrome *c* oxidase, the terminal enzyme of the respiratory chain, contains 3 Cu and 2 heme A groups as electron carriers



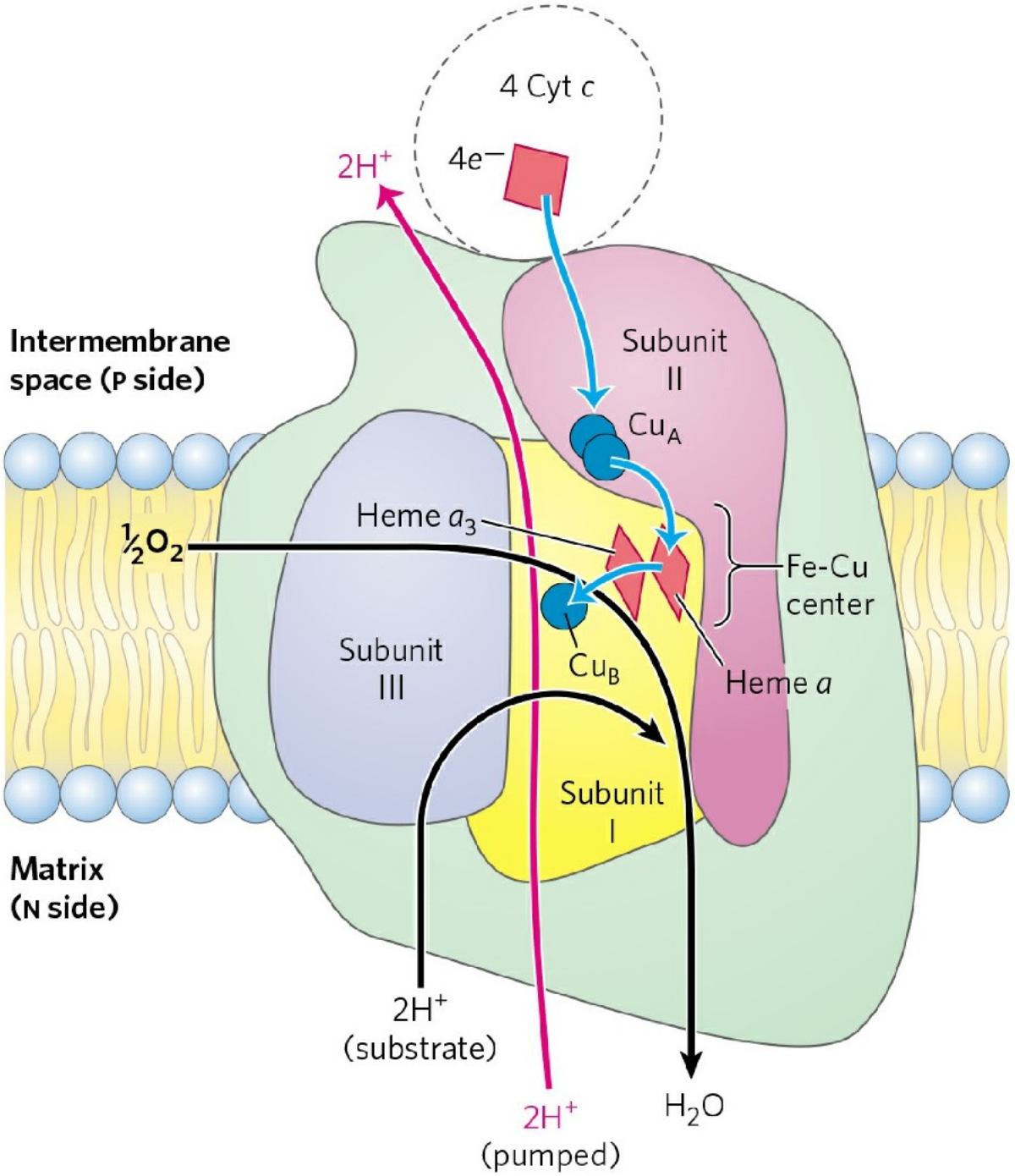

Cu_A-CuA
Cu_B
Heme *a*
Heme a_3
(a)
Copper center A
Cu_A
Cu_A
(b)

~ 10 protons are pumped per NADH, and ~6 per $FADH_2$ oxidized via the respiratory chain.




Intermembrane space (P side)
Matrix (N side)
H+
I
II
III
IV
Cyt c
e−
(2×)
4H+
2H+
QH2
Q
2e−
NADH + H+
NAD+
Succinate
Fumarate
2H+ + ½O2
H2O

7th edition


4 Cyt c
4e−
2H+
Intermembrane space (P side)
Subunit II
CuA
Heme a3
½O2
Fe-Cu center
Subunit III
CuB
Heme a
Subunit I
Matrix (N side)
2H+ (substrate)
2H+ (pumped)
H2O

磷氧比（P/O）

每消耗一个氧原子所形成的 ATP 数或每一对电子通过呼吸链传至 O_2 所产生 的 ATP 分子数。

NADH 呼吸链中 P/O 比为 2.5

$FADH_2$ 呼吸链中 P/O 比为 1.5

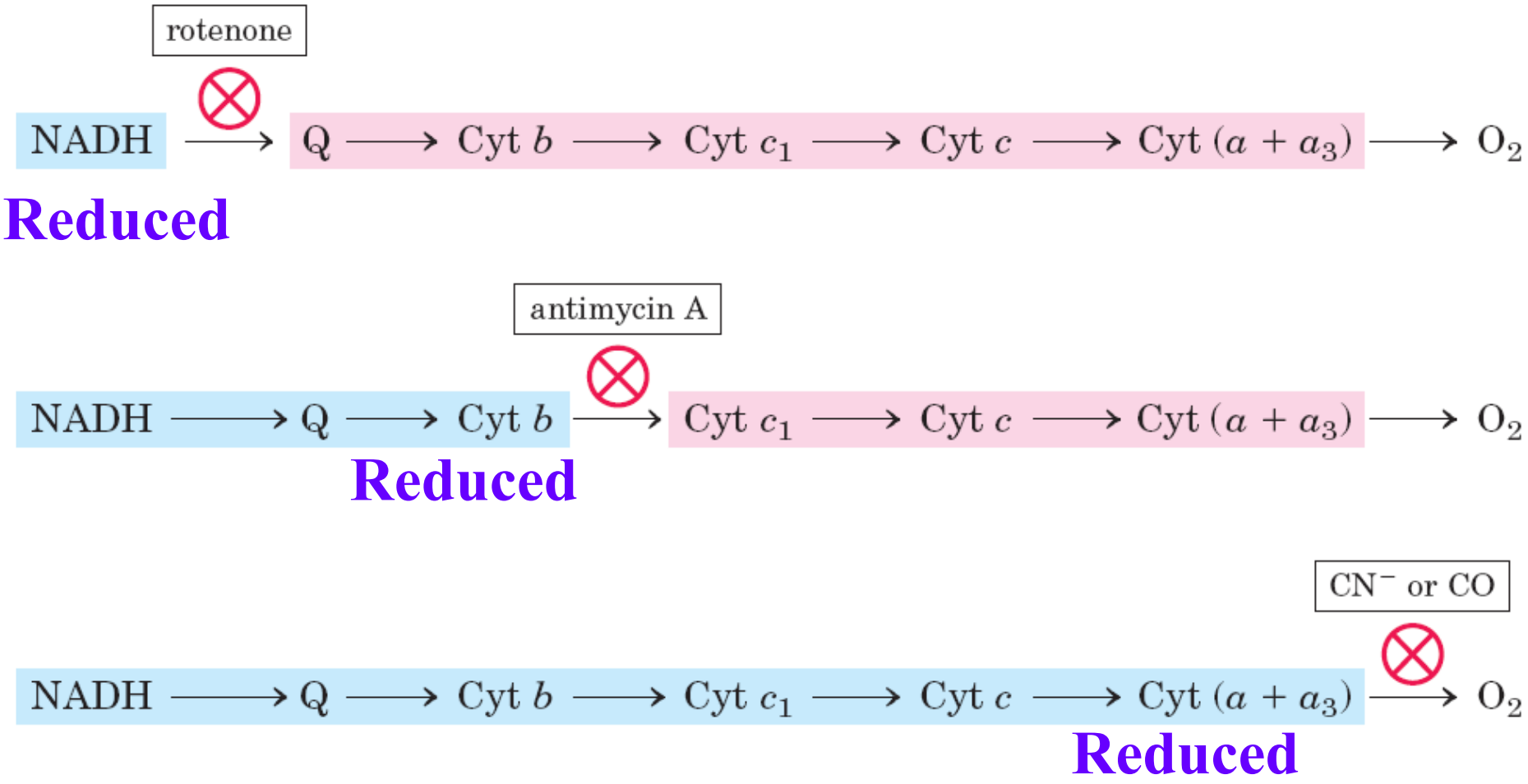
9. The order of the many electron carriers on the respiratory chain have been elucidated via various studies



Electron carriers may have an order of increasing E'^0

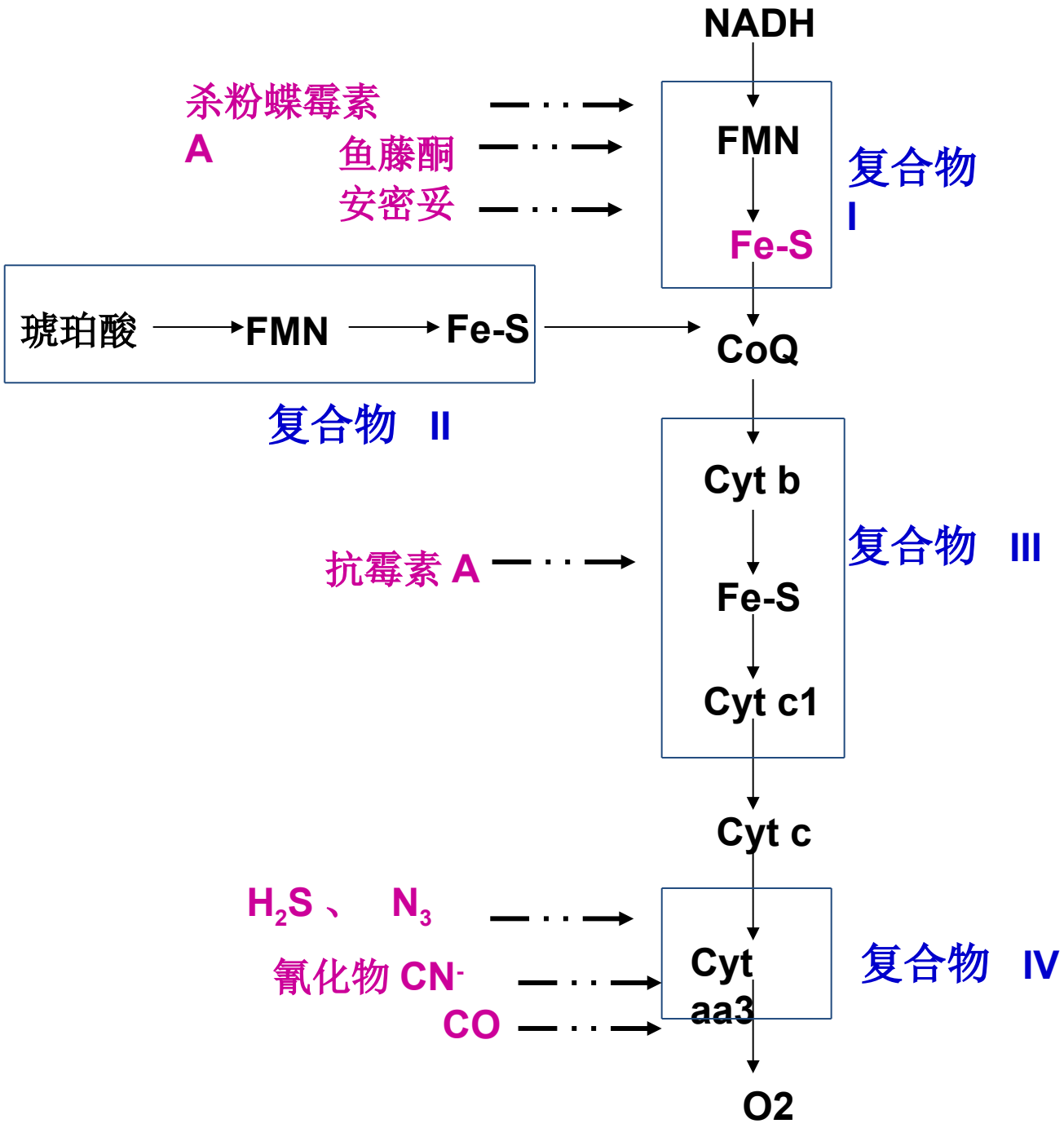
Table 18–2 Standard reduction potentials for respiratory chain and related electron carriers

Redox reaction (half-reaction)		E'_0 (V)
$2H^+ + 2e^- \longrightarrow H_2$		−0.414
$NAD^+ + H^+ + 2e^- \longrightarrow NADH$		−0.320
$NADP^+ + H^+ + 2e^- \longrightarrow NADPH$		−0.324
NADH dehydrogenase (FMN) + $2H^+ + 2e^- \longrightarrow$ NADH dehydrogenase ($FMNH_2$)	(in complex I)	−0.30
Ubiquinone + $2H^+ + 2e^- \longrightarrow$ ubiquinol		0.045
Cytochrome *b* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *b* (Fe^{2+})	(in complex III)	0.077
Cytochrome c_1 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome c_1 (Fe^{2+})	(in complex III)	0.22
Cytochrome *c* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *c* (Fe^{2+})		0.254
Cytochrome *a* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *a* (Fe^{2+})	(in complex IV)	0.29
Cytochrome a_3 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome a_3 (Fe^{2+})	(in complex IV)	0.55
$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$		0.816



Studies with specific inhibitors help to reveal the orders of the e⁻ carriers

电子传递抑制剂


NADH
杀粉蝶霉素A
鱼藤酮
安密妥
FMN
Fe-S
复合物 I
琥珀酸
FMN
Fe-S
复合物 II
CoQ
Cyt b
抗霉素 A
Fe-S
Cyt c1
复合物 III
Cyt c
H_2S、 N_3
氰化物 CN^-
CO
Cyt aa3
复合物 IV
O2

10. Electron transfer to O_2 was found to be obligatorily coupled to ATP synthesis from ADP + P_i in isolated mitochondria



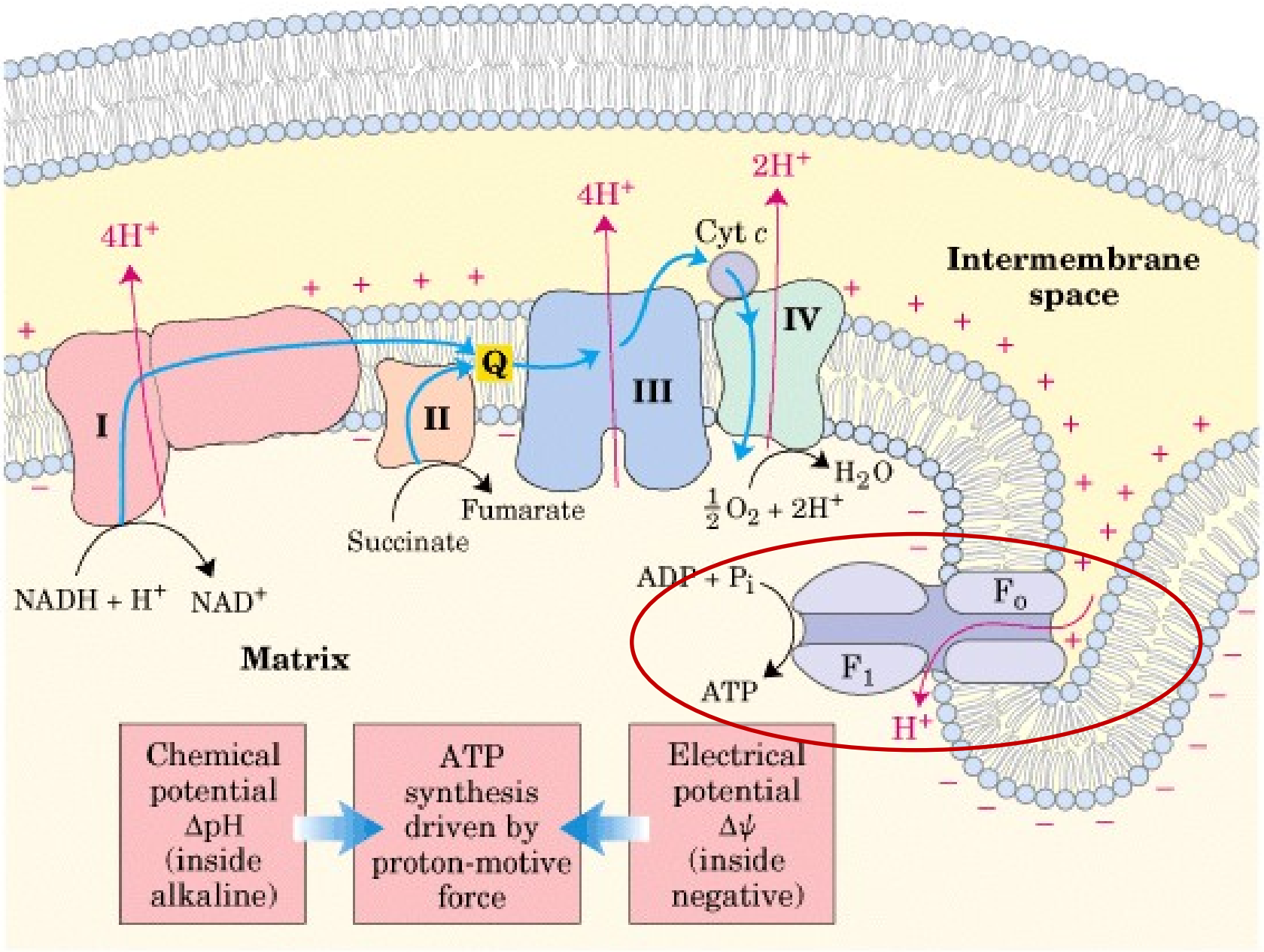
11. How is oxidation of $NADH/FADH_2$ coupled to the phosphorylation of ADP?

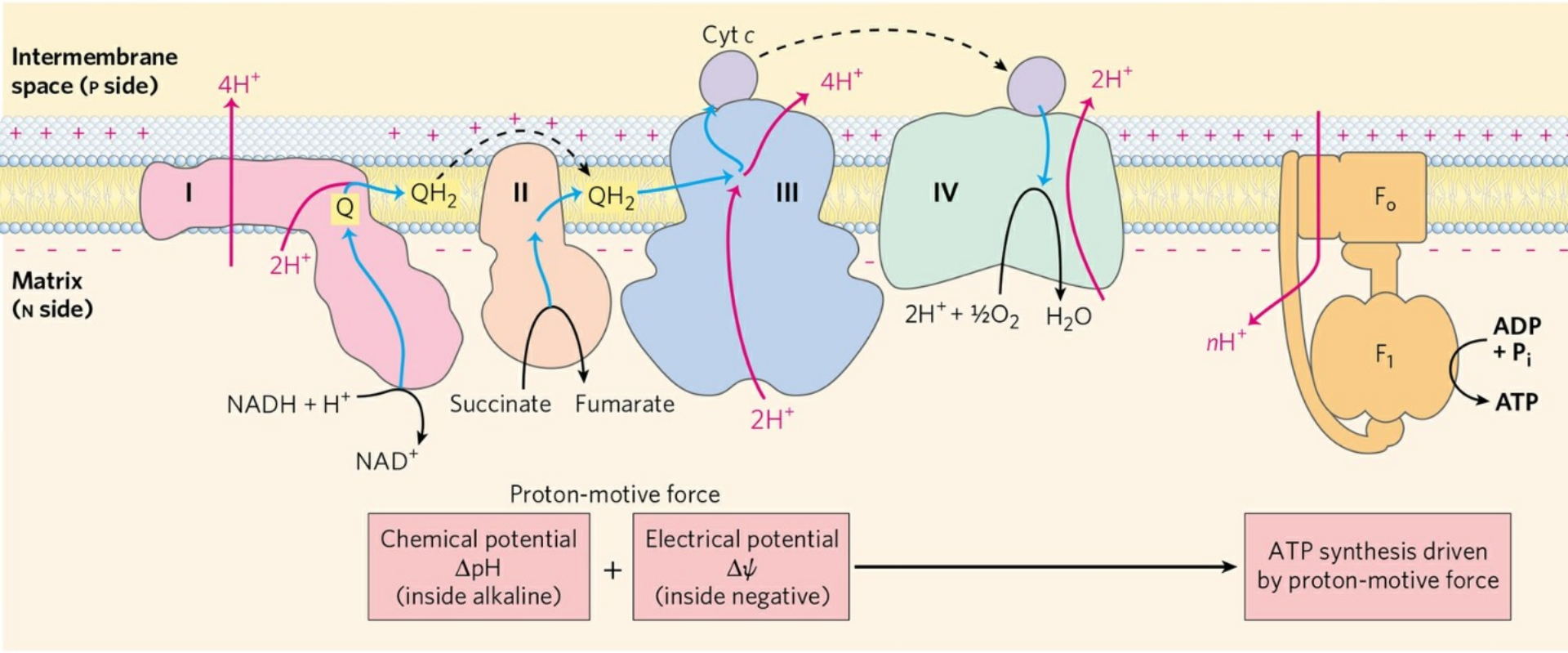


The chemiosmotic model was proposed by Peter Mitchell in 1961 to explain the coupling of electron flow and ATP synthesis

The chemiosmotic model of Mitchell:
e^- flow and ATP synthesis are separate events, coupled via a transmembrane H^+ gradient!




4H+
4H+
2H+
Cyt c
Intermembrane space
I
II
III
IV
Q
H2O
½ O2 + 2H+
Fumarate
Succinate
NADH + H+
NAD+
Matrix
ADP + Pi
ATP
Fo
F1
H+
Chemical potential ΔpH (inside alkaline)
ATP synthesis driven by proton-motive force
Electrical potential Δψ (inside negative)

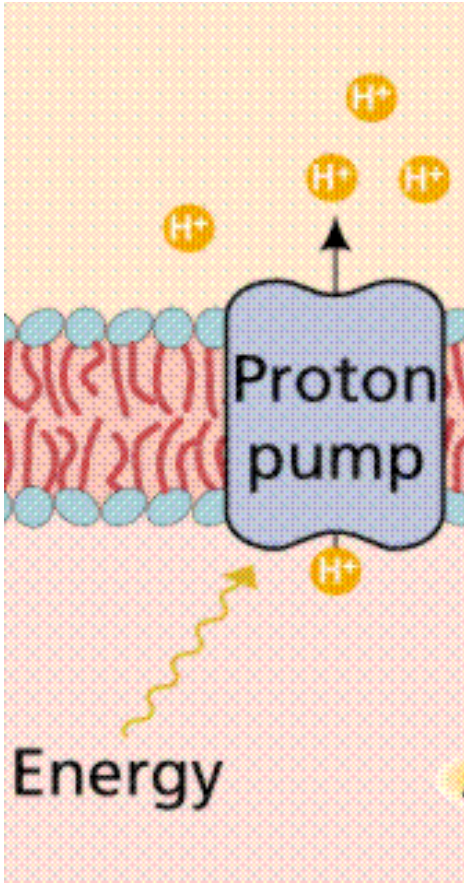

Intermembrane space (P side)
Matrix (N side)
4H+
2H+
I
Q
QH2
II
QH2
Cyt c
4H+
III
2H+
IV
2H+ + ½O2
H2O
Fo
F1
nH+
ADP + Pi
ATP
NADH + H+
NAD+
Succinate
Fumarate
Proton-motive force
Chemical potential ΔpH (inside alkaline)
+
Electrical potential Δψ (inside negative)
ATP synthesis driven by proton-motive force

化学渗透学说

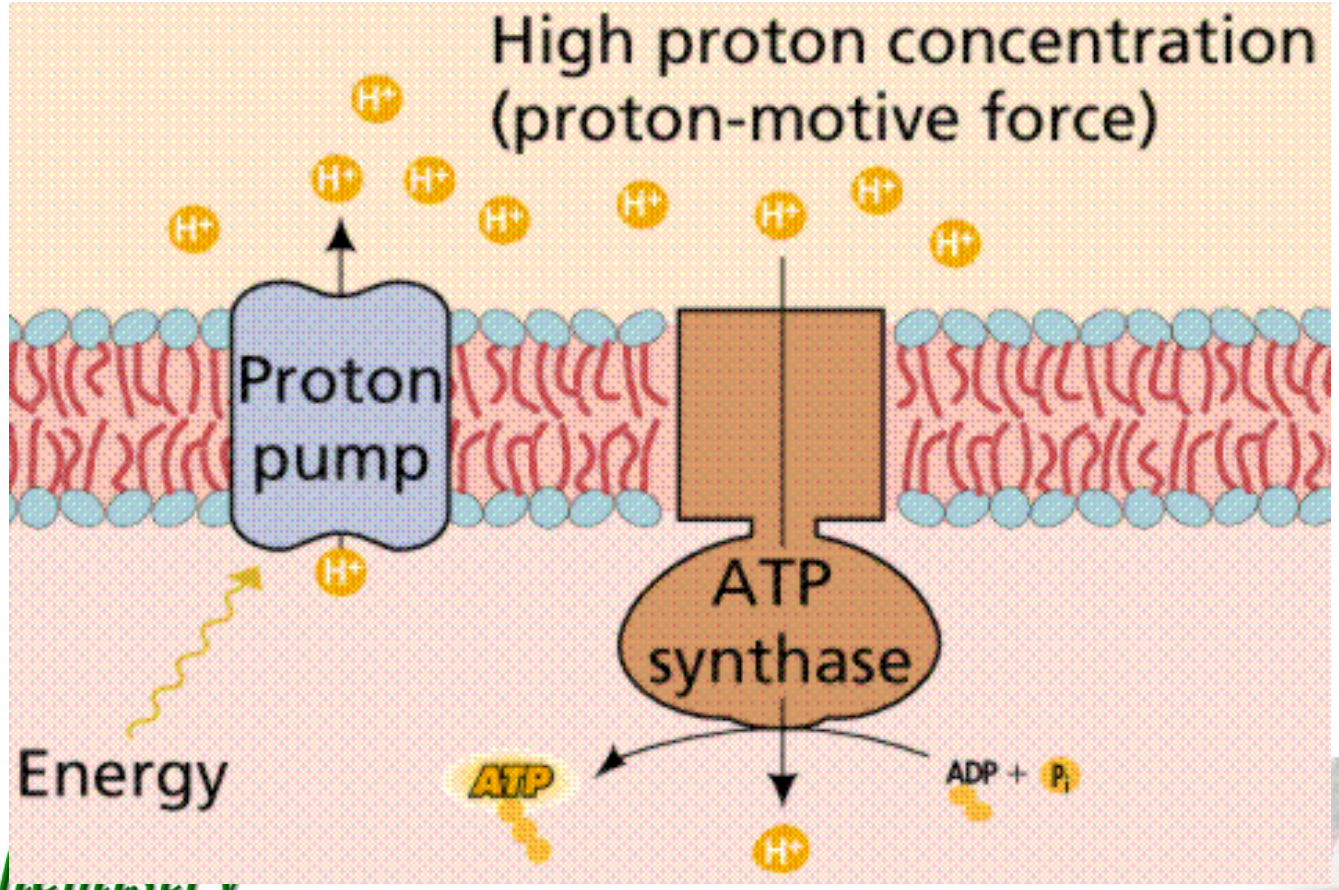
化学渗透假说的要点：

α. 线粒体内膜的电子传递链是一个质子泵；

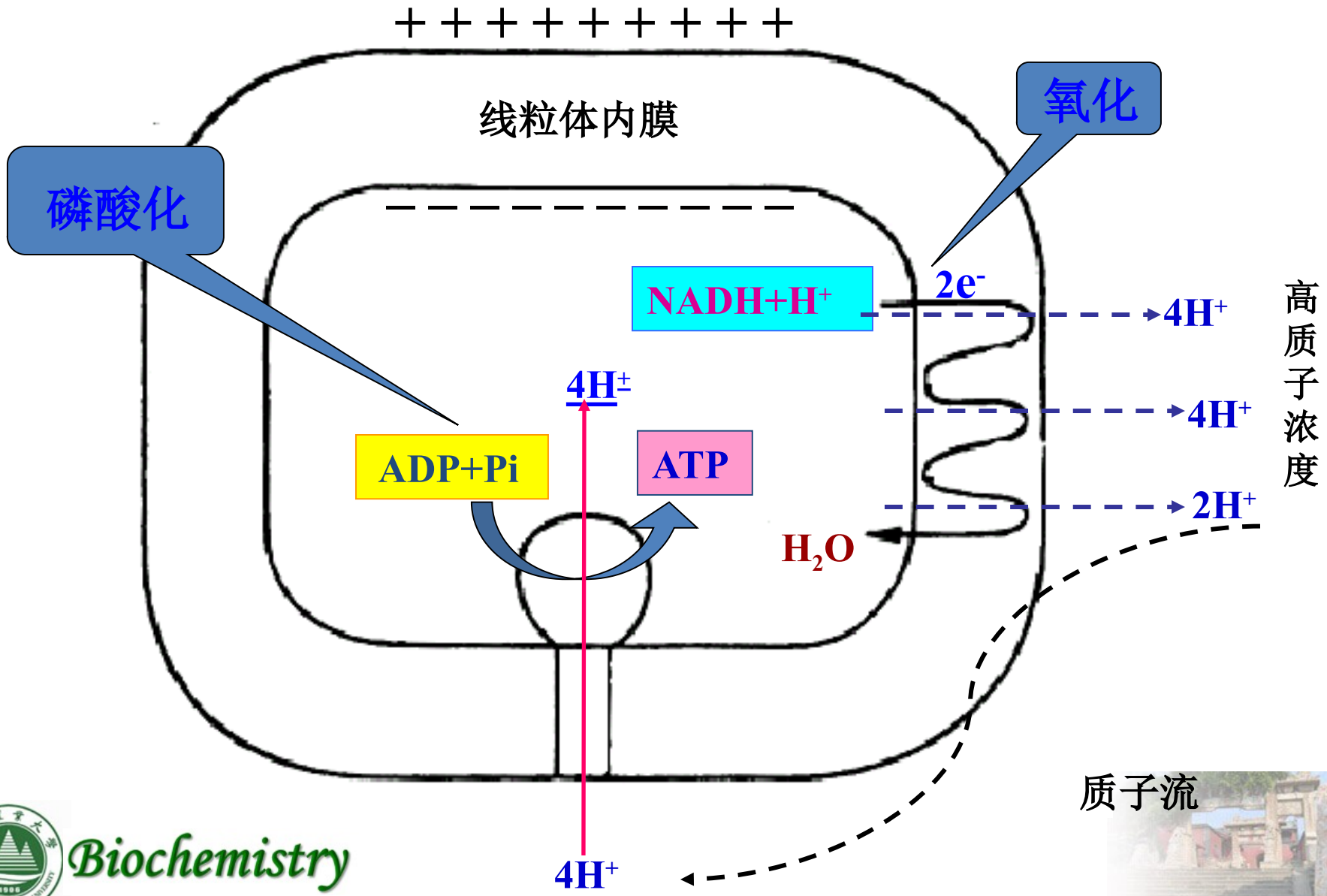
β. 在电子传递链中，电子由高能状态传递到低能状态时释放出来的能量，用于驱动膜内侧的 H^+ 迁移到膜外侧，线粒体内膜对 H^+ 是不通透的，这样，在膜的内侧与外侧就产生了跨膜质子梯度（$\Delta\pi H$）和电位梯度（$\Delta\psi$）；



χ. 在膜内外势能差（Δ πH 和Δψ）的驱动下，膜外高能质子沿着一个特殊通道（ ATΠ 酶的组成部分），跨膜回到膜内侧。质子跨膜过程中释放的能量，直接驱动 AΔΠ 和磷酸合成 ATΠ 。



化学渗透学说示意图





The Nobel Prize in Chemistry 1978

"for his contribution to the understanding of biological energy transfer through the formulation of the chemiosmotic theory"

Press release



Peter D. Mitchell

Great Britain

Glynn Research Laboratories
Bodmin, Great Britain

1920 - 1992

Biography

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• For help, info, credits or comments, see "About this project"
• Last updated by Webmaster@www.nobel.se / June 18, 1998

氧化磷酸化的解偶联和抑制

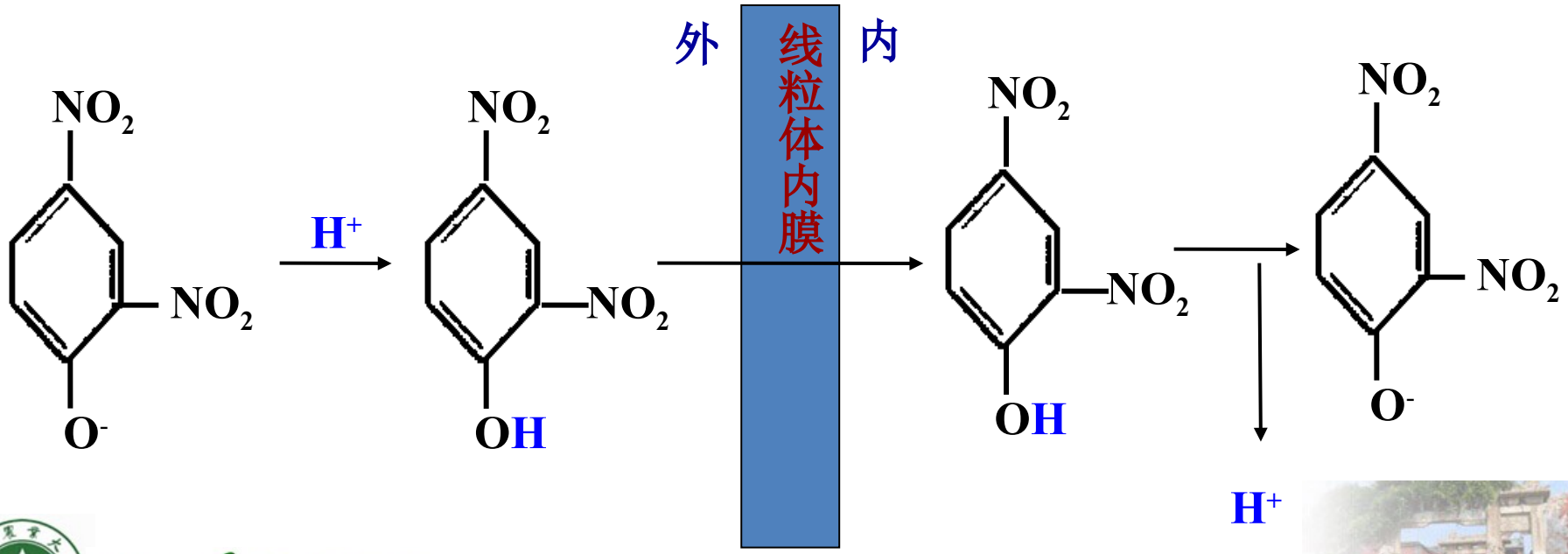
- 解偶联剂（ uncoupler)
- 氧化磷酸化抑制剂
- 离子载体抑制剂 (ionophore)

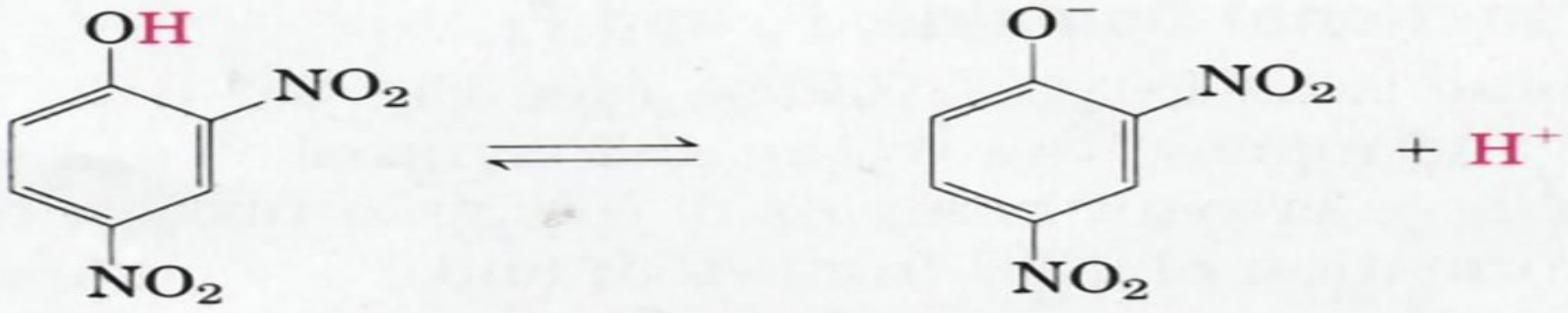


1、解偶联剂（υνχουπλερ）

使电子传递与 ΑΔΠ 磷酸化分离，只抑制 ΑΤΠ 的形成，而不抑制电子传递

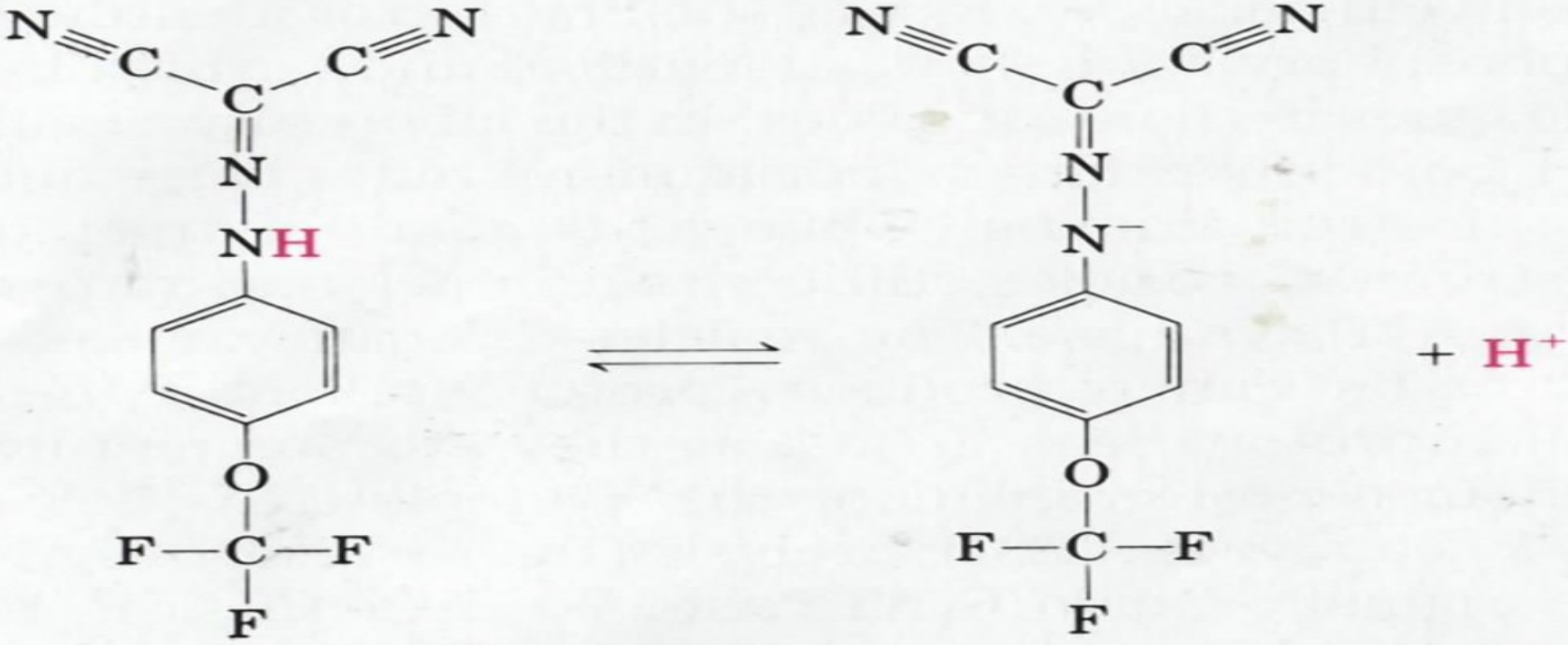
[例 1]2,4– 二硝基苯酚（δινιτροπηενολ,ΔΝΠ）：酸性条件下为脂溶性物质，自由通过线粒体膜，将膜内 H^+ 带进基质，破坏了 H^+ 梯度.





2,4-Dinitrophenol
(DNP)

proton gradient is dissipated



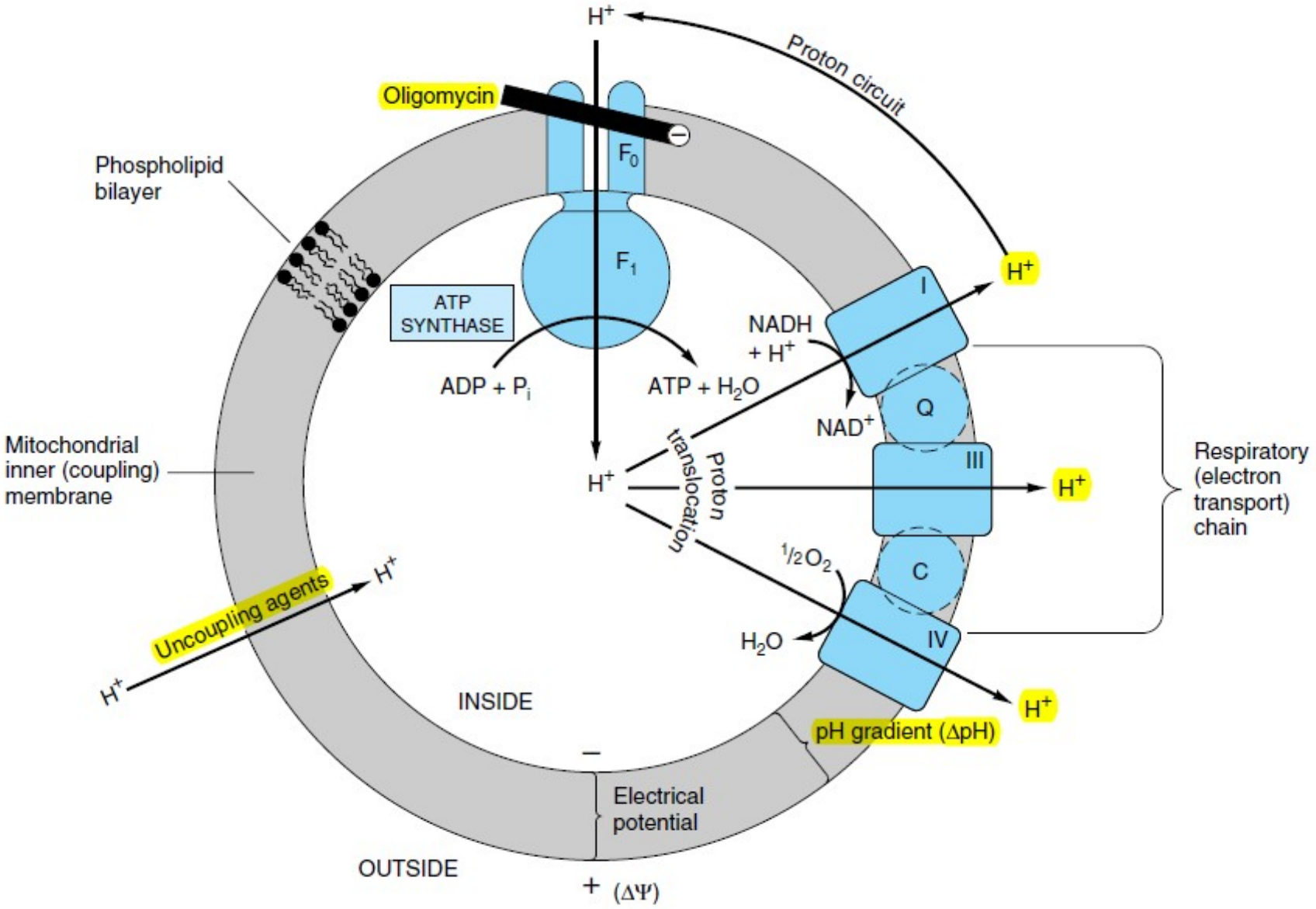
Carbonylcyanide-*p*-
trifluoromethoxyphenylhydrazone
(FCCP)

2、氧化磷酸化抑制剂

【作用】对电子传递和 ADP 的磷酸化均有抑制。

[例] 寡霉素 (Oligomycin): 与 ATP 合酶的 Φ_0 部位结合，破坏 H^+ 回流，影响呼吸链质子泵的功能，抑制电子传递。




H+
Proton circuit
Oligomycin
Phospholipid bilayer
F0
F1
ATP SYNTHASE
ADP + Pi
ATP + H2O
NADH + H+
NAD+
I
Q
III
C
IV
H+
Proton translocation
1/2 O2
H2O
Respiratory (electron transport) chain
Mitochondrial inner (coupling) membrane
Uncoupling agents
INSIDE
OUTSIDE
pH gradient (ΔpH)
Electrical potential
(ΔΨ)

3、离子载体抑制剂

【作用】与离子形成脂溶性物质，对 AΔΠ 的磷酸化有抑制。

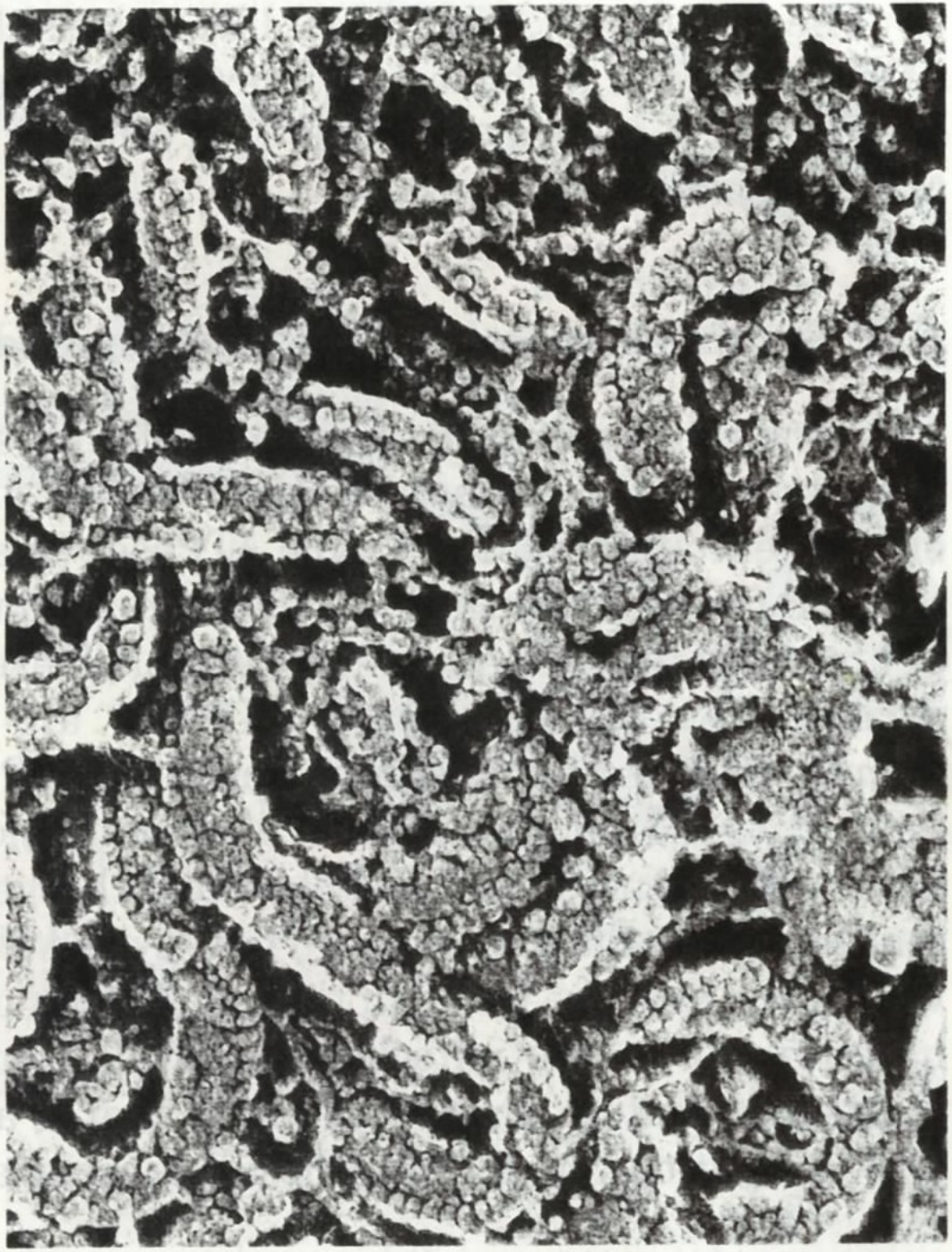
[例] 缬氨霉素 (Valinomycin) 结合 K^+，形成脂溶性复合物，于是 K^+容易透过膜。

短杆菌肽结合 K^+、Na^+， 破坏了膜两侧的电位梯度。



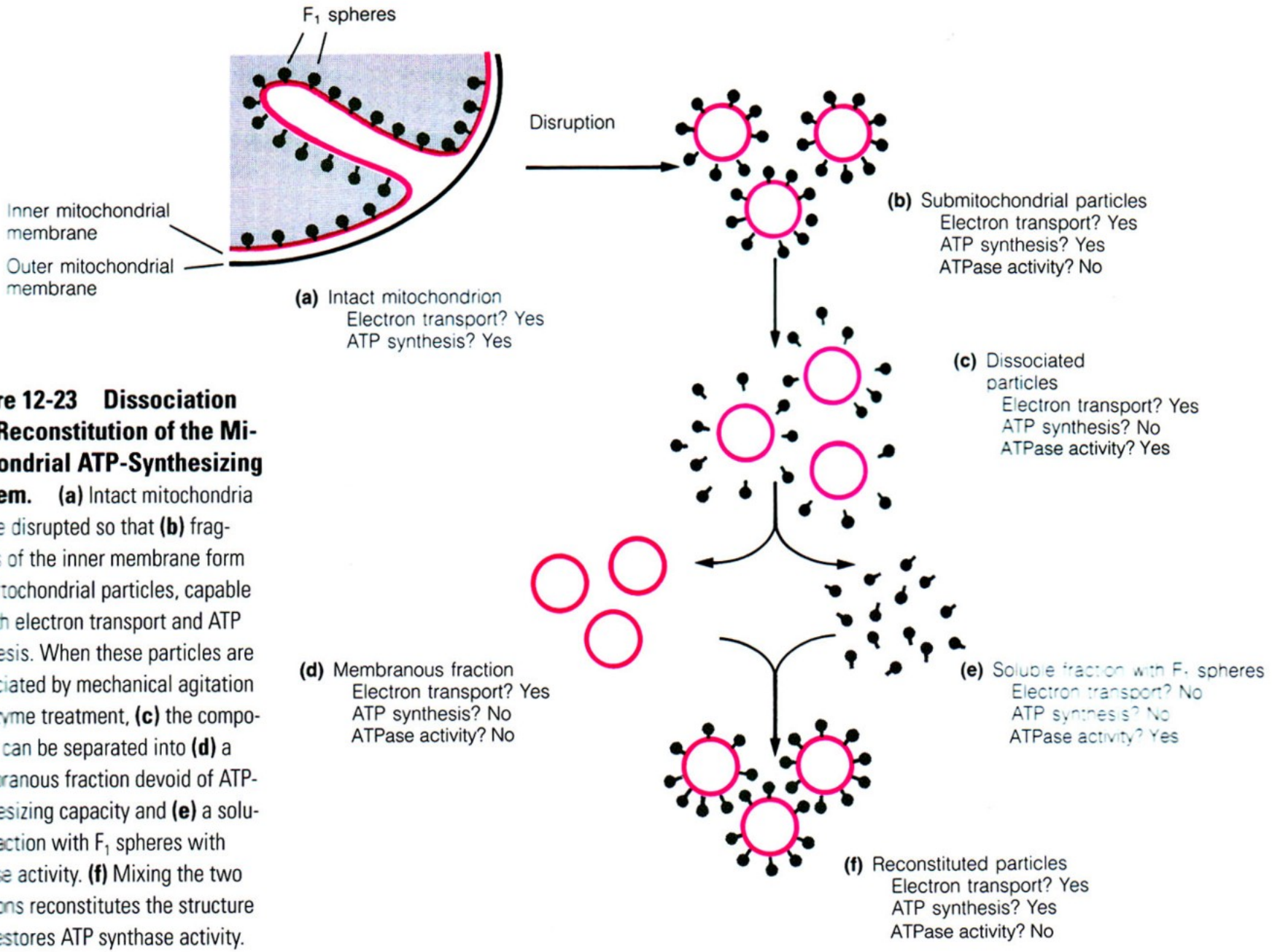
12. ATP synthase was first identified by dissociation and reconstitution studies





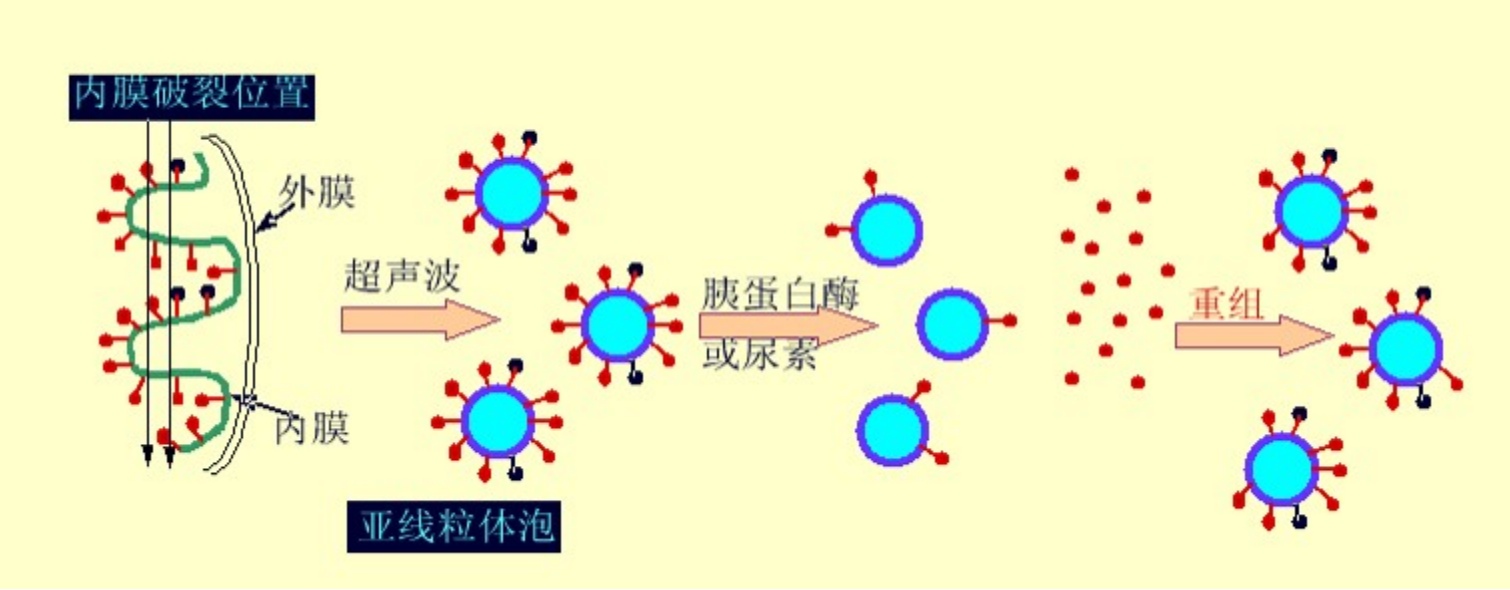
Abundant knob-like protruding structures were observed on the matrix side of the inner mitochondrial membrane by EM (Racker in 1960).





gure 12-23 Dissociation d Reconstitution of the Mi-chondrial ATP-Synthesizing stem. **(a)** Intact mitochondria be disrupted so that **(b)** frag-nts of the inner membrane form bmitochondrial particles, capable both electron transport and ATP thesis. When these particles are sociated by mechanical agitation enzyme treatment, **(c)** the compo-nts can be separated into **(d)** a mbranous fraction devoid of ATP-thesizing capacity and **(e)** a solu- fraction with F_1 spheres with Pase activity. **(f)** Mixing the two ctions reconstitutes the structure d restores ATP synthase activity.

化学渗透学说的实验证据（氧化磷酸化的重组实验）



破坏 F1 –F0 因子

有电子传递作用；
无 ATP 生成

F1-F0 重组

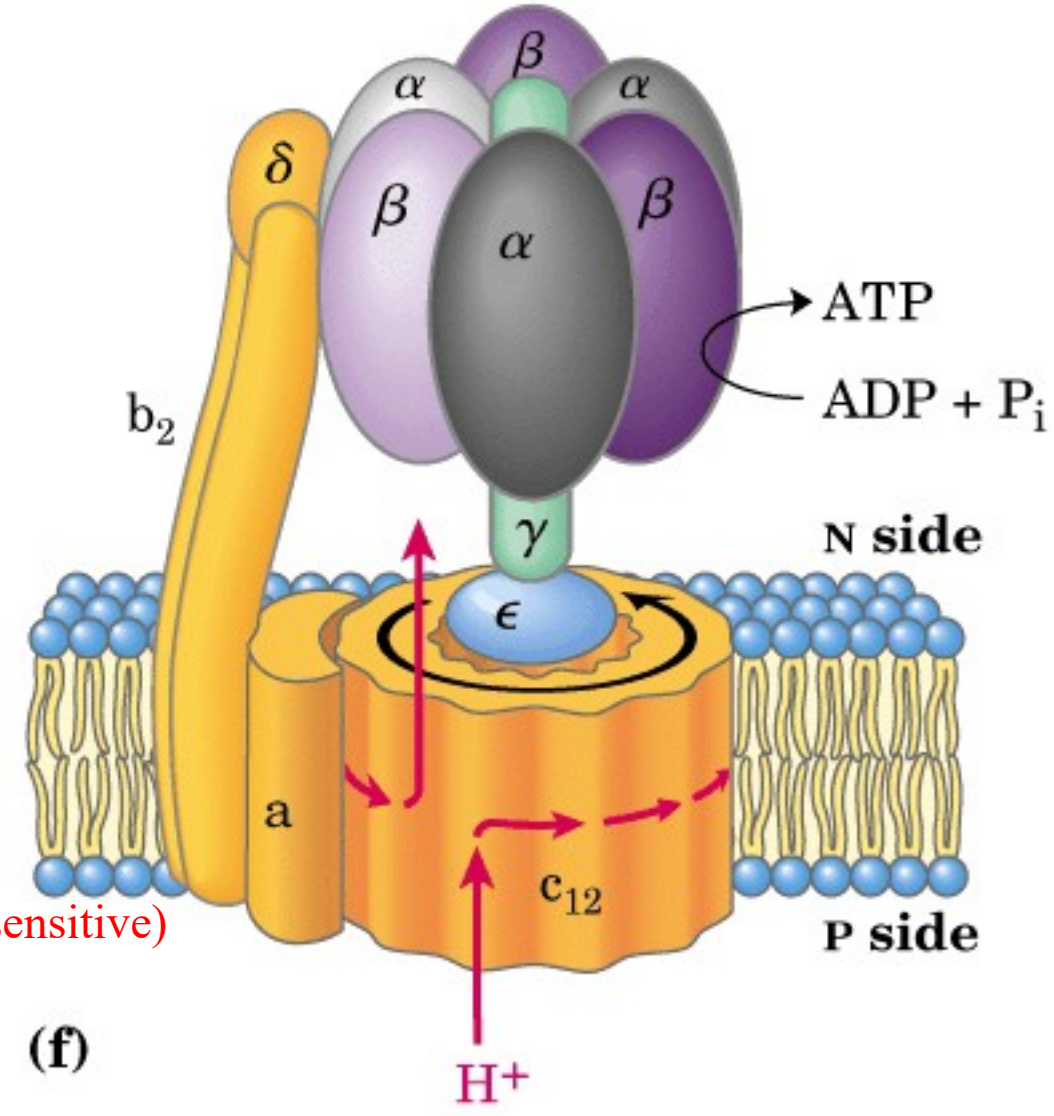
电子传递与氧化磷酸化偶联

ATP synthase comprises a proton channel (F_o) and an ATPase (F_1)

F_1: $\alpha_3\beta_3\gamma\delta\varepsilon$.

F_o: $ab_2c_{10\text{-}12}$

Fo (*o* denoting oligomycin-sensitive)



(f)

Biochemistry

The Nobel Prize in Chemistry 1997

Press release

"for their elucidation of the enzymatic mechanism underlying the synthesis of adenosine triphosphate (ATP)"

Paul D. Boyer

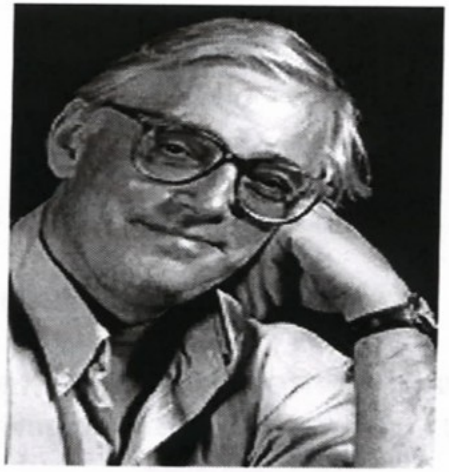
USA

University of California, Los Angeles, USA

1918 -

Autobiography
Nobel Diploma
Prize Award
Ceremony





John E. Walker

U.K.

Medical Research Council Laboratory of Molecular Biology, Cambridge, U.K.

1941 -

Autobiography
Nobel Diploma
Prize Award
Ceremony

13. Electrons in NADH generated in cytosol is shuttled into mitochondria to enter the respiratory chain



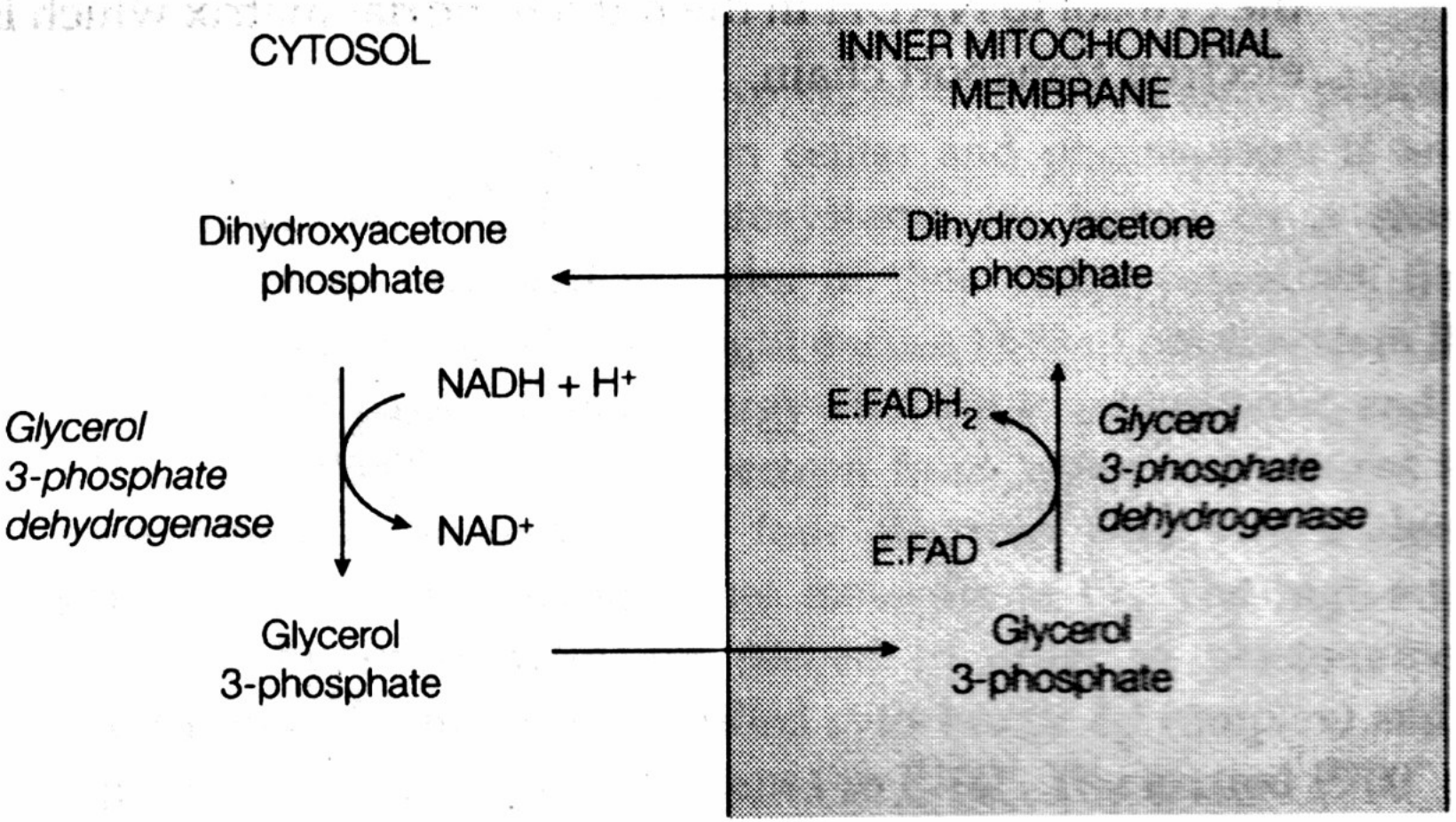
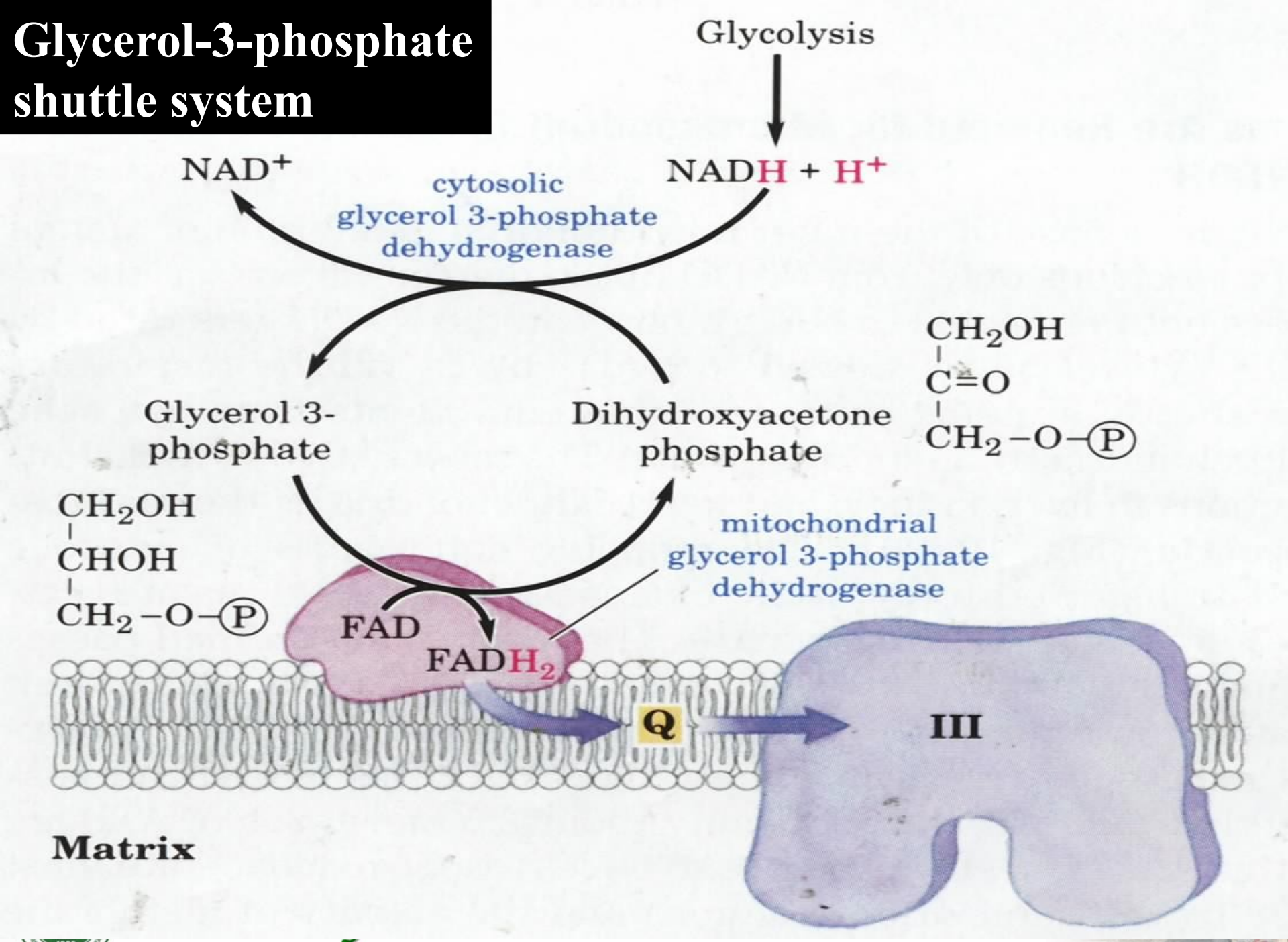
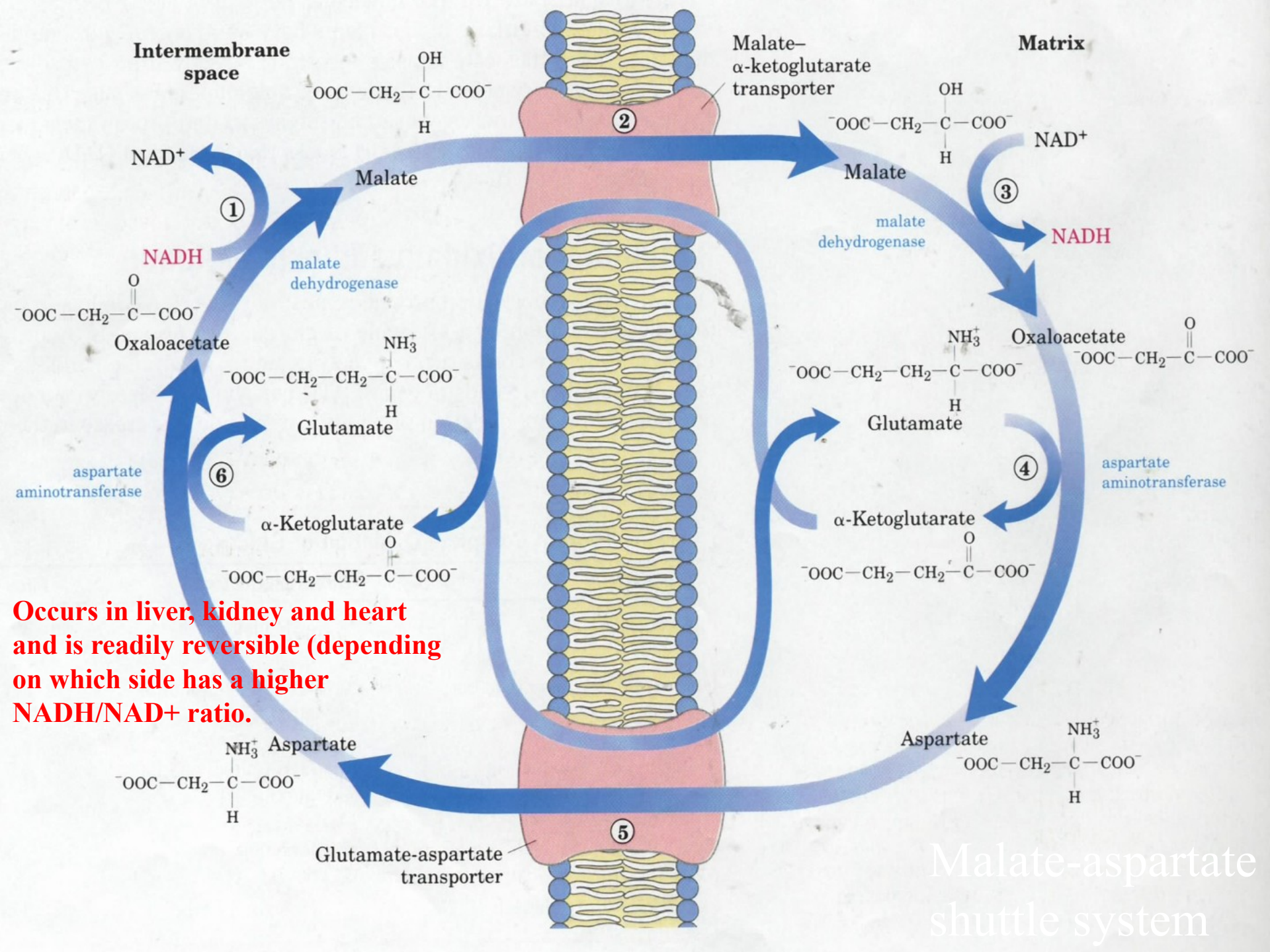


Fig. 5. The glycerol 3-phosphate shuttle.

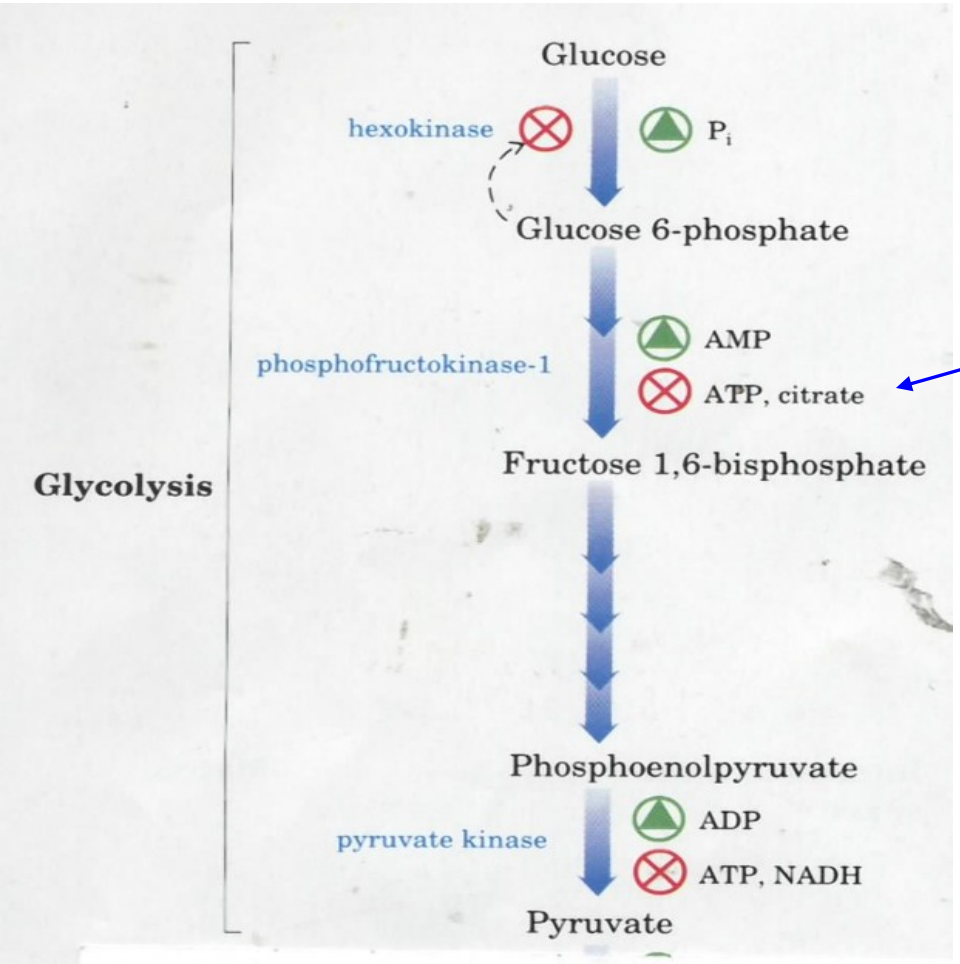
Glycerol-3-phosphate shuttle system

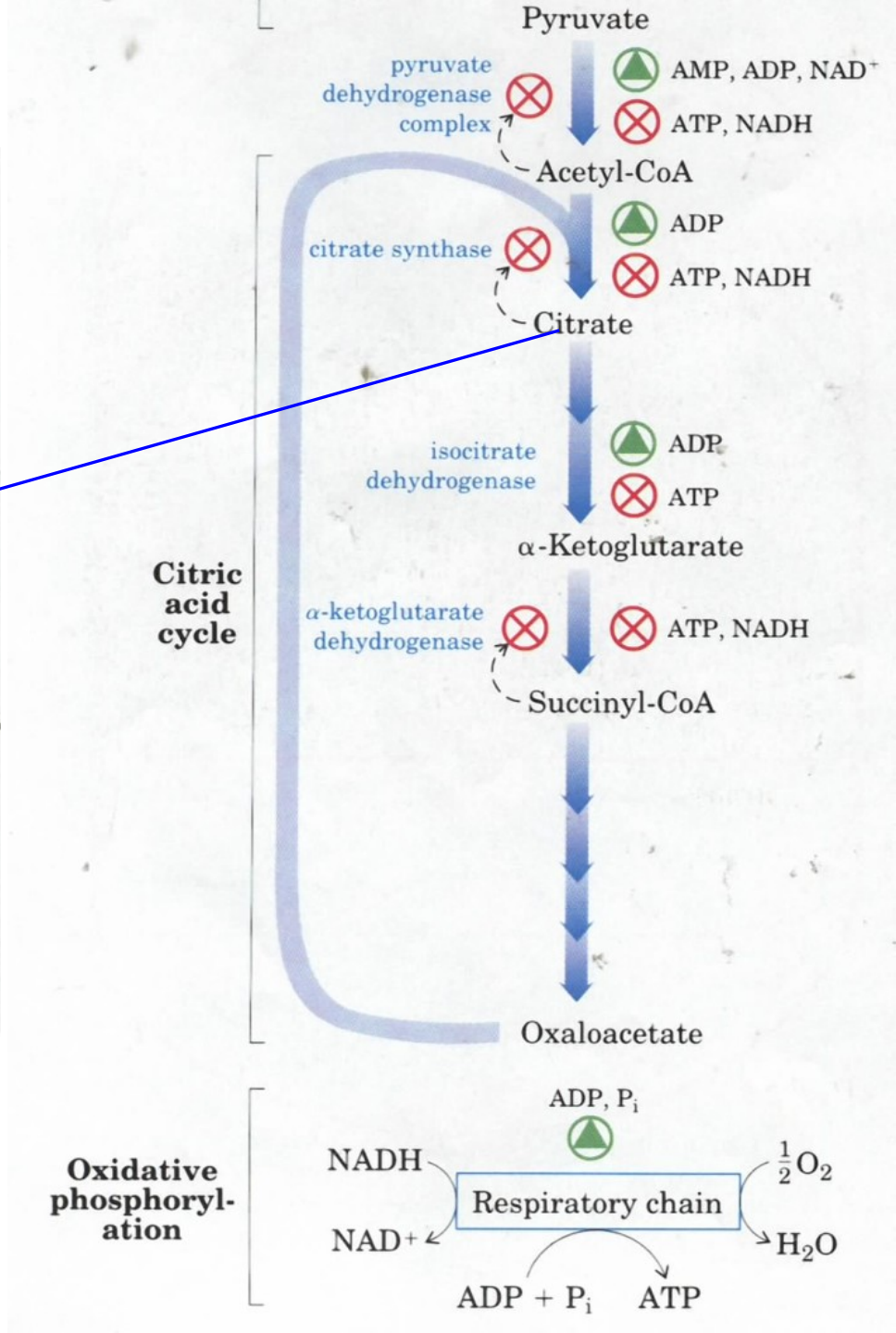



Intermembrane space
Matrix
Malate–α-ketoglutarate transporter
NAD+
NADH
Malate
malate dehydrogenase
Oxaloacetate
Glutamate
α-Ketoglutarate
aspartate aminotransferase
Aspartate
Glutamate-aspartate transporter
1
2
3
4
5
6
Occurs in liver, kidney and heart and is readily reversible (depending on which side has a higher NADH/NAD+ ratio.
Malate-aspartate shuttle system

14. The pathways leading to ATP synthesis are coordinately regulated







Biochemistry

Summary

- Bioenergy is chemical energy, studied in terms of free energy and free energy change (ΔG).
- ATP acts as the free energy carrier in cells.
- Bioenergy is mainly produced via stepwise electron flow (redox reactions) through a series of electron carriers having increasing levels of reduction potential (E).
- Electrons released from the oxidation of nutrient fuels are initially channeled to a few universal electron carriers (including NADH and $FADH_2$).



Electrons collected in NADH and $FADH_2$ are released (at different entering points) and transported to O_2 via the respiratory chain, which consists of four multiprotein complexes (I, II, III, and IV) and two mobile electron carriers (ubiquinone and cytochrome c).

The order of the many electron carriers on the respiratory chain have been elucidated via various studies, including measurements of the standard reduction potential and effects of various respiratory chain inhibitors.



Electron transfer to O_2 was found to be coupled to ATP synthesis from ADP + Pi in isolated mitochondria.

The chemiosmotic theory explains the coupling of electron flow and ATP synthesis.

ATP synthase comprises a proton channel (F_o) and a ATPase (F_1).

Electrons in NADH generated in cytosol is shuttled into mitochondria to enter the respiratory chain.

The pathways leading to ATP sysnthesis is coordinately regulated.



END

Thank you



Reduced cytochromes has three absorption bands in the visible wavelengths

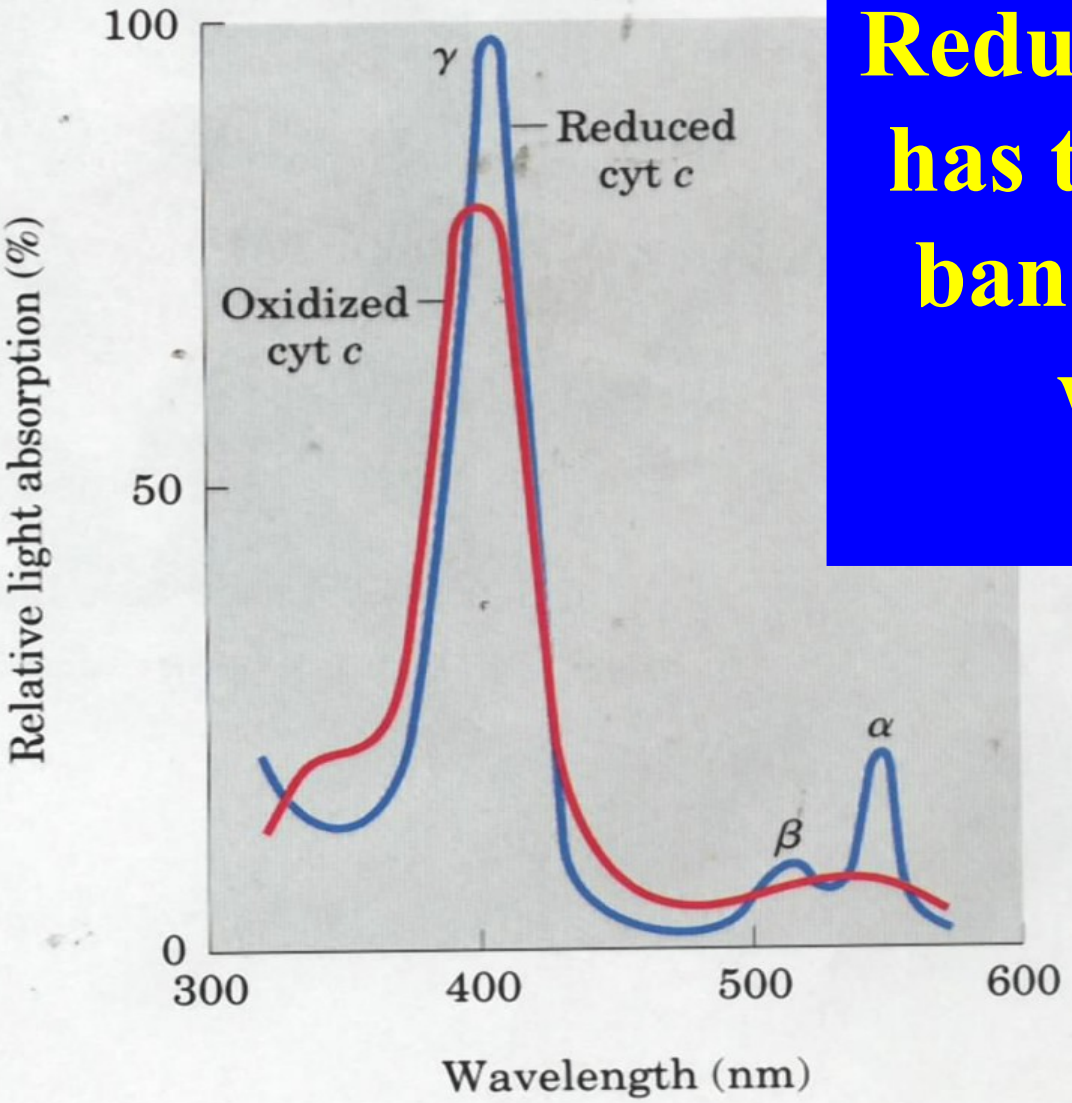
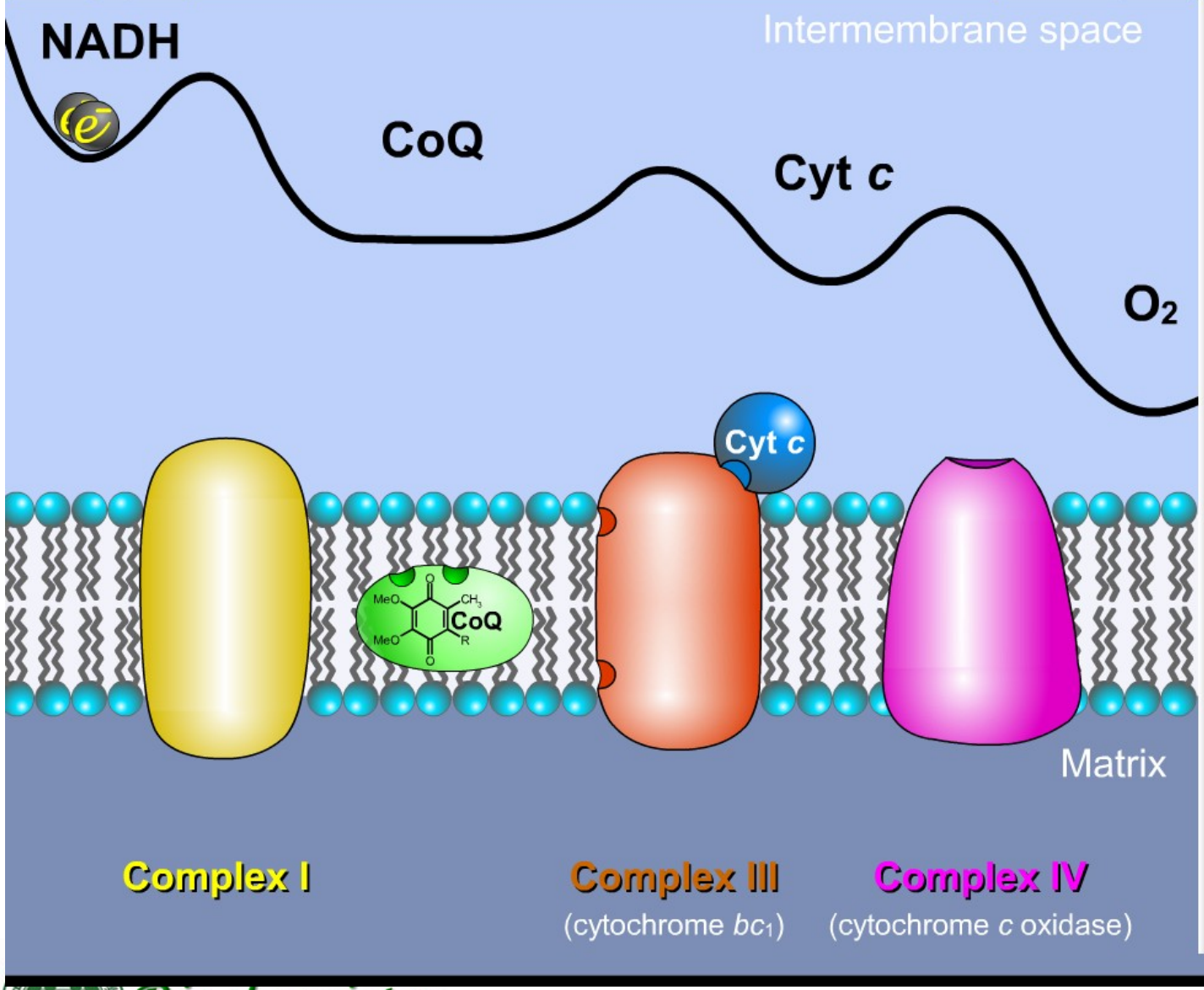


figure 19–4
Absorption spectra of cytochrome *c* in its oxidized (red) and reduced (blue) forms. Also labeled are the characteristic α, β, and γ bands of the reduced form.



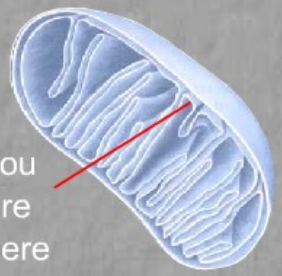
Label graph ▸ Go directly to ▸

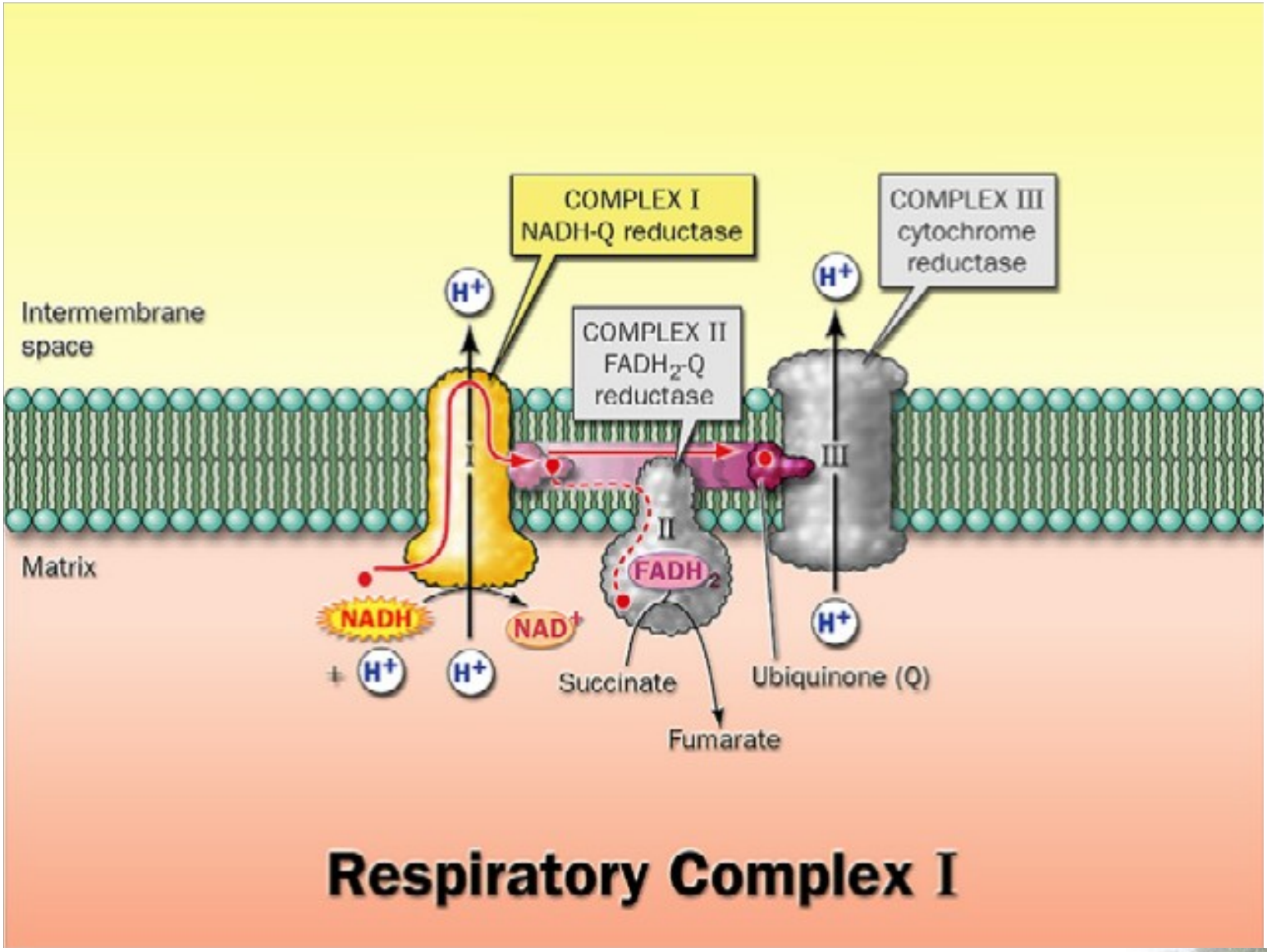


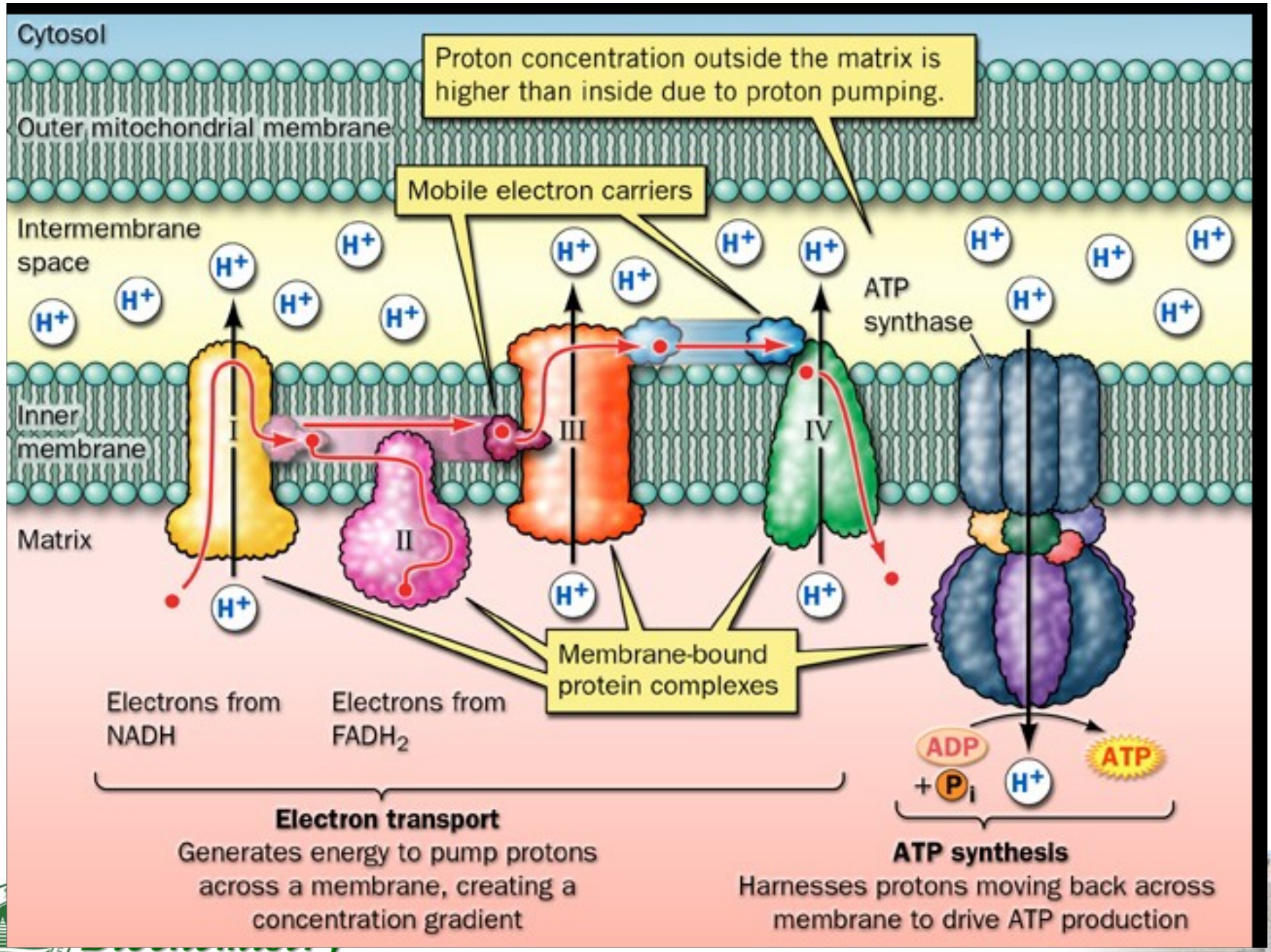
Introduction

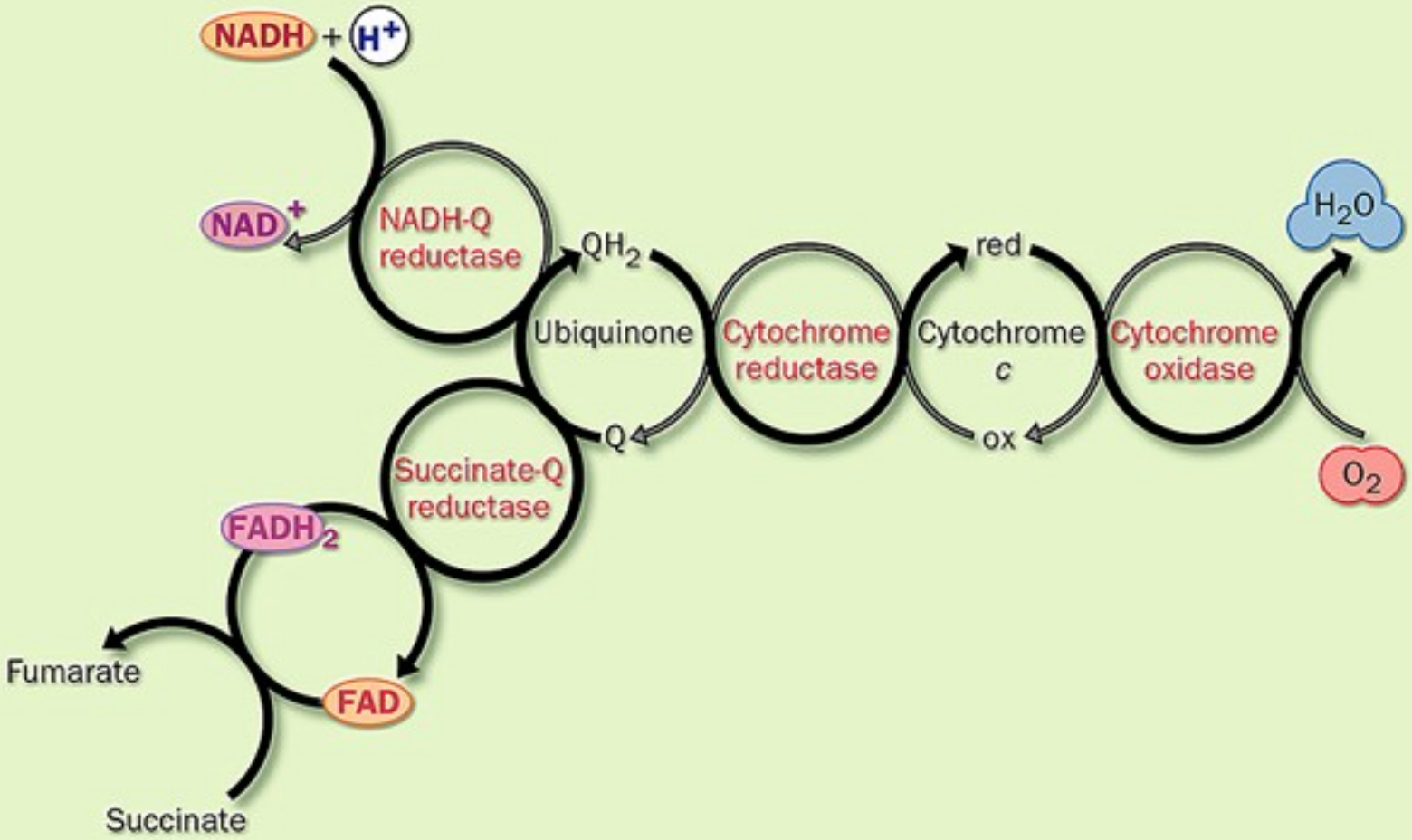
- Mitochondria use energy to produce ATP.
- The energy source is a proton gradient.
- The gradient is maintained by the oxidation of NADH and $FADH_2$.
- NADH and $FADH_2$ oxidation involve a chain of catalytic components.

play




COMPLEX I
NADH-Q reductase
COMPLEX II
FADH2-Q
reductase
COMPLEX III
cytochrome
reductase
Intermembrane
space
Matrix
H+
NADH
NAD+
FADH2
Succinate
Fumarate
Ubiquinone (Q)
I
II
III
Respiratory Complex I


Cytosol
Proton concentration outside the matrix is higher than inside due to proton pumping.
Outer mitochondrial membrane
Mobile electron carriers
Intermembrane space
H+
ATP synthase
Inner membrane
I
II
III
IV
Matrix
Membrane-bound protein complexes
Electrons from NADH
Electrons from $FADH_2$
ADP
+ P_i
ATP
Electron transport
Generates energy to pump protons across a membrane, creating a concentration gradient
ATP synthesis
Harnesses protons moving back across membrane to drive ATP production



Respiratory Chain

TABLE 19–2 Standard Reduction Potentials of Respiratory Chain and Related Electron Carriers

Redox reaction (half-reaction)	*E'° (V)*
$2H^+ + 2e^- \longrightarrow H_2$	−0.414
$NAD^+ + H^+ + 2e^- \longrightarrow NADH$	−0.320
$NADP^+ + H^+ + 2e^- \longrightarrow NADPH$	−0.324
NADH dehydrogenase (FMN) + $2H^+ + 2e^- \longrightarrow$ NADH dehydrogenase ($FMNH_2$)	−0.30
Ubiquinone + $2H^+ + 2e^- \longrightarrow$ ubiquinol	0.045
Cytochrome *b* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *b* (Fe^{2+})	0.077
Cytochrome c_1 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome c_1 (Fe^{2+})	0.22
Cytochrome *c* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *c* (Fe^{2+})	0.254
Cytochrome *a* (Fe^{3+}) + $e^- \longrightarrow$ cytochrome *a* (Fe^{2+})	0.29
Cytochrome a_3 (Fe^{3+}) + $e^- \longrightarrow$ cytochrome a_3 (Fe^{2+})	0.35
$\frac{1}{2}O_2 + 2H^+ + 2e^- \longrightarrow H_2O$	0.8166



It was widely believed that ATP synthesis occurs by chemical coupling

High energy intermediates similar to 1,3-bisphophoglycerate (which is used for substrate-level phosphorylation) was once proposed to be produced first from the electron flows on both the mitochondrial and chloroplast membranes.

Phophorylated protein intermediates (as formed in the action of phosphoglycerate mutase and phosphoglucomutase) were also hypothesized.

But neither were ever revealed despite intense efforts by many investigators over several decades.

